Scoped O.1 on the Prescribed Cone

Apex Heat Moments, a Constant-Profile Slavnov–Taylor Identity, and Domain–Readout Boundaries

Version 12; revision of `XVI-O1-CONE-v11.tex`:
closed mixed mode functional, derived D04 pairing, 40-digit M_{10}, and wall certificates

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Abstract

This is Version 12 of the scoped O.1 cone audit; it revises XVI-O1-CONE-v11. The cone and the CR and F realizations remain prescribed inputs. New in v12 (Section 39; hashes in Appendix F):

(1) A closed form for the mixed r^2 mode functional (identified, not derived): with $h = (\mu+\nu)/2$ and $d = \mu - \nu$, $F = \frac{h^3}{3} + (\frac{d^2}{6} - \frac{1}{3})h + d^2(d^2-4)\Psi(h)$, where $\Psi = \frac{h^3}{48} + \frac{h^2}{96} - \frac{5h}{288} - \frac{1}{96} - \frac{h^2(h^2-1)}{48}\psi_1(h)$. It agrees with all 414 independent quadratures to 5×10^{-28} relative. It makes the large-k asymptotics exact: only odd powers occur, and the P0 pole is $-\alpha/40$.

(2) D04 pairing and regulator. The frame rotation $e^{\pm i\theta}$ shifts k by $\pm\alpha$, which is the channel offset itself. The pairing is therefore the same-k rule, and the v11 unit-shift alternative is excluded. The regulators R_k, R_h, R_μ, R_ν differ by an explicit local counterterm, verified to 10^{-39}. XVI-P's prescription fixes local terms by normalization, so D04 is unique once R_h is declared. M_{10} is given to 40 digits: $M_{10}(\frac{3}{10}) = 0.1157438825958185096384863910924 3$, and $M_{10}^{R_h}(1) = (1 - 40\log 2 - 20\gamma)/800$.

(3) Exact $\mathbb{Q}(i)$ certificates at the chamber walls $\alpha = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}$: the ranks are 27, 33, 27 (CR) and 27, 33, 30 (F). The exceptional set in $(\frac{1}{4}, 1)$ is exactly the three walls.

Still OPEN: the weight-$\pm1, \pm2$ traces and their log/finite split, the 84 mixed weight-0 sums, the jet dictionary, the 20 jet Ward combinations, the 34-/18-dim projections, e_5, e_6, and the transverse loop momentum (feasible, not started). O.1 on the cone remains `NOT_CLOSED`; `bulk_symmetry_restored` remains false.

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1 Question, evidence, and revision contract

1.1 The scoped question

The target is transport of the separately scoped XVI-Q two-point record to a prescribed conical product. The fixed scope is:

(i) one loop and coupling order $O(\kappa^2)$;

(ii) the two-point sector with vanishing declared scalar mass;

(iii) frozen de Donder gauge;

(iv) dimensional reduction, with quasi-d loop momenta and quasi-four-dimensional vertex indices;

(v) the XVI-P p22 contract and the declared graph conventions;

(vi) a comparison of CR and F realizations without selecting either by a finite readout.

The later p23 excerpt is not silently substituted for p22. The frozen XVI-P, XVI-Q, and XVI-R records are source-qualified inputs and comparisons [18, 19, 20]. The immediate sources are the supplied v6 manuscript and its script collection [16, 17].

The central analytical question is not merely whether a conical heat coefficient exists. It is whether a complete differentiated graph, on its specified operator domain and in its specified meromorphic prescription, reduces to that coefficient and satisfies the required interacting identities.

1.2 Evidence classes

Analytical proof.
A statement proved below from the displayed definitions and explicit mathematical inputs. Its proof does not depend on a claim that a program was run.

Declared catalogue.
The explicit arrays in Section 8. Algebraic identities of these arrays are proved locally. Identification with complete cone graph residues is conditional on Assumption 7.1.

Historical executed record.
The v6 source reports executions, archived logs, and SHA-256 comparisons. Those reports are preserved as attribution to v6, not represented as a new execution in this revision.

Embedded verification program.
Appendix A contains a consolidated, corrected program whose purpose and acceptance conditions are stated explicitly. It is supplied as source, not as an executed certificate. In v8 it was extracted unchanged and executed (Section 13.2).

Executed in v8.
A statement whose evidence is a v8 run on the I-B audit box: a named script under `code/v8/`, its return code, and a log under `logs/v8/`, all hashed in Appendix B. Exact runs use $\mathbb{Q}(\mathrm{i})$ arithmetic; numerical runs are labelled corroboration. An executed statement about the declared catalogue remains conditional on Assumption 7.1.

Documentary comparison.
A version-qualified statement from another manuscript, used to keep models and targets distinct. Such a citation is not an additional graph hypothesis unless explicitly declared as one.

What is self-contained here

The heat-moment proofs, the closed Mellin calculation, the catalogue identities, the finite-channel separations, and the conditional graph-reduction theorem are self-contained at the level stated. The frozen vertex arrays and raw historical logs are not embedded in the material supplied for this revision. Consequently this single LaTeX file is not a self-contained rebuild of the complete loop calculation. The source listings make routing and reduction logic inspectable; they do not manufacture the missing input arrays.

1.3 What v7 changes

The revision makes four substantive improvements.

(a) It places the audit on the I-B host and gives a standing domain ledger.

(b) It proves the D01 Mellin value in closed form, without requiring a symbolic simplifier or numerical quadrature.

(c) It corrects several overstrong or ambiguous statements: reflection-even invariance, uniqueness of extension, mixed-key coverage, modular certificates, and numerical evidence.

(d) It promotes the continuum/preparation/instrument split and the I-C A–F separations into the release boundary.

The original coefficient values are retained. No new complete cone graph value is claimed.

1.4 What v8 changes

Version 8 keeps the v7 structure and every v7 proof. It adds executed evidence, and every executed statement names its artifact (Appendix B).

(a) The v7 listing was extracted and run. Each of the fifteen v7 corrections receives a verdict (Section 13.2).

(b) An exact, complete span certificate replaces the earlier modular and random-rank evidence. It fixes the count of 26 Ward combinations and the inventory dimension, each at named (α, branch) (Section 14).

(c) The 34-candidate basis is projected onto its twelve constant-profile coordinates. The hh part of the 18-dimensional engine map is made explicit on six of its 18 coordinates, and four of the six exact values are fixed (Section 15).

(d) Hidden mixed-channel pairs are found. As a result the v6 hh branch difference becomes conditional. A Duhamel lemma and a reduction formula for mixed traces are proved (Section 16).

(e) Evanescent pole coefficients are computed under a declared slice-DRED prescription (Section 17).

(f) The open-obligation table and the release theorem are re-stated with the labels OPEN, PARTIAL, CONDITIONAL and `CLOSED_PENDING_INDEPENDENT_REVIEW`.

No v6 number changes value. One v6 statement changes status, from unconditional on the declared catalogue to conditional, as described in (d).

1.5 Programme non-transfer

The cone and CR branch remain inputs. No Einstein dynamics, physical tensor Hessian, spacetime emergence, gravitational Page curve, or inherited quantum coefficient is constructed. XVI-N is

not discharged; the downstream XV-B admission exit is not opened. The historical H2 ending is retained version-specifically, the narrower I2 construction is not withdrawn, and the full I3/J2 characteristic maps remain separate [39, 40, 41].

The extended Hawk05 discussion supplied with the revision request concerns finite access and Page proxies. It is relevant only to the reporting boundary below; it is not imported as an apex calculation or as gravitational dynamics [38].

2 The prescribed I-B host and the typed domain ledger

2.1 Geometry

The Euclidean spectral host is

$$X_\alpha = C_\alpha \times \mathbb{R}^2, \qquad \mathrm{d}s_E^2 = \mathrm{d}r^2 + \alpha^2 r^2 \mathrm{d}\varphi^2 + \mathrm{d}y_0^2 + \mathrm{d}y_3^2, \quad 0 \le \varphi < 2\pi, \quad \alpha > 0. \tag{2.1}$$

The transverse measure is

$$\mathrm{d}V_\perp = \alpha r \, \mathrm{d}r \, \mathrm{d}\varphi.$$

The apex stratum in the completed product is

$$\mathcal{W} = \{r = 0\} \times \mathbb{R}^2.$$

For the deficit comparison we use $0 < \alpha \le 1$, with deficit $2\pi(1-\alpha)$.

The Lorentzian regular host has $r > 0$ and metric

$$\mathrm{d}s_L^2 = \mathrm{d}t^2 - \mathrm{d}r^2 - \alpha^2 r^2 \mathrm{d}\varphi^2 - \mathrm{d}z^2. \tag{2.2}$$

This is the I-B prescribed conical host, with an overall signature convention chosen to agree with the pole catalogue [21]. The causal geometry is unchanged by the opposite overall signature used in some companion descriptions.

Proposition 2.1 (The regular host is not globally hyperbolic)**.** *The spacetime* (2.2) *with its apex removed is not globally hyperbolic.*

Proof. Fix $r_* > 0$, an angle, and a longitudinal position. Choose $a > r_*$, and let p and q be the corresponding points at times $-a$ and a. For any sequence $r_j \downarrow 0$, the points at time zero and radius r_j, with the same angle and longitudinal position, lie in $J^+(p) \cap J^-(q)$: the radial spatial distance is less than a. They have no convergent subsequence in the regular host. The causal diamond is therefore not compact. □

This is the same geometric obstruction recorded in I-B-G [22, `prop:nonGH`]. A free inverse is accordingly tied to an extension and a test domain. Ordinary globally hyperbolic uniqueness must not be substituted for those data.

The apex is a worldsheet defect, not a derived horizon. There is no state-dependent metric update.

2.2 Standing host and domain ledger

Table 1: Standing comparison ledger. Companion claims are used only within the indicated source scope.

Input or comparison	Source	Permitted use	Excluded inference
Prescribed cone, deficit, and holonomy	I-B [21]	Host geometry and installed parameters	A horizon, metric update, or matter-driven sourcing law.
Scalar Friedrichs Green operator and finite speed	I-B-G, `thm:green`, `thm:speed` [22]	Free inverse and support control on its declared domain	An exact sequence on ordinary compact-spacetime tests; `rem:compact-exact` explicitly retains that gap.
Maxwell–Friedrichs photon with scalar ghost	I-B-G-S, `thm:spin1` [23]	Source-tested free reference	An interacting Slavnov–Taylor identity or discharge of a named cone Ward combination.
CR source-test selection	I-B-G-S, `def:geo`, `thm:selection`, `thm:spin2` [23]	Declared mathematical comparison branch	A physical smooth-core selection theorem.
Positive tensor-core exclusion of idealized CR	I-B-G-S-B, `cor:exclude`, `thm:contrast`, `prop:warddefect` [24]	Source-qualified comparison for its declared tensor families	An independently completed tensor proof in this paper, or a choice made by the present ST row.
Friedrichs Wick square, stress, and holonomy-odd current	I-B-VP and I-B-VP-1 [25, 26]	One-point and compact-test infrared comparison	Absence of infrared poles in massless loop graphs, or a two-point graph prescription.
Frozen-source Maxwell response	I-B-VP-4 [27]	Classical Maxwell comparison, if such a detector is declared	An interacting Ward identity.
Smooth-core finite parts and scalar/Dirac selection	I-B-VP-5-Ex1-B, Ex2, Ex2-B [28, 29, 30]	Qualifications of core limits and finite-part universality	An interacting spin-2 CR graph calculation.

The CR/F row proved below is labelled against this table: it is a constant-profile catalogue identity on two prescribed branches. It is neither the free photon theorem nor a tensor-core selection theorem.

Remark 2.2 (Sourcing is a separate equation). I-B's `prop:not-source` separates symmetry, conservation, and trace conditions from an Einstein sourcing equation [21]. For example, on a flat chart a nonzero constant symmetric conserved traceless tensor need not equal the Einstein tensor of that chart. The same refusal applies to every heat coefficient below.

2.3 Self-adjointness is not destructive access

For a radial Bessel expression

$$L_\nu = -\partial_r^2 - r^{-1}\partial_r + \nu^2 r^{-2}$$

in $L^2((0,\infty), r\,\mathrm{d}r)$, the endpoint pairing is

$$\mathfrak{b}(u,v) = \lim_{r\downarrow 0} r\big(\overline{u}\,v' - \overline{u'}\,v\big). \tag{2.3}$$

Proposition 2.3 (Closed endpoint dynamics)**.** *A self-adjoint realization A generates a unitary Schrödinger group $e^{-\mathrm{i}tA}$. On sufficiently regular domain solutions its total endpoint probability flux vanishes. A bound eigenstate is not an absorbing output channel.*

Proof. The spectral theorem gives unitarity. Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u(t)\|^2 = 2\,\mathrm{Re}\langle u(t), -\mathrm{i}Au(t)\rangle = 0.$$

The endpoint current is proportional to $\mathrm{Im}(r\overline{u}\,u')$, which vanishes in the total boundary pairing on a self-adjoint domain. An eigenstate evolves by a phase. □

These are standard endpoint facts [46, 47]. They do not identify ghosts or gauge-fixed tensor components with a positive physical probability space.

Self-adjointness alone does not imply positive-energy wave stability: a negative eigenvalue $-\sigma^2$ produces $\ddot{a} - \sigma^2 a = 0$. Conversely, reduced access can be lossy while an enlarged complete dynamics remains unitary. Neither an absorbing boundary nor an apex reservoir is introduced here.

3 Twisted scalar channels and the normalized apex trace

3.1 Channels and subtraction

For $b \in [0, 1)$, the twisted Friedrichs transverse Laplacian A_b^{F} has radial orders

$$\nu_n = \frac{|n+b|}{\alpha}, \qquad n \in \mathbb{Z}.$$

It chooses r^{ν_n} at zero, with the no-logarithm condition when $\nu_n = 0$.

Its diagonal heat kernel, relative to $\mathrm{d}V_\perp$, is

$$K_b^{\mathrm{F}}(s; r, \varphi; r, \varphi) = \frac{e^{-z}}{4\pi\alpha s} \sum_{n\in\mathbb{Z}} I_{|n+b|/\alpha}(z), \qquad z = \frac{r^2}{2s}. \tag{3.1}$$

We use the Bessel heat representation and standard modified Bessel identities [45, 46].

The infinite cone has infinite ordinary heat trace. The apex functional is therefore defined by subtraction, not by writing an unqualified L^2-trace.

Definition 3.1 (Abel-subtracted apex functional)**.** For $u, s > 0$, let

$$T_b(u) = \int_{C_\alpha} e^{-(\cosh u - 1)r^2/(2s)} K_b^{\mathrm{F}}(s; x, x)\,\mathrm{d}V_\perp, \tag{3.2}$$

$$T_{\mathrm{bulk}}(u) = \frac{\alpha}{2(\cosh u - 1)}. \tag{3.3}$$

Define

$$\mathrm{Tr}_{\mathrm{ap}}\, e^{-sA_b^{\mathrm{F}}} = \lim_{u\downarrow 0}(T_b(u) - T_{\mathrm{bulk}}(u))$$

when the limit exists.

Theorem 3.2 (Normalized twisted apex constant)**.** *For every $\alpha > 0$, $b \in [0, 1)$, and $s > 0$,*

$$\mathrm{Tr}_{\mathrm{ap}}\, e^{-sA_b^{\mathrm{F}}} = \mathcal{C}_b^{\mathrm{F}} = \frac{1}{12}\left(\frac{1}{\alpha} - \alpha\right) - \frac{b(1-b)}{2\alpha} = \frac{B_2(b)}{2\alpha} - \frac{\alpha}{12}, \tag{3.4}$$

where $B_2(b) = b^2 - b + \frac{1}{6}$.

For a compact proper-time superposition

$$f(\lambda) = \int_{s_-}^{s_+} w(s) e^{-s\lambda}\,\mathrm{d}s, \qquad 0 < s_- < s_+ < \infty, \quad w \in L^1,$$

the corresponding apex functional satisfies

$$\mathrm{Tr}_{\mathrm{ap}}\, f(A_b^{\mathrm{F}}) = \mathcal{C}_b^{\mathrm{F}} f(0). \tag{3.5}$$

Proof. For $\rho > -1$ and $u > 0$,

$$\int_0^\infty e^{-z\cosh u} I_\rho(z)\,\mathrm{d}z = \frac{e^{-\rho u}}{\sinh u}.$$

The nonnegative Bessel summands permit interchange with the radial integral. Summing the geometric series gives

$$T_b(u) = \frac{\cosh((\frac{1}{2}-b)u/\alpha)}{2\sinh u\,\sinh(u/(2\alpha))}. \tag{3.6}$$

Its Laurent expansion is

$$T_b(u) = \frac{\alpha}{u^2} + \frac{B_2(b)}{2\alpha} - \frac{\alpha}{6} + O(u^2),$$

whereas

$$T_{\mathrm{bulk}}(u) = \frac{\alpha}{u^2} - \frac{\alpha}{12} + O(u^2).$$

Subtraction proves (3.4). The change of radial variable removed s, so the result is independent of proper time.

On the compact proper-time interval the subtracted expressions are uniformly bounded for sufficiently small u. Dominated convergence and linearity give

$$\mathrm{Tr}_{\mathrm{ap}}\, f(A_b^{\mathrm{F}}) = \mathcal{C}_b^{\mathrm{F}} \int_{s_-}^{s_+} w(s)\,\mathrm{d}s.$$

This is (3.5). □

Host boundary. The background and realization are prescribed. A heat coefficient, conservation identity, or trace identity does not imply that the corresponding object sources the cone.

The untwisted result and the conical heat-kernel method are classical [42, 43, 44]. The calculation above fixes the normalization actually used here. It is not an unrestricted ordinary-trace theorem for arbitrary resolvent products or differentiated graph kernels.

3.2 An image contribution is not a diffracted-front report

Example 3.3 (A pure-image apex contribution)**.** At $\alpha = \frac{1}{2}$, $b = 0$,

$$\mathcal{C}_0^{\mathrm{F}} = \frac{1}{8}.$$

The nonidentity image in the two-fold plane quotient contributes

$$\frac{1}{4\pi s} e^{-r^2/s}.$$

Integrating over its wedge of angle π gives

$$\frac{\pi}{4\pi s}\int_0^\infty r e^{-r^2/s}\,\mathrm{d}r = \frac{1}{8}.$$

Thus an apex heat constant can be nonzero in an image representation without requiring a separately nonzero strictly diffracted-front contribution.

The image-isolated windows and short-window deficit sensitivity of I-B-ECHO concern another functional [31]. They do not identify a window ratio with $\mathcal{C}_b^{\mathrm{F}}$ or with a graph residue. This separation is part of the release theorem, not merely an informal analogy.

4 Exceptional Bessel realizations and CR/F coefficients

4.1 A one-channel replacement

For $0 < \nu < 1$, let A_ν^+ and A_ν^- be the nonnegative homogeneous regular and singular Bessel realizations in $L^2(r\,\mathrm{d}r)$. Their radial heat kernels are

$$k_\nu^\pm(s; r, r') = \frac{1}{2s} e^{-(r^2+r'^2)/(4s)} I_{\pm\nu}\left(\frac{rr'}{2s}\right).$$

The endpoint choices, rather than a periodic relabelling of an ordinary fractional holonomy, distinguish them [46].

Theorem 4.1 (Relative diagonal heat shift)**.** *For $0 < \nu < 1$,*

$$\int_0^\infty (k_\nu^-(s; r, r) - k_\nu^+(s; r, r)) r\,\mathrm{d}r = \nu. \tag{4.1}$$

The relative diagonal integral is absolutely convergent.

Proof. It equals

$$\frac{1}{2}\int_0^\infty e^{-z}\big(I_{-\nu}(z) - I_\nu(z)\big)\,\mathrm{d}z.$$

At zero the worst behavior is $z^{-\nu}$, which is integrable. At infinity use

$$I_{-\nu}(z) - I_\nu(z) = \frac{2\sin(\pi\nu)}{\pi} K_\nu(z).$$

The resulting integrand decays exponentially. With the Abel factor, the integral is

$$\frac{e^{\nu u} - e^{-\nu u}}{2\sinh u} = \frac{\sinh(\nu u)}{\sinh u}.$$

Dominated convergence gives ν. □

Host boundary. The background and realization are prescribed. A heat coefficient, conservation identity, or trace identity does not imply that the corresponding object sources the cone.

4.2 Geometric helicities

For geometric helicity S, the Friedrichs fractional label is

$$b_{\mathrm{F}}(S) = \{S\alpha\}.$$

In the comparison chamber $1/2 < \alpha < 1$, put

$$\nu = 2 - \alpha^{-1} \in (0, 1). \tag{4.2}$$

For each exceptional helicity $S = \pm 2$, CR replaces one regular channel of order ν by the singular realization.

The per-helicity coefficients are

$$c_0 = \frac{1-\alpha^2}{12\alpha}, \qquad c_1 = \frac{5\alpha^2 - 6\alpha + 1}{12\alpha},$$
$$c_2^{\mathrm{F}} = \frac{23\alpha^2 - 36\alpha + 13}{12\alpha}, \qquad c_2^{\mathrm{CR}} = \frac{23\alpha^2 - 12\alpha + 1}{12\alpha}. \tag{4.3}$$

They satisfy

$$c_1 - c_0 = \frac{\alpha - 1}{2}, \qquad c_2^{\mathrm{CR}} - c_0 = 2\alpha - 1,$$

$$c_2^{\mathrm{F}} - c_0 = \frac{(\alpha-1)(2\alpha-1)}{\alpha}, \qquad c_2^{\mathrm{CR}} - c_2^{\mathrm{F}} = \nu. \tag{4.4}$$

These follow from Theorem 3.2 at $b = 0, \alpha, 2\alpha - 1$, and from Theorem 4.1. The polynomial substitution $b = 2\alpha$ reproduces c_2^{CR}, but is justified here only through that specified channel replacement.

At $\alpha = \frac{1}{2}$, continuous evaluation gives

$$c_0 = c_2^{\mathrm{F}} = c_2^{\mathrm{CR}} = \frac{1}{8}, \qquad c_1 = -\frac{1}{8}.$$

For $0 < \alpha < \frac{1}{2}$, the Friedrichs weight-two label is 2α, not $2\alpha - 1$; the displayed F polynomial must not be extended unchanged across that chamber boundary.

Remark 4.2 (The CR flat limit is not an endpoint realization). As $\alpha \uparrow 1$,

$$c_0, c_1, c_2^{\mathrm{F}} \longrightarrow 0, \qquad c_2^{\mathrm{CR}} \longrightarrow 1.$$

This is a one-sided polynomial limit. At $\nu = 1$, $r^{-1} \notin L^2(r\,\mathrm{d}r)$ near zero, so the deficient singular branch used in the proof is no longer available in the same form. No CR operator value at the exactly smooth endpoint is inferred. Spin-dependent nonuniform smooth limits also occur in the established conical heat-kernel literature [43].

5 The weighted apex moment

Definition 5.1 (Weighted Abel subtraction). For $u, s > 0$, define

$$T_b^{(2)}(u) = \int_{C_\alpha} r^2 e^{-(\cosh u - 1)r^2/(2s)} K_b^{\mathrm{F}}(s; x, x)\, \mathrm{d}V_\perp,$$
$$T_{\mathrm{bulk}}^{(2)}(u) = \frac{\alpha s}{(\cosh u - 1)^2}.$$

Their subtracted limit defines $\mathrm{Tr}_{\mathrm{ap}}[r^2 e^{-sA_b^{\mathrm{F}}}]$.

Theorem 5.2 (Normalized r^2-weighted coefficient). *For $\alpha > 0$, $b \in [0, 1)$, and $s > 0$,*

$$\mathrm{Tr}_{\mathrm{ap}}[r^2 e^{-sA_b^{\mathrm{F}}}] = s\,\mathcal{D}_b, \qquad \mathcal{D}_b = -\frac{11\alpha}{180} + \frac{B_2(b)}{3\alpha} - \frac{B_4(b)}{6\alpha^3}, \tag{5.1}$$

where $B_4(b) = b^4 - 2b^3 + b^2 - \frac{1}{30}$. Equivalently,

$$\mathcal{D}_b = \frac{1 + 10\alpha^2 - 11\alpha^4 - 60\alpha^2 b(1-b) - 30b^2(1-b)^2}{180\alpha^3}. \tag{5.2}$$

For the proper-time superpositions of Theorem 3.2,

$$\mathrm{Tr}_{\mathrm{ap}}[r^2 f(A_b^{\mathrm{F}})] = -\mathcal{D}_b f'(0). \tag{5.3}$$

Proof. The radial variable $z = r^2/(2s)$ gives

$$\frac{T_b^{(2)}(u)}{s} = -\frac{2}{\sinh u}\frac{\mathrm{d}}{\mathrm{d}u} T_b(u).$$

To expose the constant term, write

$$T_b(u) = \frac{\alpha}{u^2} + t_0 + t_2 u^2 + O(u^4),$$

where expansion of (3.6) yields

$$t_2 = \frac{B_4(b)}{24\alpha^3} - \frac{B_2(b)}{12\alpha} + \frac{7\alpha}{360}.$$

Therefore

$$\frac{T_b^{(2)}(u)}{s} = \frac{4\alpha}{u^4} - \frac{2\alpha}{3u^2} + \frac{7\alpha}{90} - 4t_2 + O(u^2).$$

The bulk expression is

$$\frac{\alpha}{(\cosh u - 1)^2} = \frac{4\alpha}{u^4} - \frac{2\alpha}{3u^2} + \frac{11\alpha}{180} + O(u^2).$$

The pole terms cancel and the constant difference is (5.1). Expanding the Bernoulli polynomials gives (5.2).

Dominated convergence on the compact proper-time support gives

$$\mathrm{Tr}_{\mathrm{ap}}[r^2 f(A_b^{\mathrm{F}})] = \mathcal{D}_b \int s\, w(s)\, \mathrm{d}s = -\mathcal{D}_b f'(0).$$

□

Host boundary. The background and realization are prescribed. A heat coefficient, conservation identity, or trace identity does not imply that the corresponding object sources the cone.

Proposition 5.3 (Weighted singular-channel shift)**.** *Replacing one regular channel of order $0 < \nu < 1$ by its singular realization changes the weighted coefficient by*

$$\Delta\mathcal{D} = \frac{2\nu(1-\nu^2)}{3}. \tag{5.4}$$

In the weight-two comparison chamber,

$$\mathcal{D}^{\mathrm{F}}_{S=2} = \mathcal{D}_{2\alpha-1}, \qquad \mathcal{D}^{\mathrm{CR}}_{S=2} = \mathcal{D}_{2\alpha-1} + \Delta\mathcal{D} = \mathcal{D}_{2\alpha}.$$

The last expression is polynomial evaluation, not an assertion that 2α is a fractional twist.

Proof. For one channel the weighted relative Abel integral divided by s is

$$-\frac{1}{\sinh u}\partial_u\left(\frac{2\sinh(\nu u)}{\sinh u}\right).$$

Its limit is (5.4). The weighted diagonal relative integral is convergent at zero because its Bessel integrand is $O(z^{1-\nu})$, and it is exponentially convergent at infinity. Substitution in (5.2) proves the final equality. □

In particular,

$$\mathcal{D}_0 = \frac{(1-\alpha^2)(1+11\alpha^2)}{180\alpha^3}, \quad \mathcal{D}_{S=1} = \mathcal{D}_\alpha, \quad \mathcal{D}^{\mathrm{F}}_{S=2} = \mathcal{D}_{2\alpha-1}, \quad \mathcal{D}^{\mathrm{CR}}_{S=2} = \mathcal{D}_{2\alpha}.$$

All four tend to zero as $\alpha \uparrow 1$. Thus the weighted CR polynomial has no flat remnant. This vanishing does not itself establish convergence of the singular operator family at the endpoint.

Theorem 5.4 (Exact orbifold and twisted-orbifold identities)**.** *For every integer $N \geq 2$ and $j \in \{0, \ldots, N-1\}$,*

$$\mathcal{D}_{j/N}(1/N) = \frac{1}{4N}\sum_{k=1}^{N-1} \frac{\cos(2\pi jk/N)}{\sin^4(\pi k/N)}. \tag{5.5}$$

Consequently,

$$\mathcal{D}_{j/N}(1/N) = \frac{(N^2-1)(N^2+11) - 60j(N-j) - 30j^2(N-j)^2}{180N}. \tag{5.6}$$

Proof. On the rotation quotient, the k-th nonidentity image has diagonal kernel

$$(4\pi s)^{-1}e^{-r^2\sin^2(\pi k/N)/s}.$$

The character contributes the corresponding holonomy phase. Pairing conjugate images replaces the phase by its cosine. The wedge has angle $2\pi/N$, and direct integration gives

$$\frac{2\pi/N}{4\pi s}\int_0^\infty r^3 e^{-r^2\sin^2(\pi k/N)/s}\,\mathrm{d}r = \frac{s}{4N\sin^4(\pi k/N)}.$$

The identity image is precisely the bulk subtraction. The character sector has orders $|Nn+j|$, hence $\alpha = 1/N$, $b = j/N$. Theorem 5.2 proves (5.5). Substitution in (5.2) proves (5.6). □

Examples include

$$\mathcal{D}_0(1/2) = \frac{1}{8}, \qquad \mathcal{D}_{1/3}(1/3) = -\frac{4}{27}, \qquad \mathcal{D}_{1/4}(1/4) = -\frac{1}{16}, \qquad \mathcal{D}_{2/5}(1/5) = -\frac{16}{25}.$$

Remark 5.5 (Corrected literature normalization). Fursaev uses the full opening angle $\theta = 2\pi\alpha$. His coefficient $C_4^{\mathrm{Fur}}(\theta)$, in equation (2.15) of [42], is

$$C_4^{\mathrm{Fur}}(2\pi\alpha) = \frac{(\alpha^{-2}-1)(\alpha^{-2}+11)}{90}.$$

The normalization relation is

$$\boxed{\mathcal{D}_0 = \frac{\alpha}{2}C_4^{\mathrm{Fur}}(2\pi\alpha).}$$

This replaces the memory-qualified identification in v6. It follows also by inserting $f = r^2$ in that reference's weighted heat expansion. No numerical spin-two coefficient is imported from the literature.

6 The D01 derivative obstruction and its Mellin value

6.1 The actual obstruction

Let ν be (4.2). The exceptional relative radial kernel is

$$\Delta k_s(r,r') = \frac{\sin(\pi\nu)}{\pi s}e^{-(r^2+r'^2)/(4s)}K_\nu\left(\frac{rr'}{2s}\right).$$

Consider the raising expression

$$X = \partial_r - \frac{\nu}{r}, \qquad \mathcal{R}_s(r) = \left[X_r X_{r'}\Delta k_s(r,r')\right]_{r'=r}.$$

Theorem 6.1 (Local obstruction)**.** *For every $0 < \nu < 1$,*

$$\mathcal{R}_s(r) = a_0\, r^{-2\nu-2} + O(r^{-2\nu}), \qquad a_0 = \frac{2\nu^2(4s)^\nu}{s\,\Gamma(1-\nu)} > 0. \tag{6.1}$$

Thus the ordinary diagonal integral $\int_0^\epsilon r\mathcal{R}_s(r)\,\mathrm{d}r$ diverges. The expression X does not map the full singular Bessel domain into the next L^2 channel.

Proof. The leading singular part of Δk_s is

$$\frac{(4s)^\nu}{2s\,\Gamma(1-\nu)}(rr')^{-\nu}.$$

Applying X_r and $X_{r'}$ multiplies it by $4\nu^2/(rr')$, proving (6.1). Moreover

$$X(r^{-\nu}) = -2\nu r^{-\nu-1}, \qquad \int_0^\epsilon r|r^{-\nu-1}|^2\,\mathrm{d}r = \infty.$$

This excludes the unrestricted domain-preserving intertwining statement. It does not exclude all densely restricted operators or every possible distributional extension. □

6.2 A closed Mellin formula

Theorem 6.2 (Exact Mellin continuation)**.** *For* $\operatorname{Re} z > 2\nu$*, define*

$$M_s(z) = \int_0^\infty r^{1+z}\mathcal{R}_s(r)\,\mathrm{d}r.$$

With $t = z/2$*, its meromorphic continuation is*

$$\boxed{M_s(z) = \frac{\sin(\pi\nu)}{2\sqrt{\pi}}\, s^{t-1}\, \frac{t\,\Gamma(t+\nu+2)\Gamma(t-\nu)}{\Gamma(t+\frac{3}{2})}.} \tag{6.2}$$

It is regular at $z = 0$ *for* $0 < \nu < 1$*, and*

$$M_s(0) = 0, \qquad M_s'(0) = -\frac{\nu+1}{2s}. \tag{6.3}$$

Possible poles are contained in

$$\{2\nu - 2j : j \in \mathbb{N}_0\} \;\cup\; \{-2\nu - 4 - 2j : j \in \mathbb{N}_0\}.$$

Some listed poles can cancel.

Proof. Put $x = r^2/(2s)$ and $A = \sin(\pi\nu)/(\pi s)$. Direct differentiation gives

$$\mathcal{R}_s(r) = \frac{A}{2s}e^{-x}H_\nu(x),$$

where

$$H_\nu = xK - 2xK' + K' + xK'' - 2\nu(K' - K) + \frac{\nu^2}{x}K, \qquad K = K_\nu(x).$$

The Bessel equation $xK'' + K' = (x + \nu^2/x)K$ reduces this to

$$H_\nu = 2\left[(x+\nu)K - (x+\nu)K' + \frac{\nu^2}{x}K\right].$$

For $\operatorname{Re}\mu > |\operatorname{Re}\nu|$, let

$$I_\nu(\mu) = \int_0^\infty x^{\mu-1}e^{-x}K_\nu(x)\,\mathrm{d}x = \frac{\sqrt{\pi}\,\Gamma(\mu+\nu)\Gamma(\mu-\nu)}{2^\mu\Gamma(\mu+\frac{1}{2})}. \tag{6.4}$$

For completeness, use

$$K_\nu(x) = \frac{1}{2}\int_0^\infty y^{\nu-1}e^{-x(y+y^{-1})/2}\,\mathrm{d}y.$$

Integrating in x first gives

$$I_\nu(\mu) = 2^{\mu-1}\Gamma(\mu)\int_0^\infty \frac{y^{\mu+\nu-1}}{(1+y)^{2\mu}}\,\mathrm{d}y.$$

The beta integral and the gamma duplication formula give (6.4). This also identifies the $a = 1$ limit of the standard Laplace–Bessel formula in DLMF 10.43.22 [45].

In the initial convergence half-plane, integration by parts has no endpoint contribution. It yields

$$M_s(z) = A(2s)^t\big[(t+1)I_\nu(t+1) + \nu(\nu+t)I_\nu(t)\big].$$

Since

$$\frac{I_\nu(t+1)}{I_\nu(t)} = \frac{t^2-\nu^2}{2t+1},$$

the bracket is

$$\frac{t(t+\nu)(t+\nu+1)}{2t+1} I_\nu(t).$$

Substitution of (6.4) proves (6.2) on a nonempty convergence half-plane and hence by meromorphic continuation.

At $t = 0$, the gamma factors are finite for $0 < \nu < 1$, and the explicit factor t gives $M_s(0) = 0$. Using

$$\Gamma(\nu+2)\Gamma(-\nu) = -\frac{\pi(\nu+1)}{\sin(\pi\nu)}$$

gives the derivative in (6.3). The pole statement follows from the numerator gamma functions. □

6.3 What is and is not unique

Remark 6.3 (Correction of the extension claim). The *Mellin prescription* is unambiguous once its initial integral and meromorphic continuation are specified. On smooth rotationally invariant transverse tests, the local Taylor powers are even. The singular heads $r^{-2\nu-2+2j}$ therefore have no resonance at $z = 0$ for $0 < \nu < 1$. In particular, $\nu = \frac{1}{2}$ is not a resonance of this even radial test class.

This does *not* prove that the full heat-time-dependent diagonal has a unique scale-covariant extension. At fixed s, $\mathcal{R}_s$ is not a homogeneous function of r. If δ_{rad} denotes the endpoint evaluation functional relative to radial measure, then adding

$$\frac{\chi(\nu)}{s}\,\delta_{\mathrm{rad}}$$

preserves parabolic dimensional covariance but changes the integrated finite part. Thus scale covariance alone does not select the Mellin value.

The earlier script comment "unique iff 2ν is not an integer" refers to a different unrestricted half-line Taylor class and must not be used as the uniqueness theorem for smooth two-dimensional radial tests.

Corollary 6.4 (The remaining D01 prescription obligation)**.** *The forbidden composite is exactly Mellin-null under the prescription in Theorem 6.2. If the complete D01 graph reduction identifies this composite and the frozen prescription selects that continuation without additional contact terms, it supplies no correction to the declared constant-profile catalogue.*

The italicized identification and prescription statements remain part of the graph-specific contract. They are not proved by $M_s(0) = 0$.

No uniform domain theorem at $\alpha = \frac{1}{2}$ or $\alpha = 1$ follows from this interior-chamber Mellin identity. Continuity of a selected polynomial is not uniform meromorphic control of the joint endpoint family.

7 Conditional reduction of a complete graph

7.1 Constant profiles are local jets

A constant transverse profile means a local jet constant near the apex, with longitudinal Fourier dependence. It is not a globally square-integrable constant section of an infinite cone.

The helicity-zero tensor restriction is

$$h_{11} = h_{22} = f_0,$$

together with the longitudinal symmetric block. The individual coordinate directions h_{11}, h_{22}, and h_{12} are useful array diagnostics, but are not separately globally admissible constant helicity-zero tensor profiles on a generic cone.

A closed transverse loop has one spectral trace. Once that trace has been reduced to a function of one identified operator, it receives one coefficient c_S, not a product of one heat coefficient for each propagator.

Assumption 7.1 (Graph-specific reduction contract)**.** For a specified complete graph G, external jet class, routing, and branch, require:

(R1) **Complete graph data.** All vertices, propagator numerators, source conventions, phases, statistics signs, symmetry factors, and independent-coordinate normalizations are specified and matched to the frozen contract.

(R2) **Operator reduction.** The fully contracted transverse expression is reduced to a finite sum

$$f_{G,S}(A_S; l, p),$$

on identified channel domains, plus a remainder proved to have zero apex UV residue.

(R3) **Domain and trace legitimacy.** Differentiated intertwiners, integrations by parts, and trace rearrangements are valid on those domains. Every boundary or finite-part defect is included or proved absent in the complete permitted sum.

(R4) **UV/IR separation.** An auxiliary infrared prescription distinguishes the UV residue from massless infrared poles. A scaleless integral is not assigned its UV pole merely by discarding its infrared part.

(R5) **Meromorphic interchange.** The apex heat identity extends to the regulated spectral functions, and the transverse subtraction, longitudinal integration, cutoff removal, and Laurent-residue operation commute in the stated prescription.

Theorem 7.2 (Conditional spectral apex reduction)**.** *Under Assumption 7.1,*

$$\operatorname{Res}_{\mathrm{UV}} G_{\mathrm{ap}}(p) = \sum_S c_S \operatorname{Res}_{\mathrm{UV}} \left[\mu^{4-d} \int \frac{\mathrm{d}^{d-2} l_{\|}}{(2\pi)^{d-2}} f_{G,S}(0; l_{\|}, p) \right]. \tag{7.1}$$

The factors, multiplicities, phases, and statistics are included in $f_{G,S}$.

Proof. Retain the auxiliary proper-time and infrared cutoffs. By (R2), the complete graph is a finite sum of functions of identified channel operators, modulo a residue-free remainder. Apply Theorem 3.2 and the channel replacement of Theorem 4.1 to each regulated heat superposition. This replaces its transverse apex functional by $c_S f_{G,S}(0; l, p)$.

Condition (R3) ensures that no boundary or trace defect has been lost. Condition (R5) permits the stated integration and residue operations. The remainder has zero residue by (R2), giving (7.1). □

Zero transverse external momentum does not prove (R2). A mixed-domain bubble or a domain-changing differential operator need not be a function of one A_S.

The v6 source reports that the archived engine data discharge (R1), and that parts of (R3) were checked [16]. The supplied script listings allow inspection of the declared routing and prefactors. Without the frozen input files, their completeness and byte identity cannot be independently re-established from the v7 material alone. v8 replays the frozen engine (read-only copies, tree hashes in Section 13.6) but does not certify completeness. In all cases, (R2), (R4), and (R5) remain hypotheses.

7.2 Longitudinal UV normalization

Write

$$\varepsilon = 4 - d, \qquad D = d - 2 = 2 - \varepsilon.$$

Lemma 7.3 (Angular coefficient of a simple UV pole)**.** *Suppose an infrared-regular Euclidean longitudinal integrand has*

$$I(R\omega; p) = R^{-2} g(\theta; p) + O(R^{-2-\delta}), \qquad \delta > 0,$$

uniformly in angle, and the dimensional angular continuation is regular at $D = 2$. Then its $1/\varepsilon$ coefficient is

$$\frac{1}{(2\pi)^2} \int_0^{2\pi} g(\theta; p)\, \mathrm{d}\theta. \tag{7.2}$$

A prescribed Wick phase multiplies this coefficient separately.

Proof. Above a fixed large radius,

$$\int_{R_0}^{\infty} R^{D-1} R^{-2}\, \mathrm{d}R = \int_{R_0}^{\infty} R^{-1-\varepsilon}\, \mathrm{d}R = \frac{R_0^{-\varepsilon}}{\varepsilon}.$$

The remainder is UV integrable. At $D = 2$, the measure supplies $(2\pi)^{-2}\mathrm{d}\theta$. □

An $O(\varepsilon)$ angular or numerator correction does not change this simple-pole coefficient *provided there is no hidden higher pole, singular projection, or evanescent mixing that invalidates that premise.* It can change finite parts. Consistent dimensional reduction is not automatic preservation of a proposed boundary identity [48].

8 The declared constant-profile catalogue

8.1 Coordinate and normalization conventions

Use the independent-coordinate order

$$0 \leftrightarrow 00, \quad 1 \leftrightarrow 01, \quad 2 \leftrightarrow 02, \quad 3 \leftrightarrow 03, \quad 4 \leftrightarrow 11,$$
$$5 \leftrightarrow 12, \quad 6 \leftrightarrow 13, \quad 7 \leftrightarrow 22, \quad 8 \leftrightarrow 23, \quad 9 \leftrightarrow 33.$$

For longitudinal momentum,

$$p_\mu = (q_0, 0, 0, q_3), \qquad p^\mu = (q_0, 0, 0, -q_3), \qquad P^2 = q_0^2 - q_3^2.$$

Let c_2 denote either c_2^{F} or c_2^{CR}, and put

$$A = c_0 + 2c_1 + 4c_2, \quad B = -c_0 + 2c_1, \quad W = \frac{1}{2}(c_0 + 2c_1 + c_2). \tag{8.1}$$

The hat in $\widehat{B}_G$ denotes a declared pole record. It does not remove Assumption 7.1. The historical statement that the catalogue matches the v4 spectral logs is attributed to the v6 execution record [16].

8.2 Scalar sector

Graph	Declared coefficient of $1/(4-d)$
D04	$-\dfrac{\kappa^2}{4\pi} c_0 P^2$
D05a	$-\dfrac{\kappa^2}{4\pi}(c_2 - c_0) P^2$
Sum	$-\dfrac{\kappa^2}{4\pi} c_2 P^2$

The supplied Q02 comparison gives symmetry factors 1 for D04 and 1/2 for D05a and D12 [19, 16]. Those source values are not re-fitted to the displayed poles.

8.3 Tensor bubbles

Every entry in Table 2 is multiplied by $\kappa^2/(16\pi)$. Transposes are understood, together with the $1 \leftrightarrow 2$ images.

Table 2: Declared tensor-bubble entries. The $(5,5)$ entries were added to the printed catalogue in v6.

Graph and key	Coefficient with $\kappa^2/(16\pi)$ removed
D11: $(4,4),(4,7),(7,7)$	$\frac{1}{2}c_0P^2$
D03: $(0,0),(0,9),(9,9)$	$-c_0P^2$
D03: $(3,3)$	$4c_0P^2$
D03: $(0,4)$	$c_0(q_0^2-3q_3^2)$
D03: $(3,4)$	$4c_0q_0q_3$
D03: $(4,9)$	$-c_0(3q_0^2-q_3^2)$
D03: $(4,4),(7,7)$	$-(c_0+4c_1)P^2$
D03: $(4,7)$	$-c_0P^2$
D03: $(5,5)$	$-8c_1P^2$
D01: $(0,0),(0,9),(9,9)$	c_0P^2
D01: $(3,3)$	$-4c_0P^2$
D01: $(4,4),(7,7)$	$(c_0-4c_1+c_2)P^2$
D01: $(4,7)$	$(c_0+c_2)P^2$
D01: $(3,4)$	$-4(c_0+2c_2)q_0q_3$
D01: $(4,9)$	$Aq_0^2-Bq_3^2$
D01: $(0,4)$	$-Bq_0^2+Aq_3^2$
D01: $(5,5)$	$-8c_1P^2$

On the physical constant scalar jet,

$$B_{f_0f_0'} = B_{44} + B_{47} + B_{74} + B_{77}.$$

No extra off-diagonal multiplicity is applied to an array already differentiated in independent coordinates.

8.4 The D02 and D12 tadpoles

The nonzero D02 coordinate entries, including their transposes, are the following, with common factor $\kappa^2 W/\pi$:

$$\begin{aligned}
&(01)(01),(02)(02) : q_3^2, && (13)(13),(23)(23) : q_0^2,\\
&(12)(12) : P^2, && (11)(22) : -\tfrac{1}{2}P^2,\\
&(01)(13),(02)(23) : -q_0q_3, && (03)(11),(03)(22) : q_0q_3,\\
&(00)(11),(00)(22) : -\tfrac{1}{2}q_3^2, && (11)(33),(22)(33) : -\tfrac{1}{2}q_0^2.
\end{aligned}$$

There are five diagonal and eighteen off-diagonal entries, hence 23 nonzero coordinate entries. The declared D12 array is

$$\widehat{B}_{\mathrm{D12}} = 0.$$

These are catalogue statements, not a claim that every massless scaleless prescription has been independently audited.

8.5 Ghost and source–ghost arrays

Define

$$R = 3c_0 + 4c_1, \qquad T = 4c_0 + 3c_1, \qquad V = c_1 - c_0. \tag{8.2}$$

For rows and columns $0, 3$,

$$\widehat{B}_{\mathrm{D06}} = \frac{\mathrm{i}\kappa^2}{8\pi} RP^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{8.3}$$

For source rows $00, 03, 33$ and upper ghost columns $0, 3$,

$$\widehat{\Gamma}_{\mathrm{D13}} = \frac{\mathrm{i}\kappa^2}{16\pi} \begin{pmatrix} -2Tq_0 & 2Vq_3 \\ -Rq_3 & Rq_0 \\ -2Vq_0 & 2Tq_3 \end{pmatrix}. \tag{8.4}$$

The transverse source rows $11, 22$, for these two ghost columns, are declared zero in the v6 reduction. This is the precise source-trace datum used below.

8.6 Vector-type keys are excluded, not zero

The supplied spectral script logs and skips mixed-channel monomials. The v6 assessment locates them on:

- D01 vector-type tensor keys involving $01, 02, 13, 23$;
- D06 transverse ghost keys $11, 22$;
- D13 keys $(01)c^1, (02)c^2, (13)c^1, (23)c^2$.

Mandatory import boundary

Mixed-helicity monomials occur on vector-type keys that are excluded from the released cone catalogue, not set to zero. Any later sheet importing this catalogue must preserve this sentence. A partial same-channel sum on an excluded key is not a complete graph value.

There is a programming distinction here: the historical `continue` statement skips a *monomial*, not necessarily the whole affected key. The corrected replay in Appendix A quarantines the entire key whenever a nonzero mixed-channel pole is found.

8.7 Endpoint checks

Proposition 8.1 (Catalogue checks)**.** *Within the comparison chamber and its stated continuous endpoints:*

(i) *all displayed F entries vanish at* $\alpha = 1$*;*

(ii) *CR and F entries agree at* $\alpha = \frac{1}{2}$*;*

(iii) *the displayed rational coefficients are continuous at* $\alpha = \frac{1}{2}, \frac{2}{3}, \frac{4}{5}$*;*

(iv) *D06 and D13 are branch independent;*

(v) *D04+D05a has the scalar sum printed above.*

Proof. At $\alpha = 1$, $c_0 = c_1 = c_2^{\mathrm{F}} = 0$. At $\alpha = \frac{1}{2}$, $c_2^{\mathrm{CR}} = c_2^{\mathrm{F}}$. There is no zero denominator at the three specified openings. The ghost arrays contain no c_2, and the scalar sum is immediate. □

These checks are not uniform domain estimates and do not prove $(\mathrm{H}_{\mathrm{reg}})$.

9 Reflection-even invariant structure of the hh array

9.1 The symmetry class

Let h, h' be the longitudinal symmetric 2×2 blocks, and let f_0, f_0' be the transverse trace coordinates. Write

$$\operatorname{tr} h = \eta^{ab} h_{ab}, \qquad h \cdot h' = h^{ab} h'_{ab}, \qquad php = p^a p^b h_{ab}, \quad a, b \in \{0, 3\}.$$

The reflection-even invariant structures quadratic in p are

$$\begin{aligned} S_1 &= P^2 \operatorname{tr} h \operatorname{tr} h', & S_2 &= P^2 h \cdot h', \\ S_3 &= (php) \operatorname{tr} h' + \operatorname{tr} h(ph'p), & & \\ S_6 &= P^2(f_0 \operatorname{tr} h' + \operatorname{tr} h\, f_0'), & S_7 &= (php) f_0' + f_0(ph'p), \\ S_8 &= P^2 f_0 f_0'. & & \end{aligned} \tag{9.1}$$

Proposition 9.1 (Six-dimensional reflection-even sector)**.** *A slot-symmetric, reflection-even $O(1,1)$-invariant bilinear of the displayed fields, quadratic in p, is a unique linear combination of the six structures in* (9.1)*.*

Proof. Reflection evenness excludes a single Levi-Civita tensor; a pair reduces to metric contractions. For two longitudinal symmetric tensors, the metric contractions are generated by

$$P^2 \operatorname{tr} h \operatorname{tr} h', \quad P^2 h \cdot h', \quad (php) \operatorname{tr} h' + \operatorname{tr} h(ph'p),$$

and the symmetrized contraction $p\,hh'\,p$. For the mixed matrices $H = \eta^{-1}h$, $H' = \eta^{-1}h'$, the polarized Cayley–Hamilton identity is

$$HH' + H'H = (\operatorname{tr} H)H' + (\operatorname{tr} H')H + \big(\operatorname{tr}(HH') - \operatorname{tr} H \operatorname{tr} H'\big) I.$$

It removes the last contraction. A longitudinal tensor paired with a scalar gives S_6, S_7; two scalars give S_8. Independence follows by comparing the coefficients of the independent field coordinates and q_0, q_3. □

Remark 9.2 (Why $SO(1,1)$ alone is insufficient)*.* Without reflection evenness, pseudoscalar terms such as

$$(\epsilon^{ab} p_a h_{bc} p^c)\, f_0' + f_0 (\epsilon^{ab} p_a h_{bc}' p^c)$$

are allowed by proper longitudinal Lorentz transformations. Thus the unqualified assertion that six structures classify *all* $SO(1,1)$-invariant bilinears is too strong. The frozen metric-contraction sector used here is reflection even.

9.2 Coefficients

Write the restricted graph bilinear as

$$\widehat{B}_G(h, f_0; h', f_0') = \frac{\kappa^2}{\pi} \sum_i \beta_i^G S_i.$$

The coefficient table is

Graph	β_1	β_2	β_3	β_6	β_7	β_8
D01	$-c_0/16$	$c_0/8$	0	$-c_0/8 - c_1/4 - c_2/2$	$c_0/4 + c_2/2$	$c_0/4 - c_1/2 + c_2/4$
D02	0	0	0	W	$-W$	$-W$
D03	$c_0/16$	$-c_0/8$	0	$3c_0/8$	$-c_0/4$	$-c_0/4 - c_1/2$
D11	0	0	0	0	0	$c_0/8$
D12	0	0	0	0	0	0
Sum	0	0	0	$3(c_0 + c_1)/4$	$-c_0/2 - c_1$	$-3c_0/8 - 2c_1 - c_2/4$

Proposition 9.3 (Catalogue decomposition and branch content)**.** *The table is an exact identity of the declared coordinate arrays, for arbitrary formal c_0, c_1, c_2. For the complete hh sum:*

(i) *the purely longitudinal structures cancel;*

(ii)
$$\frac{\kappa^2}{\pi}(\beta_6 + \beta_7) = -\frac{\kappa^2 V}{4\pi};$$

(iii) *the CR-minus-F difference is*

$$\Delta\widehat{B}^{hh} = -\frac{\nu\kappa^2}{4\pi}P^2 f_0 f_0'. \tag{9.2}$$

Proof. Substitute the definitions of S_i and compare with Table 2 and the D02 entries. This is finite polynomial coefficient comparison; no operator reduction is used.

Summing the rows gives the final row of the table. Its first three coefficients vanish, and

$$\beta_6 + \beta_7 = \frac{1}{4}(c_0 - c_1).$$

Only the total β_8 depends on c_2, with coefficient $-1/4$. Since $c_2^{\mathrm{CR}} - c_2^{\mathrm{F}} = \nu$, (9.2) follows. □

For D01 alone the branch difference has additional S_6, S_7 parts. The D02 difference is

$$\frac{\nu\kappa^2}{2\pi}(S_6 - S_7 - S_8),$$

and cancels those parts. D12 is zero in the declared catalogue. Thus the complete branch comparison includes the tadpoles.

The difference (9.2) is gauge invariant on this restricted lattice because the permitted longitudinal gauge variation has $\delta f_0 = 0$. It is a two-point branch difference, not a stress tensor or a physical core selection.

10 Ghost equation and the constant-profile ST row

10.1 The ghost equation

For a symmetric source tensor Y, define the longitudinal de Donder contraction

$$\begin{aligned}(\mathsf{F}_L Y)_0 &= \frac{q_0}{2}(Y_{00} + Y_{33}) - q_3 Y_{03},\\ (\mathsf{F}_L Y)_3 &= q_0 Y_{03} - \frac{q_3}{2}(Y_{00} + Y_{33}).\end{aligned} \tag{10.1}$$

The full contraction is

$$(\mathsf{F}Y)_\rho = p^\mu Y_{\mu\rho} - \frac{1}{2}p_\rho \eta^{\mu\nu} Y_{\mu\nu},$$

so

$$(\mathsf{F}Y)_0 = (\mathsf{F}_L Y)_0 + \frac{q_0}{2}(Y_{11} + Y_{22}), \quad (\mathsf{F}Y)_3 = (\mathsf{F}_L Y)_3 + \frac{q_3}{2}(Y_{11} + Y_{22}).$$

Theorem 10.1 (Declared projected ghost identity)**.** *The arrays* (8.3)–(8.4) *satisfy*

$$\widehat{B}_{\mathrm{D06}} = -2\eta_{\parallel}\mathsf{F}_L\widehat{\Gamma}_{\mathrm{D13}}, \qquad \eta_{\parallel} = \mathrm{diag}(1,-1). \tag{10.2}$$

With the declared zero transverse D13 source rows for columns $0, 3$, $\mathsf{F} = \mathsf{F}_L$ *on this matrix.*

Proof. Since $T + V = R$, direct multiplication gives

$$\mathsf{F}_L\widehat{\Gamma}_{\mathrm{D13}} = \frac{\mathrm{i}\kappa^2}{16\pi}RP^2\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}.$$

Multiplication by $-2\eta_{\parallel}$ proves the identity. The full-contraction statement uses exactly the two declared transverse source rows. □

The identification with a graph antighost equation additionally uses the source normalization, Fourier convention, and Assumption 7.1. This is not a test of the full interacting basis.

10.2 Independent-coordinate weights

The ghost is upper-indexed, and

$$\delta_c h_{\mu\nu} = p_\mu c_\nu + p_\nu c_\mu, \qquad c_\nu = \eta_{\nu\nu} c^\nu.$$

For an invariant quadratic action written in independent coordinates,

$$\frac{\partial S_{\mathrm{inv}}}{\partial h_{(mn)}} = (2 - \delta_{mn}) E^{mn}(h). \tag{10.3}$$

Thus the lower-index 03 component carries

$$2E^{03} = -2E_{03}.$$

The sign is not an optional positive multiplicity.

The supplied v4 scan varied only $w \in \{\frac{1}{2}, 1, 2\}$, so it omitted $w = -2$ [17]. This diagnoses its reported failure, without interpreting that historical residual as an anomaly.

Supplied input 10.2 (Tree normalization, not loop fitting)**.** In the frozen conventions, the tree source is

$$\frac{\partial S}{\partial K'_{(mn)}} = \mathrm{i} A_{\mathrm{GH}} (p_m c_n + p_n c_m), \qquad A_{\mathrm{GH}} = \frac{1}{2},$$

and the invariant quadratic normalization relative to the displayed E is $N = 1$. Accordingly,

$$\lambda_{\mathrm{pred}} = \frac{N}{\mathrm{i} A_{\mathrm{GH}}} = -2\mathrm{i}. \tag{10.4}$$

These values are attributed to the frozen tree data and the v6 normalization check, not derived from the loop row [18, 16].

The optional tree routine in Appendix A checks the quadratic normalization on nonzero de Donder configurations if the frozen engine is supplied. It is not represented as run here.

10.3 An explicit algebraic closure theorem

On the axisymmetric lattice write

$$h = (h_{00}, h_{03}, h_{33}, f_0), \qquad c = (c^0, c^3), \qquad p \cdot c = q_0 c^0 + q_3 c^3.$$

Use the lower-index Einstein expression

$$\begin{aligned} E_{\mu\nu}(h) = {} & P^2 h_{\mu\nu} - p_\mu p^\rho h_{\rho\nu} - p_\nu p^\rho h_{\rho\mu} + p_\mu p_\nu \operatorname{tr}_4 h \\ & + \eta_{\mu\nu} \big(p^\rho p^\sigma h_{\rho\sigma} - P^2 \operatorname{tr}_4 h \big). \end{aligned} \tag{10.5}$$

This normalization is fixed for the present row. It is not a physical tensor-core Hessian theorem.

Let

$$\begin{aligned} \mathcal{B}(c, h) &= \widehat{B}^{hh}(\delta_c h, h), \\ \mathcal{G}(c, h) &= \sum_{(mn),\rho} \widehat{\Gamma}^{\mathrm{D13}}_{(mn),\rho} c^\rho (2 - \delta_{mn}) E^{mn}(h), \end{aligned} \tag{10.6}$$

where

$$\widehat{B}^{hh} = \mathrm{D01} + \mathrm{D02} + \mathrm{D03} + \mathrm{D11} + \mathrm{D12}$$

is the declared lattice array.

Theorem 10.3 (Constant-profile hh/Kc identity)**.** *For arbitrary formal c_0, c_1, c_2,*

$$\mathcal{B}(c,h) = -\frac{\kappa^2 V}{2\pi} P^2 f_0(p\cdot c), \qquad \mathcal{G}(c,h) = \frac{\mathrm{i}\kappa^2 V}{4\pi} P^2 f_0(p\cdot c). \tag{10.7}$$

Hence

$$\mathcal{B} + \lambda_{\mathrm{pred}}\mathcal{G} = 0. \tag{10.8}$$

For CR and F with $1/2 < \alpha < 1$ and nonzero coupling, $-2\mathrm{i}$ is the unique constant λ for which $\mathcal{B} + \lambda\mathcal{G}$ vanishes as a polynomial in all the declared fields, ghosts, and momenta.

Proof. The longitudinal gauge variation satisfies

$$\delta_c f_0 = 0, \qquad \mathrm{tr}(\delta_c h) = 2(p\cdot c), \qquad p(\delta_c h)p = 2P^2(p\cdot c).$$

By Proposition 9.3, only $\beta_6 + \beta_7 = (c_0 - c_1)/4$ contributes. This proves the formula for $\mathcal{B}$.

In two longitudinal dimensions the longitudinal part of (10.5) vanishes when $f_0 = 0$. Its contributions from the transverse trace are

$$E_{00} = -2q_3^2 f_0, \quad E_{03} = -2q_0 q_3 f_0, \quad E_{33} = -2q_0^2 f_0.$$

After raising and independent-coordinate weighting they become

$$E_{00}^{\mathrm{coord}} = -2q_3^2 f_0, \quad E_{03}^{\mathrm{coord}} = 4q_0 q_3 f_0, \quad E_{33}^{\mathrm{coord}} = -2q_0^2 f_0.$$

Multiplying by (8.4) and using $R - T = V$ gives the formula for $\mathcal{G}$. Equation (10.8) follows from (10.4).

For either physical branch, $V = (\alpha - 1)/2 \neq 0$ in the open comparison chamber. The polynomial $P^2 f_0(p\cdot c)$ is not identically zero. Thus the scalar equation $-1/2 + \lambda\mathrm{i}/4 = 0$ forces $\lambda = -2\mathrm{i}$. □

This proof establishes the identity of the declared arrays without fitting λ or invoking an unavailable log. Its complete-graph interpretation remains conditional.

Corollary 10.4 (Matter row and UNIT)**.** *The declared matter row D11+D12 is transverse on this lattice. For the UNIT substitution $c_0 = c_1 = c_2 = 1$, both $\mathcal{B}$ and $\mathcal{G}$ vanish separately. UNIT therefore cannot determine λ.*

Proof. D11 contributes only S_8, and D12 is zero. For UNIT, $V = 0$, so (10.7) gives the result. □

The normalized quadratic convention is important: if $Q(h) = \frac{1}{2}\widehat{B}^{hh}(h,h)$, then $\delta Q = \widehat{B}^{hh}(\delta h, h)$. An extra factor two would change the printed convention constant.

10.4 The three identity classes

Class	Status and target
Free source-tested	The Maxwell–Friedrichs identity $C_1 G^{\pm}_{(1)} = G^{\pm}_{(0)} C_1$ is a free-domain comparison from I-B-G-S [23]. It is not counted as a cone interacting-basis test.
Constant-profile interacting catalogue	The ghost row and Theorem 10.3 are exact identities of the stated restricted arrays, with graph identification conditional. No certified map assigns them to a named member of the programme's 26 combinations.
Full interacting basis	The 34-dimensional projection and 26 Ward combinations are the O.1 target. They have not been constructed or evaluated here. No missing combination is reported as zero.

The historical UNIT expectation also needs care. An identity holding only after full loop integration need not survive restriction to $l_\perp = 0$. For rapidly decreasing F,

$$\int \mathrm{d}l\, \mathrm{d}^2k\, \partial_{k_1}(k_1 F(l,k)) = 0,$$

but its restriction at $k = 0$ is $F(l,0)$, whose longitudinal integral need not vanish. Theorem 10.3 avoids this shortcut by direct algebra.

11 Heat moments, one-point fields, and missing two-point data

11.1 The object ledger

Object	Definition or source	What it does not establish
Apex heat moment	$\mathcal{C}_b, \mathcal{D}_b$, Theorems 3.2 and 5.2	A stress tensor, Landauer heat, or complete differentiated graph.
One-point field observable	Friedrichs Wick square, stress, and current in I-B-VP/I-B-VP-1 [25, 26]	A massless loop UV/IR theorem or two-point normalization.
Two-point graph pole	A graph satisfying (R1)–(R5)	A physical core selection or an Einstein source law.
Two-point branch difference	Equation (9.2)	A one-point stress or a tensor-core limit.

11.2 Constant-profile restriction is not injective

Proposition 11.1 (A transverse-jet kernel)**.** *On a smooth radial external-jet class containing constant and quadratic transverse jets, restriction of apex bilinear functionals to constant profiles need not be injective.*

Proof. Consider

$$L(F,G) = \int_{\mathcal{W}} (F\Delta_\perp G + G\Delta_\perp F)\, \mathrm{d}^2 y.$$

It vanishes for two constant transverse jets. For F constant near the apex and $G = r^2 g(y)$, $\Delta_\perp r^2 = 4$, so suitable longitudinal tests give a nonzero value. Thus the restriction has a nontrivial kernel. □

This does not declare a complete physical counterterm basis. It proves that the displayed restriction cannot determine every transverse-derivative coefficient.

11.3 The second scalar value is partial

For a quadratic jet, (5.3) gives a weighted spectral moment only after an operator reduction has justified that insertion. Within that reduction:

- the supplied D05a weighted longitudinal expression has large-radius behavior $O(R^{-4})$ in $D = 2$, hence no logarithmic UV term under Lemma 7.3;
- D04 requires mixed-channel traces with transverse position operators between different channel realizations. It is not determined by a function of one A_S.

Accordingly the retained status is

$$\text{D05a part} = 0 \text{ in the stated reduced model}, \qquad \text{D04 part} = \mathsf{OPEN}.$$

The second scalar basis value is not identified with its D05a part. Neither the weighted heat theorem nor the jet/curvature language of the tensor-core comparison computes D04. v8 status: still OPEN; see Section 15 and Lemma 16.3 for the ingredients supplied.

11.4 Evanescent data and the engine map

The simple-pole observation after Lemma 7.3 is a pole-level statement under its stated hypotheses. The $\tilde{g}$-leg arrays needed to determine evanescent coefficients are not supplied. Those coefficients remain open. v8 computes them under a declared slice-DRED prescription (Section 17); that is CONDITIONAL, not an identification with the frozen export.

The six reflection-even structures classify the executed constant-profile restriction. They do not supply its convention map into the programme's 18-dimensional engine basis, which also sees transverse derivatives and non-axisymmetric data. That map and the unseen components are open, not zero. v8 supplies the six constant-profile coordinates of the *hh* block exactly (Section 15); twelve remain open.

11.5 What an exact span certificate requires

The 34-dimensional basis and associated counts are retained as supplied metadata, not reconstructed from the coordinate table. A complete certificate needs the declared external-jet class, the full matrix, its coefficient field, column completeness, and the appropriate relation quotient. v8 supplies these for the declared inventory at named angles and branches (Section 14).

Lemma 11.2 (What modular rank can certify)**.** *Let M be a specified rational matrix, and let p avoid all denominators. Then*

$$\operatorname{rank}_{\mathbb{F}_p}(M \bmod p) \le \operatorname{rank}_{\mathbb{Q}} M.$$

A nonzero modular minor is an exact certificate that the corresponding rational minor is nonzero. If this lower bound equals an independently established upper bound, it certifies the exact rational rank.

Proof. Clear denominators by an integer factor not divisible by p. An integer determinant nonzero modulo p is nonzero as an integer. Apply this to a maximal nonzero modular minor. □

Thus modular arithmetic is not intrinsically incapable of an exact certificate. The unresolved issue here is the absence of the complete specified matrix and its completeness and upper-rank arguments, not a general prohibition on modular methods. For rational-function matrices, a good specialization can prove generic nonvanishing, but does not settle exceptional parameter values without additional analysis.

I-B-G's compact-test exactness warning is a useful free-field comparison: an identity on one spectral test domain is not automatically exactness on another test space [22, `rem:compact-exact`]. That comparison is not a proof of a loop span statement.

12 The continuum–preparation–instrument boundary

12.1 Three objects and three targets

Any proposed finite report of an apex coefficient must separately specify

$$\boxed{\text{defined continuum coefficient}} \longrightarrow \boxed{\text{acquisition or supplied preparation}} \longrightarrow \boxed{\text{complete finite instrument}}. \tag{12.1}$$

The arrows are obligations, not constructions supplied by the diagram. This paper studies the first object. A program printing a rational expression is not an apparatus acquiring an unknown continuum amplitude.

Following the distinction recorded in the supplied Oai290-A companion, the reporting target must also be typed [36]:

(i) recovery of an arbitrary unknown quantum code, including references;

(ii) reconstruction or estimation of one named observable;

(iii) consistency with a separately declared geometric model.

A supplied finite report of $\mathcal{D}_b$ is not automatically even an acquisition theorem for target (ii), and is not target (i). No external catalogue family supplies an apex graph premise.

The no-acquisition warning attributed to X-J-ST-2, and the window/report distinction in I-B-ECHO, preserve precisely this boundary [35, 31].

12.2 The complete destructive instrument

Let Q be a finite measured register with orthonormal basis $\{|\tau_j\rangle\}$, and let A contain every retained quantum register. Define

$$D_j = \langle\tau_j|_Q \otimes I_A, \qquad \mathcal{M}(X) = \sum_j |j\rangle\langle j|_R \otimes D_j X D_j^\dagger.$$

Proposition 12.1 (Record and retained residual)**.** *The map $\mathcal{M}$ is completely positive and trace preserving, and*

$$\operatorname{Tr}_R \mathcal{M}(X) = \operatorname{Tr}_Q X.$$

These identities hold with every untouched finite reference. The output contains A, not a returned quantum Q-factor. Positive-probability branches retain their actual normalized conditional states; zero branches have no assigned normalized state.

Proof. The recorded Kraus operators are $|j\rangle_R \otimes D_j$, and

$$\sum_j D_j^\dagger D_j = I_{QA}.$$

Writing $X = \sum_{j,k} |\tau_j\rangle\langle\tau_k| \otimes X_{jk}$ gives output blocks X_{jj}, whose sum is $\operatorname{Tr}_Q X$. The same expansion includes a reference. □

This is the complete I-B/En09-A instrument, restated locally [21, 33]. Consumption refers to the declared interface, not physical annihilation of all apparatus degrees of freedom.

12.3 I-C A and D: a classical report is not code recovery

For channels use half-diamond distance

$$d_\diamond(\Phi, \Psi) = \frac{1}{2}\|\Phi - \Psi\|_\diamond, \qquad e_{\mathrm{rec}}(\mathcal{C}) = \inf_{\mathcal{D}} d_\diamond(\mathcal{D}\mathcal{C}, \mathrm{id}).$$

Theorem 12.2 (Sharp record-only obstruction)**.** *Let $C \simeq \mathbb{C}^d$, $d \geq 2$, and suppose the complete accessible output is classical, with no input-correlated quantum side resource available to the decoder:*

$$\mathcal{C}(X) = \sum_j \operatorname{Tr}(F_j X)|j\rangle\langle j|, \qquad F_j \geq 0, \quad \sum_j F_j = I_C.$$

For every deterministic CPTP decoder $\mathcal{D}$,

$$F_e(\mathcal{D}\mathcal{C}) \leq \frac{1}{d}, \qquad d_\diamond(\mathcal{D}\mathcal{C}, \mathrm{id}_C) \geq 1 - \frac{1}{d}. \tag{12.2}$$

The distance bound is sharp over this class. If the implemented complete channel is within half-diamond distance δ of such a classical channel, its optimal recovery error is at least $\max\{0, 1 - d^{-1} - \delta\}$.

Proof. The decoded channel has measure-and-prepare form

$$\mathcal{D}\mathcal{C}(X) = \sum_j \operatorname{Tr}(F_j X)\sigma_j.$$

Its normalized Bell-input state is separable. For product density operators a, b,

$$\operatorname{Tr}[\Phi_d(a \otimes b)] = \frac{1}{d}\operatorname{Tr}(a^{\mathsf{T}} b) \leq \frac{1}{d}.$$

Convexity gives the fidelity bound. Testing the Bell projector gives the trace-distance lower bound, and the diamond distance dominates this referenced test.

For sharpness, computational dephasing is

$$\Delta_d = \frac{1}{d}\sum_{k=0}^{d-1} \mathrm{Ad}_{Z^k}, \quad Z = \sum_{j=0}^{d-1} e^{2\pi \mathrm{i} j/d}|j\rangle\langle j|.$$

Separating its identity term gives the matching upper bound $1 - 1/d$. The approximate statement follows from contraction under the decoder and the triangle inequality. □

This is the entanglement-breaking mechanism [49]; its operational forms are recorded in I-C, En07-A/B/C, and the supplied Hawk05 sources [32, 34, 38]. Appending classical heat labels or classical postprocessing does not change the premise. A retained input-dependent quantum register does change it.

The distinct Haar-average pure-state infidelity bound is $(d-1)/(d+1)$, using $F_{\mathrm{av}} = (dF_e + 1)/(d+1)$ [50]. It must not replace the whole-code diamond constant.

12.4 I-C B and F: logs are records, not a Page history

For a complete retained classical history,

$$H(Y_{\le n+1}) = H(Y_{\le n}) + H(Y_{n+1} \mid Y_{\le n}) \ge H(Y_{\le n}).$$

For a classical–quantum output,

$$S(YK) = H(Y) + \sum_y p_y S(\rho_{K|y}).$$

If only the classical record survives, its accessible observable algebra is commutative.

These are the relevant I-C B/F distinctions [32]. A particular list of SHA-256 digests is deterministic integrity data unless a probability model for that list is separately introduced. It is not an entropy of radiation, and an append-only execution transcript is not a Page history of the loop.

The finite Page means in the Hawk05 sources have fixed-input ensemble quantifiers. Whole-code recovery has arbitrary referenced-input quantifiers. No Page mean is used to infer (12.2), a Ward identity, or a gravitational endpoint [38].

12.5 I-C C: spectral heat is not thermodynamic heat

The symbols $\mathcal{C}_b$ and $\mathcal{D}_b$ denote spectral functionals. They are not Landauer costs.

If a complete non-bath system M, a fixed bath Hamiltonian H_B, inverse temperature $\beta > 0$, an initially independent Gibbs bath τ_B, and a joint unitary are separately declared, then

$$\beta Q_B = S_{\mathrm{n}}(\rho_M) - S_{\mathrm{n}}(\omega_M) + I_{\mathrm{n}}(M:B)_\omega + D_{\mathrm{n}}(\omega_B \| \tau_B). \tag{12.3}$$

Here entropies use natural logarithms and $Q_B = \operatorname{Tr} H_B(\omega_B - \tau_B)$. The equality follows from unitary entropy conservation and the definition of Gibbs relative entropy [51]. It does not require an entropy-decreasing reset merely to be valid. A strictly positive entropy decrease of the complete non-bath system implies positive heat because the last two terms are nonnegative.

No such bath, Hamiltonian, or reset is declared in this loop calculation. Thus neither the equality's physical hypotheses nor a positive reset cost is supplied by an apex heat coefficient.

12.6 I-C E: a destructive prefix selects no apex domain

A finite destructive prefix does not choose between CR, F, a positive tensor-core family, or a different boundary model. These are operator-domain specifications. Theorem 10.3 closing on both branches is not a physical branch-selection rule.

Likewise, fixed-resolution neutral-core report convergence does not imply equality or non-injectivity of exact continuum functionals. The I-C/I-B-ECHO comparison concerns fixed reports, specified continuous preparations, and complete finite instruments [32, 31].

Readout release boundary

Only the first object in (12.1) is an analytical target of this paper. A shredder instrument neither acquires the missing coefficient nor selects its domain. An echo window is not an apex residue. No readout, entropy, or thermal comparison discharges O.1.

13 Static debugging and artifact integrity

13.1 Corrections identified from the supplied listings

Issue	Correction in this revision
Six "$SO(1,1)$" structures	Require reflection evenness; otherwise Levi-Civita pseudoscalars are allowed.
Unique scale-covariant D01 extension	Replace by uniqueness of the specified meromorphic continuation. Parabolic covariance alone allows endpoint contact terms; see Remark 6.3.
Mellin value established by a routine name	Replace with the explicit formula and proof (6.2).
Mixed monomial skipped by `continue`	Quarantine the entire affected physical-branch key. Keep a coverage status and the nonzero mixed terms.
UNIT mixed terms	A genuine UNIT momentum-slice replay must retain all channel pairs. On the released lattice, where the reported mixed terms are absent, the algebraic UNIT closure is unchanged.
Missing log key interpreted as computed zero	Require an evaluated-key/coverage ledger. Legacy absence supports at most a comparison with a declared zero, not proof that the key was evaluated.
`weighted_checks.py` uses numerical `nsimplify` for an "exact" check	Use exact arithmetic in a cyclotomic quotient. Rational reconstruction from a numerical approximation is not an exact identity proof by itself.
Printed diagnostics without assertions	New acceptance checks raise exceptions on mismatch. A printed `ALL_MATCH` is not a substitute for an acceptance predicate and process status.
`eval` on serialized component keys	Use literal key parsing and a restricted arithmetic-expression parser. This is an input-integrity improvement, not a graph proof.
"Radial" Ward scan	The historical point $(X, Y) = (\mathtt{rr}, 0.2)$ has $r = \sqrt{\mathtt{rr}^2 + 0.04} \geq 0.2$, so it does not approach the apex. A true radial scan must scale both coordinates at fixed angle. No apex asymptotic theorem is inferred from that scan.
Finite resonance or random-rank scans	Distinguish exact arithmetic on a specified finite matrix from proof of candidate completeness and polynomial identities on the full jet class.
Modular rank described as only heuristic	Apply Lemma 11.2; exactness fails here because the complete matrix/certificate is absent, not because modular minors can never certify rank.
Memory-qualified C_4 normalization	Use the explicit relation in Remark 5.5.

Numerical quadrature without interval or rigorous truncation bounds remains corroboration, not a certified error enclosure. Converting high-precision numbers to ordinary Python `float` or `complex` also limits the precision actually retained.

13.2 v8 execution of the v7 program and response to its corrections

The listing of Appendix A was extracted byte for byte as `code/v8/cone_v7.py` and run unchanged. The results were:

- `selftest`: return code 0. All assertions passed, covering the cyclotomic orbifold checks for $N = 2, \ldots, 8$, the catalogue algebra, the ST row, and rank 6 for the six-structure decomposition.
- legacy log comparison: PASS on all 18 graph–branch pairs. Keys absent from the legacy logs are listed rather than read as zero.
- tree normalization: $N = 1$, $A_{\mathrm{GH}} = 1/2$, $\lambda_{\mathrm{pred}} = -2\mathrm{i}$.
- frozen-engine replay: 27 runs (graph $\times$ {CR, F, UNIT}), all with return code 0. The coverage comparison passed on all 18 physical pairs, and the physical branch values differ from `logs/v4` by exactly zero.

The replay quarantines mixed keys. These are D01 $(1,1)$, $(1,6)$, $(2,2)$, $(2,8)$, $(6,1)$, $(6,6)$, $(8,2)$, $(8,8)$; D06 $(1,1)$, $(2,2)$; and D13 $(1,1)$, $(2,2)$, $(6,1)$, $(8,2)$. On the UNIT lattice, those keys gain the mixed-pair terms that the historical `continue` discarded. For example, D01 $(1,1)$ gains $-\kappa^2 q_3^2/\pi$, D06 $(1,1)$ gains $\mathrm{i}\kappa^2 P^2/(2\pi)$, and D13 $(1,1)$ gains $\mathrm{i}\kappa^2 q_0/(4\pi)$. Table 4 gives the point-by-point verdicts.

Table 4: Verdicts on the v7 static-debugging corrections (executed in v8).

#	v7 correction	Verdict	Executed check and effect on v6 numbers
1	Six structures classify only the reflection-even sector	wording, agreed	The rank-6 assertion passes. No number changes.
2	Contact term and uniqueness of extension	wording, agreed	No number changes.
3	Mellin value should not rest on a routine name	agreed	The closed form (6.2) is checked exactly; the value is unchanged.
4	Mixed monomial skipped by `continue`	real defect (`spectral_apex.py`, line 134)	No historical physical value was affected, because the v4 logs hold no partial values on mixed keys. v8 strengthens the point: the quarantine must test pair densities, not pole totals (Section 16).
5	UNIT records omit mixed terms	real defect	The UNIT records change on 14 keys. The UNIT statements in v6 were lattice-only, so they are unaffected.
6	A missing key read as a computed zero	real defect	The v7 program's explicit `key_status` coverage was run on every key. No number changes.
7	`nsimplify` in `weighted_checks.py`	real defect	The exact cyclotomic check passes. No number changes.
8	Unasserted diagnostics	real defect	The v7 program asserts them, and every assertion passes. No number changes.
9	`eval` on keys	real defect (`st_scan`, `st_test_const`, `st_v6`)	A literal parser gives identical results.
10	Ward scan not at a radial point	real defect	The true radial scan was run. The geometric residual is $\leq 10^{-28}$ for all $r \geq 0.003$. The Friedrichs residual grows from 0.0039 at $r = 1$ to 0.101 at $r = 0.003$, whereas the historical reading was a plateau near 0.019. v6 printed no Ward-scan number.

#	v7 correction	Verdict	Executed check and effect
11	Finite resonance and random-rank scans	real limitation	Replaced by the complete exact certificates of Section 14.
12	Modular rank called only heuristic	addressed	Every exact rank equals the historical modular rank, and Lemma 11.2 applies.
13	C_4 normalization	algebra verified	$(\alpha/2)(\alpha^{-2}-1)(\alpha^{-2}+11)/90 = \mathcal{D}_0$ holds identically. The equation-number attribution to the source remains from memory (†).
14	Coarse channel marker	verified	`projector_check` passes on the frozen engine.
15	Tree-hash ordering	valid in principle	The locale is `C.UTF-8`, and the shell digest is reproducible. The frozen-tree digests are unchanged (Section 13.6).

v6 numbers changed: none. One v6 status changes: the *hh* branch difference becomes conditional (Section 16).

13.3 The coarse channel marker needs its own check

The historical spectral code labels a tensor propagator entry by whether its independent coordinates are longitudinal, mixed, or transverse. That rule is not a general helicity projector theorem.

For the four-dimensional de Donder numerator, the transverse trace–trace block vanishes, while the transverse trace can couple to the longitudinal scalar block. In that particular case the coarse marker can agree with the projected propagator. The corrected replay verifies

$$P_S \Pi = \Pi_S$$

against explicit coordinate projectors before using the marker. It does not export this four-dimensional check to unsupplied evanescent-index arrays.

13.4 The XVI-O1-ST sign witness

The supplied comparison attributes to XVI-O1-ST-v5 a cross-artifact witness: a typeset D01 sixth coefficient $-4/3$, a regression print $+4/3$, and a residual $(8/3)e_6$ identified there as a transcription discrepancy [37].

The arithmetic is immediate:

$$\left(\frac{4}{3} - \left(-\frac{4}{3}\right)\right) e_6 = \frac{8}{3} e_6 .$$

Its identification as a transcription rather than a computational error remains source-qualified: that manuscript's complete artifacts are not supplied here. The e_6 coordinate belongs to that comparison's basis and must not be silently identified with a present cone coordinate.

Selecting the source's preferred sign does not rerun a loop, clear every predecessor sign audit, or update XVI-O. A matching cone catalogue is not a universal cross-artifact sign certificate.

13.5 Historical hashes and new manifests

The following are *reported v6 labels*, not freshly recomputed hashes:

Tree	Reported SHA-256
XVI-P	`ec965c15ae25cd3bb9cc2b3a134e966b 19a1803c78d1fe5bf7554da2a8b3aaa6`
XVI-Q	`0df9a0d10409d31ac0c2d0a47197e3f6 e03c7ae258ec5fb775137f66cb0493c4`
XVI-R	`57187bc6825f5c6ae88633b7279c6c0b e57f7c4918fdbc525da4aaf958331edb`

The original shell tree hash depends on its filename ordering, exclusions, and command semantics. The new Python manifest routine records relative names and byte hashes without `find|xargs` filename ambiguity. Its tree digest uses a different explicit serialization and is *not* expected to equal the historical shell digest.

A hash binds supplied bytes to an expected digest. It does not prove a theorem, establish authorship, authenticate an unavailable file, or show that a log was generated by a claimed execution. Before/after equality also requires both actual manifests, not merely two printed matching strings.

13.6 v8: recomputed frozen-tree digests

The historical shell serialization was recomputed in v8 (locale `C.UTF-8`) on the read-only trees, both before any v8 run and after the last run:

Tree	When	SHA-256 (shell serialization)
XVI-P	before/after	`ec965c15ae25cd3bb9cc2b3a134e966b` `19a1803c78d1fe5bf7554da2a8b3aaa6`
XVI-Q	before/after	`0df9a0d10409d31ac0c2d0a47197e3f6` `e03c7ae258ec5fb775137f66cb0493c4`
XVI-R	before/after	`57187bc6825f5c6ae88633b7279c6c0b` `e57f7c4918fdbc525da4aaf958331edb`

They equal the reported v6 labels above, before and after (`logs/v8/tree_hashes_v8.json`). The v7 Python manifest serialization gives different digests, as v7 predicted: XVI-P `acd88f27...5b3c7` (1209 files), XVI-Q `c13b12d1...37586` (879 files), XVI-R `d8b9ed80...f4015` (457 files). No XVI-P, XVI-Q or XVI-R file was edited.

13.7 Computational scope of the embedded program

Appendix A consolidates the parts of the supplied scripts needed by this paper:

- heat-moment algebra and exact orbifold checks;
- declared catalogue construction;
- invariant decomposition, ghost equation, and ST identity;
- coverage-aware comparison with historical or new logs;
- optional frozen-engine momentum-slice replay and tree normalization;
- byte-level manifests.

The older exploratory jet/cohomology, kernel, scratch, and document-assembly scripts remain historical material in [17]; they are not copied as redundant submission machinery or claimed as completed certificates. The consolidated program is a replacement verification implementation, not a verbatim archive of every v1–v6 file. Its inclusion does not change their historical hashes.

14 v8: an exact complete span certificate

14.1 What is certified

The apex functionals of the historical inventory (`apex_jets.py`) are bilinear in finitely many jet atoms and polynomial in the external momentum. v8 ports them mechanically to exact Gaussian-rational arithmetic (`code/v8/apex_exact.py`; the port reproduces every historical modular count) and evaluates them on *all* relevant unit atom pairs over a complete 6×6 momentum grid, enough to determine every polynomial of the occurring degree. Therefore the evaluated columns are complete, and the row spaces computed are the exact row spaces.

Theorem 14.1 (Exact span certificate; executed)**.** *For each* $(\alpha, \textit{branch})$ *in Table 5, over* $\mathbb{Q}(\mathrm{i})$ *and for the functionals as implemented in the declared inventory: the gauge-invariant candidate ranks, the ST rank,* $\dim Z$, $\dim B$, $B \subset Z$ *and* $\dim H^0 = \dim Z - \dim B$ *are as displayed. Each rank is certified by a nonzero minor (hash and indices in* `apex_cert_*.json`*) and each kernel by verification on every complete column.*

Table 5: Complete exact certificates (v8).

(α, br)	N	ST rows	ST rank	$\dim Z$	$B \subset Z$	$\dim B$	$\dim H^0$	Ward	on K	s
$(3/10, \mathrm{CR})$	34=18+8+6+2	8640	26	8	yes	6	2	26	6	292
$(3/10, \mathrm{F})$	34=18+8+6+2	8640	26	8	yes	6	2	26	6	302
$(7071/10^4, \mathrm{CR})$	34=18+8+6+2	8640	26	8	yes	6	2	26	6	316
$(7071/10^4, \mathrm{F})$	38=21+8+7+2	14400	29	9	yes	6	3	29	6	554

Here "Ward" is the number $N - \dim Z$ of Ward-constrained combinations; at the 34-candidate points this is the v6 count 26, now exact. The ghost-number $+1$ cohomology vanishes at the sampled exact point $(7071/10^4, \mathrm{CR})$ (26/26 images closed; $H^1 = 0$).

14.2 The count 34 is angle and branch dependent

The historical modular inventory, which the exact port reproduces wherever both exist, gives:

α	branch	N	$\dim Z$	$\dim B$	$\dim H^0$	$N - \dim Z$
3/10	CR	34	8	6	2	26
3/10	F	34	8	6	2	26
41/100	CR	34	8	6	2	26
41/100	F	34	8	6	2	26
1/2	CR	42	9	7	2	33
1/2	F	42	9	7	2	33
309/500	CR	34	8	6	2	26
309/500	F	38	9	6	3	29
2/3	CR	35	8	6	2	27
2/3	F	39	9	6	3	30
7071/10000	CR	34	8	6	2	26
7071/10000	F	38	9	6	3	29
4/5	CR	34	8	6	2	26
4/5	F	38	9	6	3	29
8501/10000	CR	34	8	6	2	26
8501/10000	F	38	9	6	3	29

Thus $N = 34$, $\dim Z = 8$, $N - \dim Z = 26$ hold on CR at every tested non-resonant angle and on F for $\alpha < 1/2$; F at $\alpha > 1/2$ gives 38 with $\dim H^0 = 3$ (an additional F-branch singular exponent enters), and $\alpha = 1/2$, $2/3$ are resonant. The "34-dimensional basis" and "26 Ward combinations" of v6/v7 are therefore qualified statements; only the rows of Table 5 are exact.

14.3 Scope of the certificate

The certificate concerns the declared inventory and its functionals, at the listed rational angles. It does not prove a statement for general α (two points do not certify a rational function), does not identify the inventory with the complete interacting basis, and does not evaluate any graph. Within that scope it is CLOSED_PENDING_INDEPENDENT_REVIEW.

15 v8: Ward combinations, the 34- and 18-dimensional maps, and $\phi\phi$

15.1 Constant-profile support of the Ward combinations

Let K be the set of constant-profile coordinates, meaning jets with $m = 0$. Let π_K be the coordinate projection onto K, and let Ann_K be the annihilator of $\pi_K Z$ inside the span of the

K-supported functionals. In every exact run we found $|K| = 12$, with HH 6, CB 2, CK 3 and PP 1. The ST functionals supported on K span a space of dimension 6. The $m = 0$ rows alone already span Ann_K. Further, $\operatorname{rank} \pi_K Z = 6$ and $\operatorname{rank} \pi_K B = 5$. Also $\dim(Z \cap \ker \pi_K) = 2$, except at $(7071/10^4, \mathrm{F})$, where it is 3. The K-supported Ward combinations split by block support as

$$\underbrace{2}_{hh \text{ only}} + \underbrace{2}_{hh\text{+CK}} + \underbrace{2}_{\text{CB+CK}} = 6.$$

No combination is supported on CB, CK or PP alone. This decomposition is identical at every computed angle and branch.

15.2 The hh dictionary and the two exact Ward checks

The engine's constant-profile hh coordinates are related to the Wick-rotated longitudinal block by

$$x_{00} = -h_{\tau\tau}, \quad x_{03} = \pm \mathrm{i}\, h_{\tau z}, \quad x_{33} = h_{zz}, \quad f = \tfrac{1}{2}(h_{11} + h_{22}), \qquad q_0 = \pm \mathrm{i} q_\tau,\ q_3 = q_z.$$

Both sign choices give a solvable linear system, and the two give the same map. With $\Gamma_{\mathrm{sym}} = 4\Gamma$, the six engine coordinates y are

$$y = (-b_1,\, -2b_3,\, -b_2,\, b_6,\, b_7,\, -\tfrac{1}{4} b_8)$$

in units of κ^2/π, where b_j are the coefficients of the structures (9.1) (from $E = MS$ with $4b = M^{\mathsf{T}} y$, recorded in `hh_map_*.json`). The two hh-only Ward combinations are

$$W_1 = b_1 + 2b_3, \qquad W_2 = b_2.$$

Proposition 15.1 (Exact hh Ward checks; executed)**.** *In exact arithmetic, at $\alpha = 3/10$ and at $\alpha = 7071/10000$ and on both branches, the declared hh catalogue sum satisfies $W_1 = W_2 = 0$. The individual graphs do not. At $\alpha = 3/10$, for example,*

$$(W_1, W_2)^{\mathrm{D01}} = \left(-\tfrac{91}{5760},\ \tfrac{91}{2880}\right) = -(W_1, W_2)^{\mathrm{D03}},$$

and D02, D11 and D12 give zero.

So the check is not trivially satisfied: it tests a cancellation between D01 and D03. Like every catalogue statement, it is conditional on Assumption 7.1.

15.3 The remaining Ward combinations

The two hh+CK and two CB+CK combinations contain the v7 ST row and the projected ghost equation restricted to $m = 0$ (Theorems 10.1 and 10.3). This identification is CONDITIONAL, because v8 builds no CK/CB coordinate dictionary. The $26 - 6 = 20$ combinations that involve $m > 0$ jets have not been evaluated. They are OPEN, not zero.

15.4 The 18-dimensional engine map

The HH sector of the inventory has rank 18 on the 34-candidate branches, and rank 21 at $(7071/10^4, \mathrm{F})$. The dictionary above fixes the six constant-profile coordinates exactly, on all four computed branches. The other twelve involve transverse derivatives or non-axisymmetric jets, and their convention map is OPEN. The map is therefore PARTIAL: 6 of 18 coordinates.

15.5 The six exact values

Write the non-PP part of a closed class as $\sum_{i=1}^{6} e_i B_i$, using the exact basis of B stored in `apex_cert_*_vectors.json`. The hh constant-profile data give $\operatorname{rank}(\pi_{K\cap hh} B) = 4$, and the resulting system is consistent:

(α, branch)	e_1	e_3	$e_2 + 2e_5$	$e_4 + e_5$
$(3/10, \mathrm{CR})$	$7/40$	0	$7/120$	$-131/1920$
$(3/10, \mathrm{F})$	$7/40$	0	$7/120$	$63/640$
$(7071/10^4, \mathrm{CR})$	$2929/40000$	0	$-32842877/282840000$	$-78439319/4525440000$
$(7071/10^4, \mathrm{F})$	$2929/40000$	0	$-32842877/282840000$	$-136599773/1508480000$

At both angles, $e_1 = (1-\alpha)/4$. This is observed at two points, not proved for general α. The CR/F difference appears only in $e_4 + e_5$, where it equals $\nu/8$ exactly: $-1/6$ at $\alpha = 3/10$ and $2071/28284$ at $\alpha = 7071/10^4$, with $\nu = 4142/7071$. This matches the hh-sum difference $b_8^{\mathrm{CR}} - b_8^{\mathrm{F}} = -\nu/4$, which is consistent with (9.2). The coefficients e_5 and e_6 are not determined. They need CK/CB constant data, which in turn need a dictionary, and $m > 0$ data, because $\dim(Z \cap \ker \pi_K) = 2$ (and 3 at $(7071/10^4, \mathrm{F})$). The status is PARTIAL, and only in the stated basis.

15.6 The $\phi\phi$ values

The first $\phi\phi$ value is the catalogue PP coefficient $-c_2\kappa^2 P^2/(4\pi)$ on each branch. Its branch difference is $-\nu\kappa^2 P^2/(4\pi)$, and the engine coordinate is $y_{PP} = \kappa^2 c_2/(4\pi)$. For the second value, the D05a part is 0 in the reduced model of Section 11. The D04 part is still OPEN, for the following reasons.

- At constant profile, the replay of D04 has a single key $(0, 0)$. It has no hidden mixed pair and no evanescent coefficient, and its UNIT value is $-\kappa^2 P^2/(4\pi)$.
- The quadratic-profile part needs the r^2-weighted mixed-channel trace. v8 proves only the ingredients for it (Lemma 16.3); the mode sum and the r^2 assembly were not done.

16 v8: mixed channels, hidden pairs, and a Duhamel lemma

16.1 Feynman densities of channel pairs

For a bubble whose two lines carry channels s and t, let x be the Schwinger fraction of the t-line. The identity

$$\frac{1}{l^2(l-p)^2} = \int_0^1 \frac{\mathrm{d}x}{\big((l-xp)^2 + x(1-x)P^2\big)^2}$$

turns the pole of the slice numerator into a polynomial density $\rho_{(s,t)}(x)$. By construction, $\int_0^1 \rho_{(s,t)}\,\mathrm{d}x$ equals the circle pole of that pair. This identity was checked exactly for every bubble pair of every graph (`slice_D*.json`, field `rho_integrates_to_circle`). The cone value of the pair is

$$\int_0^1 \rho_{(s,t)}(x)\, C_{s,t}(x)\,\mathrm{d}x, \qquad C_{s,t}(x) = \operatorname{Tr}_{\mathrm{ap}}\big[e^{-(1-x)A_s} e^{-xA_t}\big].$$

For $s = t$ this is the circle pole times $\operatorname{Tr}_{\mathrm{ap}} e^{-A_s}$.

Definition 16.1 (Hidden mixed pair)**.** A pair with $s \neq t$ is *hidden* if its circle pole vanishes but $\rho_{(s,t)} \not\equiv 0$.

A selection rule based on totals, circle $\neq 0$, never sees a hidden pair. The historical lattice records value each pair as its circle pole times one channel coefficient, so a hidden pair receives the value zero.

16.2 The complete list (executed)

The slice run `logs/v8/slice/hidden_mixed.json` found hidden pairs in exactly two graphs.

- **D01, 13 keys.** Eight of them, $(1,1)$, $(1,6)$, $(2,2)$, $(2,8)$, $(6,1)$, $(6,6)$, $(8,2)$, $(8,8)$, were already quarantined; their pairs $(0,1)$ and $(1,0)$ have $\rho \propto (2x-1)$. The other five are **new**: $(4,4)$, $(4,7)$, $(5,5)$, $(7,4)$, $(7,7)$. Their pairs $(0,2)$ and $(2,0)$ have

$$\rho^{(4,4)}_{(0,2)} = -\frac{\mathrm{i}(q_0^2-q_3^2)(6x^2-10x+3)}{8\pi}, \qquad \rho^{(4,4)}_{(2,0)} = -\frac{\mathrm{i}(q_0^2-q_3^2)(6x^2-2x-1)}{8\pi}.$$

  The $(5,5)$ densities are 4 times these, $(7,7)$ equals $(4,4)$, and $(4,7)$, $(7,4)$ carry the opposite sign.

- **D03, 8 keys.** These are the vector-type keys, already excluded.

D02, D04, D05a, D06, D11, D12 and D13 have no hidden pairs.

Proposition 16.2 (Status of the v6 branch difference; executed and proved)**.** *The new D01 keys enter the hh sum, because the indices 4, 5 and 7 are the transverse components 11, 12 and 22. Since $\rho_{(2,0)}(1-x) = \rho_{(0,2)}(x)$ and $C_{2,0}(x) = C_{0,2}(1-x)$ by cyclicity, their total cone value is*

$$2\int_0^1 \rho_{(0,2)}(x)\, C_{0,2}(x)\,\mathrm{d}x = -2\int_0^1 R(x)\, C'_{0,2}(x)\,\mathrm{d}x.$$

Here $R(x) = \int_0^x \rho_{(0,2)}$, which is proportional to $x(x-1)(2x-3)$. This total is generically nonzero, and on the CR and F branches it is built from different channel spectra. Therefore the printed v6 value $-\nu\kappa^2 P^2 f_0 f_0'/(4\pi)$ is unchanged as the lattice evaluation of the declared catalogue. As a statement about the cone graph, however, it is CONDITIONAL *on the vanishing, or the separate evaluation, of this mixed-channel term.*

Proof. The symmetry of the densities is read off the executed list. The cyclicity follows from the cyclicity of the regulated apex trace. Integration by parts uses $R(0) = R(1) = 0$, which holds because $\int_0^1 \rho_{(0,2)} = 0$. □

16.3 A per-mode Duhamel identity

Lemma 16.3 (Duhamel identity for Bessel channels)**.** *In one angular mode, let $A_\mu = -\partial_r^2 - r^{-1}\partial_r + \mu^2 r^{-2}$ on its Friedrichs domain, with $\mu, \nu > 0$. With the bulk-subtracted mode trace,*

$$(\mu^2-\nu^2)\int_0^1 \mathrm{Tr}\big[e^{-xA_\nu}\, r^{-2}\, e^{-(1-x)A_\mu}\big]\mathrm{d}x = \frac{\mu-\nu}{2},$$

and $C'_{\mu,\nu}(x) = (\mu^2-\nu^2)\,\mathrm{Tr}[e^{-(1-x)A_\mu} r^{-2} e^{-xA_\nu}]$. In particular $C(0) = \mathrm{Tr}\, e^{-A_\mu}$ and $C(1) = \mathrm{Tr}\, e^{-A_\nu}$ after the same subtraction.

Proof. Differentiate $x \mapsto e^{-(1-x)A_\mu} e^{-xA_\nu}$ and use $A_\mu - A_\nu = (\mu^2-\nu^2)r^{-2}$. Integrate over $x \in [0,1]$, and use the per-mode value $-\mu/2$ (up to a μ-independent constant) of the subtracted trace of e^{-A_μ}. □

Numerical corroboration. We used Richardson extrapolation in the radial cutoff. For $\mu - \nu = 1$ the result was 0.49993 against 0.5, and for $\mu - \nu = 0.85$ it was 0.84990 against 0.85 (`duhamel_mode_check2.json`). This is corroboration, not an enclosure.

In the Cartesian frame, all channels share the angular phase set, and the Bessel orders are $|k/\alpha + S|$. The CR branch flips the $k = -1$ mode of $S = \pm 2$. Lemma 16.3 reduces $C'_{0,2}$ to a mode sum of r^{-2}-sandwiched traces. v8 does *not* evaluate that mode sum, and does not do the r^2-weighted D04 assembly. Both remain OPEN.

17 v8: evanescent coefficients under a declared prescription

The frozen $\tilde{g}$-leg arrays are still not supplied. v8 therefore declares a prescription and computes under it.

Definition 17.1 (Slice-DRED prescription)**.** On the zero-transverse-momentum slice, the UV degree -2 angular average uses $\hat{g}_L = \eta_L - \tilde{G}$. The symbols $\tilde{G}_{00}, \tilde{G}_{03}, \tilde{G}_{33}$ are formal, and $d = 2$ at pole level. The average is $\langle P \rangle = (l^2)^j \hat{\Delta}^j P/(2^j j! \prod_i (d+2i))$. The *evanescent coefficients* of a key are the coefficients of the $\tilde{G}$-dependent part of its pole.

Two checks were run for every pair of every graph. The $\tilde{G} = 0$ part reproduces the circle pole of the replay, and the bubble densities integrate to it. After combining with c_s and excluding quarantined and evanescent-mixed keys, the counts are as follows (`evanescent_combined.json`).

	D01	D02	D03	D04	D05a	D06	D11	D12	D13	total
CR	21	0	26	0	0	4	25	0	10	86
F	21	0	26	0	0	4	25	0	10	86
UNIT	34	0	34	0	0	6	25	0	14	113

Beyond the earlier quarantine, D01 also excludes its five evanescent-mixed keys. On the CR branch, D03, D06, D11 and D13 carry an overall factor $(\alpha - 1)$. For example,

$$\mathrm{D13}\,(4,0): \quad \frac{\mathrm{i}\kappa^2(\alpha^2-1)}{192\pi\alpha}(\tilde{G}_{00}q_0 - 2\tilde{G}_{03}q_3 + \tilde{G}_{33}q_0),$$

so the coefficient is $-c_0/16$ times the bracket. **Status:** CONDITIONAL on the declared prescription. These values are not identified with the frozen engine's $\tilde{g}$ export. Agreement at $\tilde{G} = 0$ is a consistency check, not a proof of identity preservation.

18 v9: what is new, and the response to the priority list

v9 is built mechanically from the read-only v8 source (`code/build_v9.py`). Every v8 statement is kept unless this section or Table 17 says it has changed. v9 has new evidence class marks: *Executed in v9* means a script under `code/v9/`, a log under `logs/v9/`, and a SHA-256 in Appendix C. The frozen trees XVI-P, XVI-Q and XVI-R were hashed before and after the v9 runs. Both digests equal the v8 values (Section 23). Files v1–v8 were not edited.

Table 6: v9 verdicts on the nine priority items.

#	Item	Verdict	Evidence / obstruction
1	Hidden mixed D01 terms	computed; v6 *restored*	Proposition 19.2: pure TT, vanishes on the lattice. $\Delta J(\alpha)$ at 14 angles (Table 7). IR log: Proposition 19.3.
2	r^2-weighted mixed sum, D04 part of 2nd $\phi\phi$	OPEN, precise obstruction	Section 22.1: the jet densities are not exported, and same-k pairing is not IR-summable. No closed form is claimed.
3	CK/CB dictionary; e_5, e_6; closed forms	dictionary computed; e_5, e_6 *excluded with proof* at constant profile	Theorem 20.2: all 6 constant-profile Ward functionals hold; $\ker \pi_K B$ is one-dimensional.
4	20 jet-level Ward combinations, full projection	OPEN, precise obstruction	Section 22.2.
5	18-dim engine map beyond 6	OPEN, precise obstruction	Section 22.2.

#	Item	Verdict	Evidence
6	Evanescent vs frozen $\tilde{g}$	*excluded*, precise obstruction	Proposition 22.1.
7	Vector-type keys	log parts computed; finite parts OPEN	Proposition 22.2.
8	Generic-α rank certificate	computed on chambers; exceptional set PARTIAL	Theorem 21.2.
9	New sheets	read; one newer sheet	Section 23; `new_oct18g/` written.

Changes to v8 results. (a) The v8 status "CONDITIONAL on the hidden D01 term" for the v6 hh branch difference is lifted. The value is unchanged and is now proved on the lattice (Proposition 19.2). (b) The four CK/CB-supported Ward combinations, which v8 identified only conditionally, are now checked exactly on the catalogue. This uses two fitted, α-independent ghost normalizations (Theorem 20.2). (c) The v8 open item "e_5 and e_6" is resolved negatively: the constant-profile data determine only $e_6 - e_2$ and $e_5 + e_6/2$ beyond the v8 combinations. (d) The incomplete-beta reduction used in an internal v8/v9 draft module for the F-branch $k = -1$ mode (`I_F`) contained an exponent error ($x^{1-\mu}$ for $x^{\mu-1}$). It was not used in any printed v8 number. It was removed, and every v9 value uses the closed-form quadrature `I_gen`. No other v8 number changes.

19 v9: the hidden D01 pairs

19.1 Closed form of the mixed trace

Let A_μ be the radial Bessel operator of order $\mu \geq 0$ on the cone channel, and let

$$g(x;\mu,\nu) = \mathrm{Tr}\big[e^{-(1-x)A_\mu}\, r^{-2}\, e^{-xA_\nu}\big], \qquad 0 < x < 1,$$

with $d = \mu - \nu$ and $h = (\mu+\nu)/2$.

Proposition 19.1 (Mixed trace; derived and executed)**.**

$$g(x;\mu,\nu) = \frac{\Gamma(h)\Gamma(h+1)}{4\,\Gamma(\mu+1)\Gamma(\nu+1)}\; {}_2F_1\big(\tfrac{d}{2}+1, \tfrac{d}{2}; \mu+1; x\big)\; {}_2F_1\big(1-\tfrac{d}{2}, -\tfrac{d}{2}; \nu+1; 1-x\big).$$

In particular $g(x;\mu,\mu) = 1/(4\mu)$*, and* $\int_0^1 g\,\mathrm{d}x = 1/(2(\mu+\nu))$.

Derivation. Expand both Bessel kernels in their power series and integrate the radial variables term by term. This gives $x^\mu(1-x)^\nu$ times $F_4(h, h+1; \mu+1, \nu+1; x^2, (1-x)^2)$. Since $h + (h+1) - (\mu+1) + 1 = \nu + 1$, Bailey's reduction $F_4(a,b;c,a+b-c+1;X(1-Y),Y(1-X)) = {}_2F_1(a,b;c;X)\,{}_2F_1(a,b;a+b-c+1;Y)$ applies with $X = x$ and $Y = 1-x$. Euler's transformation of each factor removes the prefactor $x^\mu(1-x)^\nu$ and gives the stated parameters. The integral identity is an independent check of the v8 Duhamel lemma, Lemma 16.3. `code/v9/g_formula_check.py` confirms both the closed form and the integral against direct quadrature of the kernels. The residual, 4.7×10^{-3}, is the truncation tail of the quadrature at $R = 60$, not a discrepancy. □

19.2 The hidden pairs are pure TT

With $\hat{\rho} = 6x^2 - 10x + 3$ and $\hat{R} = x(1-x)(3-2x)$, we have $\hat{R}' = \hat{\rho}$. For a mode pair (μ,ν), integration by parts gives

$$I(\mu,\nu) = \int_0^1 \hat{\rho}\, f = -(\mu^2 - \nu^2) \int_0^1 \hat{R}\, g.$$

`code/v9/modefun.py` evaluates this in two independent ways for $\nu = \mu \pm 2$: an exact Gamma series summed by Euler–Maclaurin, and direct quadrature of Proposition 19.1. The two agree to 30 digits.

Proposition 19.2 (TT structure; proved)**.** *On the D01 keys* $(4,4),(4,7),(7,4),(7,7),(5,5)$*, the hidden* $(0,2)/(2,0)$ *pairs contribute*

$$-\frac{\kappa^2 P^2 J}{8\pi}\Big[(h_{11}-h_{22})(h'_{11}-h'_{22})+4h_{12}h'_{12}\Big],$$

with $J=\sum_{\text{modes}} I$ *in the stated mode list. On the v6 lattice* ($h_{11}=h_{22}=f$*, no* h_{12}*) the bracket is identically zero. Hence the v6 hh branch difference* $-\nu\kappa^2P^2f_0f_0'/(4\pi)$ *holds without the v8 condition.*

Proof. The key-by-key coefficients come from `logs/v8/slice/hidden_mixed.json` and the catalogue normalization $(P^2\kappa^2/16\pi)/(\mathrm{i}(q_0^2-q_3^2)/8\pi)$. Key $(4,4)$ receives $-\kappa^2P^2J/(8\pi)$; keys $(4,7)$ and $(7,4)$ carry the opposite sign, and key $(5,5)$ carries four times key $(4,4)$. Collecting terms gives the bracket. The lattice projection of `invariant_data` uses $\{0{:}0, 3{:}1, 9{:}2, 4{:}3, 7{:}3\}$. It sets $h_{11}=h_{22}$ and drops index 5, so it maps the bracket to zero for every value of J. □

19.3 Infrared structure and the finite branch difference

Proposition 19.3 (IR logarithm; executed asymptotics)**.** *As* $\mu\to\infty$*,* $I(\mu,\mu+2)=\frac{1}{3}+\frac{1}{30\mu}-\frac{1}{10\mu^2}+\cdots$ *and* $I(\mu,\mu-2)=-\frac{1}{3}+\frac{1}{30\mu}+\frac{1}{10\mu^2}+\cdots$ *(coefficients up to* μ^{-8} *are in* `logs/v9/asym02.json`*). Under same-k pairing, the constant terms cancel between* k *and* $-k$*, but the* $1/\mu$ *terms add to* $\alpha/(15k)$*. Therefore J diverges like* $(\alpha/15)\log\Lambda$*. In general, a pair* (s,t) *contributes* $\frac{\alpha}{2}(s-t)^2\,x(x-1)\log\Lambda$ *to* $C_{s,t}(x)$*. This is the frame/flux mismatch of a long-range* $r^{-2}L$ *perturbation.*

So the absolute TT values are not pure apex coefficients. Their finite part depends on the scheme (for example k-ζ). The CR and F branches differ only in the $S=2$, $k=-1$ mode, whose order is $-\nu_0$ (CR) or $+\nu_0$ (F), with $\nu_0=2-1/\alpha$. The difference

$$\Delta J(\alpha)=I(1/\alpha,\,1/\alpha-2)-I(1/\alpha,\,2-1/\alpha)$$

is therefore finite and does not depend on the regularization. The corresponding catalogue difference is $-\kappa^2P^2\Delta J/(8\pi)$ times the TT bracket, and it is invisible on the lattice.

Table 7: The finite TT branch difference (executed, `code/v9/hidden02_run.py`; CR mode by series and by quadrature, agreement $< 10^{-30}$).

α	ν_0	I_{CR} $(k=-1)$	I_{F} $(k=-1)$	ΔJ
7/20	−0.857143	−0.301772223719917	−0.316747429419663	0.014975205699746
2/5	−0.500000	−0.292350993934024	−0.321408664644675	0.029057670710650
9/20	−0.222222	−0.281195058597902	−0.300043868930418	0.018848810332516
1/2	0.000000	−0.268135533878013	−0.268135533878013	0.000000000000000
11/20	0.181818	−0.253008191060374	−0.233034968697703	−0.019973222362671
3/5	0.333333	−0.235654644431579	−0.198192074766226	−0.037462569665354
2/3	0.500000	−0.208792354336879	−0.154666895890071	−0.054125458446808
7071/10000	0.585773	−0.190296473359310	−0.130398179653525	−0.059898293705785
3/4	0.666667	−0.168755307135675	−0.106489500975910	−0.062265806159765
4/5	0.750000	−0.141051936421474	−0.080941692454300	−0.060110243967174
17/20	0.823529	−0.110436566041365	−0.057725471830814	−0.052711094210551
9/10	0.888889	−0.076793924011713	−0.036635399844723	−0.040158524166990
19/20	0.947368	−0.040015650826980	−0.017460622324418	−0.022555028502562
99/100	0.989899	−0.008266790265495	−0.003364124456341	−0.004902665809154

At $\alpha=1/2$, $\nu_0=0$ and $\Delta J=0$ exactly. The sign of ΔJ is the sign of $-\nu_0$ at every computed angle. As $\alpha\to1$, the CR mode degenerates ($\mu+\nu\to0$); the values decrease toward zero ($\Delta J(99/100)\approx-4.9\times10^{-3}$). PSLQ was not attempted. We claim no elementary closed form, since each value is an integral of a product of two ${}_2F_1$ factors.

20 v9: the CB/CK dictionary and the six constant-profile Ward functionals

20.1 Dictionaries

We use the same method as the *hh* map (Section 15). Unit $m = 0$ atoms $L = (1,0), (0,1)$ for $c, \bar{c}$, and LL/PM atoms for K, are evaluated by `eval_sym` (reflection only, so $\Gamma_{\rm sym} = 2\Gamma$) on the complete 6×6 momentum grid. A dictionary is accepted only if the catalogue bilinear (D06 for $\langle \bar{c}\, c\rangle$, D13 for $\langle Kc\rangle$) lies *exactly* in the span of the evaluated K-candidates on all columns (`code/v9/cbck_map.py`, `code/v9/ck_free.py`).

Proposition 20.1 (CB and CK dictionaries; executed, exact)**.** *With $q_0 = \pm \mathrm{i} q_\tau$, $q_3 = q_z$:*

(i) CB: exactly the four sign choices with opposite i*-signs for $\bar{c}_0$ and c_0 are accepted. All four give $y_{\rm CB} = \mathrm{i}(-R/8,\, 0)\, \kappa^2/\pi$, with $R = 3z_0 + 4z_1$.*

(ii) CK: with the K components left free, the solution is unique: $K^{00} = -LL_{\tau\tau}$, $K^{03} = \mp 2\mathrm{i} LL_{\tau z}$, and $y_{\rm CK} = \mathrm{i}\big((1-\alpha)/16,\, R/8,\, 0\big)\kappa^2/\pi$. The PM scale is undetermined because the catalogue's PM keys are computed zeros.

Both hold at $(3/10, \mathrm{CR})$, $(3/10, \mathrm{F})$, $(7071/10^4, \mathrm{CR})$ and $(7071/10^4, \mathrm{F})$. Here $R = (17\alpha - 7)(\alpha - 1)/(12\alpha)$.

20.2 The six functionals

The annihilator of $\pi_K Z$ is printed in `apex_cert_*_vectors.json` and is identical at all four points. In the order $K = (0, 1, 2, 5, 6, 12, 18, 19, 26, 27, 29, 32)$ it consists of $W_1 = y_0 + y_1$, $W_2 = y_2$, $W_3 = -2y_0 + y_{19}$, $W_4 = -2\mathrm{i}y_0 + 4\mathrm{i}y_5 + 4\mathrm{i}y_6 + y_{26}$, $W_5 = -2\mathrm{i}y_{18} + y_{27}$, and $W_6 = 2\mathrm{i}y_0 + y_{29}$. (v8 quoted only the real parts of these vectors.)

Theorem 20.2 (Ghost-sector Ward check; executed, exact)**.** *Put $\lambda_{\rm CB} = -4\mathrm{i}$ and $\lambda_{\rm CK} = -8$, fixed once and independent of α and of the branch. Rescale the CB and CK coordinates by these factors. Then all six W_i vanish on the catalogue at all four certified points. Solving $\sum_i e_i \pi_K B_i = y_K$ on all twelve coordinates gives*

$$e_3 = 0, \quad e_1 = \tfrac{1-\alpha}{4}, \quad e_6 - e_2 = -\tfrac{R}{2}, \quad e_5 + \tfrac{e_6}{2} = \frac{\alpha^2 - 1}{48\alpha},$$

$$e_4 - \tfrac{e_6}{2} = \frac{119\alpha^2 - 120\alpha + 25}{192\alpha}\ (\mathrm{CR}), \qquad e_4 - \tfrac{e_6}{2} = \frac{7(17\alpha - 7)(\alpha - 1)}{192\alpha} = \tfrac{7R}{16}\ (\mathrm{F}).$$

The kernel of $\pi_K B$ is exactly $\mathrm{Span}\{(0, 1, 0, \tfrac{1}{2}, -\tfrac{1}{2}, 1)\}$*. Therefore e_5 and e_6 are not determined separately by constant-profile data.*

Proof. `code/v9/e_solve.py` carries out the linear algebra exactly; its output is `logs/v9/e_solve.json`. The checks have different strengths. W_3 and W_6 hold because both of their entries vanish. W_5 defines $\lambda_{\rm CB}$, since both of its terms are proportional to R. W_4 is a genuine check. Its *hh* part is $4\mathrm{i}(b_6 + b_7) = \mathrm{i}(c_0 - c_1) = \mathrm{i}(1-\alpha)/2$, coming from D01–D03 and D11–D12. Its CK part is $\lambda_{\rm CK}\, \mathrm{i}(1-\alpha)/16$, coming from D13. The two have the same α-shape, so one constant $\lambda_{\rm CK} = -8$ works at every α and on both branches. The e-values follow from the twelve equations. The closed forms are the rational identities displayed above, checked at the four points. □

Scope. The normalizations λ are fitted, not derived from the catalogue's BRST conventions. The closed forms hold on the certified points. For general generic α they follow from the closed forms of z_0, z_1, z_2 together with the observed α-independence of $\pi_K B$ and of the annihilator, which was verified at four points but not proved. Status: PARTIAL for general α, exact on the four points. The v8 identity $e_1 = (1-\alpha)/4$ is now obtained twice, from the *hh* block and from the CK coordinate 26.

21 v9: chambers and a generic-α rank certificate

The structural key of a jet (C, n, m) is $(C, n, \epsilon, j, \mathrm{sgn}(n/\alpha + s))$, where $m = \epsilon\nu + 2j$. Two angles lie in the same *chamber* when every gh0 ($T = 4$) and gh-1 ($T = 3$) sector has the same set of structural candidates. In a chamber, every ST row entry is a fixed rational function of α.

Proposition 21.1 (Walls; executed and proved)**.** *In* $(1/4, 1)$ *the walls are exactly* $\{1/3, 1/2, 2/3\}$*, on both branches. CR has the same structural signature on all four chambers. F has one signature on* $(1/4, 1/3) \cup (1/3, 1/2)$ *and a different one, with* $N = 38$ *at the certified points, on* $(1/2, 2/3) \cup (2/3, 1)$*.*

Proof. A non-generic candidate requires $\pm 2n/\alpha \in \mathbb{Z}$ for some admissible pair $(n, -n)$, or a sign or ϵ change. The jet budget gives $\nu \leq 6$ and $n \leq 4$. So a wall has the form $\alpha = 2n/M$ with $M \leq 12$, or $\alpha = n/|s|$. Every such rational has denominator at most 12. `code/v9/chamber_scan.py` evaluates the structural signature at every rational with denominator at most 40 in $(1/4, 1)$, which is exhaustive by this bound. It finds only the three walls listed (`logs/v9/chamber_cache2.json`). □

Theorem 21.2 (Generic rank over $\mathbb{Q}(\alpha)$)**.** *On every CR chamber in* $(1/4, 1)$*, and on the F chambers in* $(1/4, 1/2)$*, the ST rank over* $\mathbb{Q}(\alpha)$ *is exactly* 26*, with* $\dim Z = 8$ *generically. On the F chambers in* $(1/2, 1)$*, the generic rank* r *satisfies* $29 \leq r \leq 30$*.*

Proof. Lower bound. The rank is lower semicontinuous under specialization within a chamber. So it suffices to have one point per chamber with a nonzero minor of the required size. Exact $\mathbb{Q}(\mathrm{i})$ minors exist at $3/10$ and $7071/10^4$ (Theorem 14.1). A nonzero minor mod p also implies a nonzero rational minor, and the modular runs at $41/100$, $309/500$, $4/5$, $8501/10^4$ and at the scan points of Table 8 provide such minors. *Upper bound.* The identity $\sigma_0\sigma_{-1} = 0$ holds identically, so $B \subset Z$. We have $s\phi = 0$, so $PP \subset Z$. B has no PP components, and $\dim B \geq 6$ generically, by lower semicontinuity from the certified points. Hence $\dim Z \geq 8$, which gives $r \leq N - 8$: 26 for $N = 34$ and 30 for $N = 38$. The ninth cocycle seen at the F points with $N = 38$ is not shown to persist generically, so $29 \leq r \leq 30$ there. □

Table 8: Modular scan (`code/v9/apex_scan.py`, an unchanged copy of the historical driver with a new output directory; prime $10^9 + 9$, randomized complete-rank sampling). Every rational $p/q \in (1/4, 1)$ with $q \leq 20$.

branch	result	angles
CR	$(1/4, 1/3)$: $N = 34$, rank 26, $\dim Z = 8$, $\dim B = 6$, $\dim H^0 = 2$	10 angles
CR	$(1/3, 1/2)$: $N = 34$, rank 26, $\dim Z = 8$, $\dim B = 6$, $\dim H^0 = 2$	21 angles
CR	$(1/2, 2/3)$: $N = 34$, rank 26, $\dim Z = 8$, $\dim B = 6$, $\dim H^0 = 2$	21 angles
CR	$(2/3, 1)$: $N = 34$, rank 26, $\dim Z = 8$, $\dim B = 6$, $\dim H^0 = 2$	41 angles
CR	wall $1/3$: $N = 35$, rank 27, $\dim Z = 8$, $\dim H^0 = 2$	wall
CR	wall $1/2$: $N = 42$, rank 33, $\dim Z = 9$, $\dim H^0 = 2$	wall
CR	wall $2/3$: $N = 35$, rank 27, $\dim Z = 8$, $\dim H^0 = 2$	wall
F	$(1/4, 1/3)$: $N = 34$, rank 26, $\dim Z = 8$, $\dim B = 6$, $\dim H^0 = 2$	10 angles
F	$(1/3, 1/2)$: $N = 34$, rank 26, $\dim Z = 8$, $\dim B = 6$, $\dim H^0 = 2$	21 angles
F	$(1/2, 2/3)$: $N = 38$, rank 29, $\dim Z = 9$, $\dim B = 6$, $\dim H^0 = 3$	21 angles
F	$(2/3, 1)$: $N = 38$, rank 29, $\dim Z = 9$, $\dim B = 6$, $\dim H^0 = 3$	41 angles
F	wall $1/3$: $N = 35$, rank 27, $\dim Z = 8$, $\dim H^0 = 2$	wall
F	wall $1/2$: $N = 42$, rank 33, $\dim Z = 9$, $\dim H^0 = 2$	wall
F	wall $2/3$: $N = 39$, rank 30, $\dim Z = 9$, $\dim H^0 = 3$	wall
Completed: 192 of 192 runs (an incomplete scan is reported as such, never extrapolated).		

Exceptional set. Inside a chamber, the rank drops exactly on the common zero set of the maximal minors. That set is finite, but the minor polynomial was not computed: it would need a symbolic-α elimination of unknown degree. So the exceptional set is $E = \{1/3, 1/2, 2/3\} \cup E_{\mathrm{minor}}$. The scan shows that E_{minor} contains no rational with denominator at most 20, but E_{minor} itself is OPEN.

22 v9: precise obstructions

22.1 The D04 part of the second $\phi\phi$ value

The D04 internal lines are a graviton, with channel weights pw, and ϕ ($S = 0$). At constant profile only z_0 contributes, and the same-channel part is the x-independent D-moment $\mathrm{Tr}[r^2 e^{-A}]$. The second $\phi\phi$ value needs the r^2-weighted mixed traces $\mathrm{Tr}[e^{-(1-x)A_\mu} r^2 e^{-xA_\nu}]$ with transverse-momentum numerators and the H_I intertwining. There are two obstructions:

(a) the zero-transverse-momentum slice does not export these jet-level densities, so the integrand is not available as data;

(b) the frame/flux mismatch of Proposition 19.3 carries over with an extra r^2. Same-k pairing is then not IR-summable. The mismatch gives the log of the r^{-2} case there, and power counting gives a power divergence with the r^2 weight (*argued, not executed*).

So a rational reconstruction in α has nothing well-defined to reconstruct until a scheme for the mixed same-k sum is declared. Status: OPEN; no closed form is claimed, and no checks at $\alpha = 1$ or $1/2$ are claimed.

22.2 Jet-level Ward combinations and the engine map

Twenty of the 26 Ward combinations, and twelve of the eighteen hh engine coordinates, involve $m > 0$ jets (r^2 profiles) or transverse P/M components. The dictionary method of Sections 15 and 20 would need the catalogue's values on $m = 2$ and weight-± 1 atoms. The declared catalogue has no such values: it is the zero-transverse, constant-profile slice. They would come from the r^2/D_b-weighted traces, which are obstructed as in (b) above. The full 34-dimensional projection is therefore available only on the twelve K coordinates, where all six functionals now hold (Theorem 20.2). That projection has rank 6, and its angle and branch dependence enters only through z_0, z_1, z_2. Status: OPEN.

22.3 Evanescent coefficients versus the frozen $\tilde{g}$ rule

Proposition 22.1 (Excluded, with obstruction; executed)**.** *The frozen engine (`XVI-P/code/p5_entry/dred.py`, `eta_trace_Gt`, read-only) admits exactly one evanescent pattern, $c\sum_r \eta^{rr}\tilde{G}_{rr} \to c(4-d)$, and declares every other pattern open. Of the 285 v8 slice-DRED keys (`logs/v8/slice/evanescent_combined.json`), 164 contain $\tilde{G}$ monomials of degree 2 or 3. All of them contain only the longitudinal $\tilde{G}_{00}, \tilde{G}_{03}, \tilde{G}_{33}$, and only 31 have a linear part of partial-trace form $c(\tilde{G}_{00} - \tilde{G}_{33})$ (`code/v9/evan_match.py`). The v8 coefficients therefore lie outside the class the frozen rule can evaluate. A match would need the transverse $\tilde{G}_{11}, \tilde{G}_{22}$ components, which the slice does not export, and a rule for non-trace monomials.*

22.4 Vector-type keys

Proposition 22.2 (IR logs of the hidden keys; executed)**.** *Let $L_{\mathrm{key}} = \sum_{(s,t)}(s-t)^2 \int_0^1 \rho_{(s,t)} x(x-1)$ over D01 and D03 (`code/v9/vector_logs.py`), and put $c = \mathrm{i}P^2/(30\pi)$. Then the vector keys $(1,6),(2,8),(6,1),(8,2)$ have $L = 0$. The keys $(1,1)$ and $(2,2)$ have $L = +c$, and $(6,6)$ and $(8,8)$ have $L = -c$. The TT keys give $-c, +c, +c, -c, -4c$ on $(4,4),(4,7),(7,4),(7,7),(5,5)$.*

The vector-key log is $c\,P^2 \sum_{a=1,2} \eta^{\mu\nu} h_{\mu a} h'_{\nu a}$ up to key normalization. It does not cancel within D01+D03, so the vector keys inherit the scheme dependence of Proposition 19.3. Whether profile-dependent Ward partners remove it cannot be decided from the slice. The finite parts of the vector keys stay OPEN and are excluded, not zero.

23 v9: frozen-tree digests and the new registry sheets

The tree digests of XVI-P, XVI-Q and XVI-R, in the shell serialization of Section 13.6, were recomputed before the first v9 run and after the last one:

Tree	before / after	SHA-256
XVI-P	equal	`ec965c15ae25cd3bb9cc2b3a134e966b`
		`19a1803c78d1fe5bf7554da2a8b3aaa6`
XVI-Q	equal	`0df9a0d10409d31ac0c2d0a47197e3f6`
		`e03c7ae258ec5fb775137f66cb0493c4`
XVI-R	equal	`57187bc6825f5c6ae88633b7279c6c0b`
		`e57f7c4918fdbc525da4aaf958331edb`

They are equal to the v8 values. The v8 source and PDF were hashed before and after as well (`logs/v9/v8_hashes_before.sha`, `logs/v9/v8_hashes_after.sha`).

Registry. The registry was fetched again on 18 October 2026. It lists 258 identifiers, one more than the 257 in `new_oct18f/`: the new identifier is `I-C-Ex2`. It was stored under `new_oct18g/` with SHA-256 sums. The six sheets of `new_oct18f/` and `I-C-Ex2` were read as cited inputs (abstracts and all theorem statements; `logs/v9/sheets_extract.txt`):

- *I-B-G-S-B* [8] is the only sheet that bears on a v9 input. For a declared class of nonnegative tensor connection forms, it proves Mosco and strong-resolvent selection of the componentwise Friedrichs realization, and states that for neutral spin 2 and $\alpha > 1/2$ the limit excludes the curvature-regular branch. This bears on the CR/F branch input: it favours F *within that declared core class*. It does not derive the graviton operator, and the sheet itself stresses that a smooth metric and its holonomy do not fix the perturbation operator. So both branches remain inputs here, and the CR results are kept as conditional on CR.
- *I-B-ECHO-v2* [9] gives the scalar apex-diffracted Green operator and finite report instruments. It is not an apex-residue input; compare Example 3.3.
- *X-J-ST-2* [10] gives the free photon reference (Maxwell–Friedrichs with a Friedrichs scalar ghost) and charged scalar coefficients. It is a free, source-test reference and imports no interacting graviton Ward identity.
- *Hawk05-ST, I-C-v2, I-C-Ex-v2, I-C-Ex2* [11, 12, 13, 14] are finite access and recovery bounds $(1 - 1/d,\ 1 - k/d)$. They belong to the instrument side of the boundary of Section 12 and are not used for O.1.

24 What v10 adds, and what it does not

v10 keeps every v9 statement [1]. The v9 statements that v10 amends are listed in Section 31. The new items, with their status, are:

Table 9: v10 verdicts (all executed; hashes in Appendix D).

Item	Status
ΔJ closed form	CR mode *derived*. F mode *identified* by PSLQ and verified at 9 fresh angles ($\leq 1.5 \times 10^{-52}$), not derived. Section 25.
$\lambda_{\mathrm{CB}}, \lambda_{\mathrm{CK}}$	*Derived* from tree operators and the v9 dictionaries (Theorem 26.1).
Exceptional points inside chambers	*None*: rank 26 on all $N = 34$ chambers, 29 on F above $\frac{1}{2}$. Conditional on the a priori degree bound (Theorem 27.1).
Ninth cocycle on F above $\frac{1}{2}$	*Resolved*. It is present at every angle; it is not a constant vector, while the other eight are.

Transverse-jet catalogue	PARTIAL. Exact slice jets for all 9 graphs. Same-channel r^2 part dressed in two declared schemes; mixed and weight $\neq 0$ parts OPEN (Section 28).
D04 part of second $\phi\phi$ value	PARTIAL. The same-channel part is exactly 0. The full jet term is mixed $(1,0)$, with constant density $\kappa^2/4\pi$. Per-mode traces are computed; the mode sum is not executed.
20 jet Ward combinations; 34-/18-dim maps; e_5, e_6	OPEN. Precise obstruction in Section 29. No residual is claimed, since none was evaluated.
Vector-key finite parts	Computed in a declared k-ζ scheme at 4 angles (Table 11).
Registry	Unchanged (258 ids); no new sheet directory written.

25 v10: closed form of the finite TT branch difference

Proposition 25.1 (CR mode; derived)**.** *Let $\mu = 1/\alpha$ and $\beta = \mu - 1$. Then*

$$I(\mu, \mu-2) = \frac{(\mu-1)(6\mu^2+3\mu-2)}{6} - (\mu-1)^3(\mu+1)\,\psi_1(\mu-1).$$

The same expression gives every $k \le -2$ mode with $\mu = m/\alpha$.

Derivation. In the Gamma series of `code/v9/modefun.py`, use $(n+1)!\,\Gamma(\mu+1)/\Gamma(\mu+1+n) = \mu(\mu-1)B(n+2,\beta)$. Each sum becomes a log-Beta integral. Continue it analytically, keeping the Laurent data at the coincident poles. With $T = \psi_1(\beta)$, the intermediate sums are $G_2 = (T\beta-1)/\beta$, $G_3 = (-T\mu^2+3T\mu-2T+\mu-\frac{3}{2})/\beta$ and $G_4 = (6T\mu^3-36T\mu^2+66T\mu-36T-6\mu^2+27\mu-25)/(12\beta)$. Collecting them gives the formula (`code/v10/cr_mode_closed.py`). At $\alpha = 3/5$, 10000/7071, 4/7 and 99/100 it matches the 34-digit series to $\le 6\times 10^{-35}$. □

Proposition 25.2 (F mode; identified, not derived)**.**

$$I(\mu, 2-\mu) = \frac{(\mu-1)(59\mu^3-57\mu^2-62\mu+48)}{36} - (\mu-1)^3(\mu+1)\frac{\pi^2}{6}.$$

PSLQ [3] on the basis $\{I, 1, \pi^2\}$ found integer relations at $\mu \in \{6/5, 5/4, 4/3, 7/5, 3/2, 8/5, 5/3\}$. The values were computed to 55 digits by direct quadrature of Proposition 19.1 (`code/v10/f_mode_pslq.py`). The relations interpolate to the displayed polynomial. That polynomial was then checked at 7/4 and 9/5, which were not used in the fit. This is an identification, and its status is *verified, not derived.*

Corollary 25.3 (ΔJ in closed form)**.**

$$\Delta J(\alpha) = \frac{(\mu-1)(-59\mu^3+93\mu^2+80\mu-60)}{36} - (\mu-1)^3(\mu+1)\Big[\psi_1(\mu-1) - \frac{\pi^2}{6}\Big], \qquad \mu = \tfrac{1}{\alpha}.$$

It vanishes at $\alpha = \frac{1}{2}$ ($\psi_1(1) = \pi^2/6$, and the polynomial has a zero at $\mu = 2$). As $\alpha \to 1$, $\Delta J = -\frac{1}{2}(\mu-1) + O((\mu-1)^2)$, consistent with the v9 value $\Delta J(99/100) \approx -4.9\times 10^{-3}$.

Table 10: Fresh-angle verification (`code/v10/dJ_closed_verify.py`; quadrature at 60 digits, and 80 digits for a precision check).

$\mu = 1/\alpha$	$\lvert I_{\mathrm{F}}^{\mathrm{quad}} - I_{\mathrm{F}}^{\mathrm{closed}}\rvert$	$\lvert I_{\mathrm{CR}}^{\mathrm{quad}} - I_{\mathrm{CR}}^{\mathrm{closed}}\rvert$	ΔJ (closed)
$\sqrt{2}$	1.0×10^{-56}	2.5×10^{-56}	-0.0598989753454146726
10000/7071	5.1×10^{-57}	1.5×10^{-56}	-0.0598982937057849765
137/100	2.3×10^{-56}	2.8×10^{-56}	-0.0616430256621234476
193/100	6.1×10^{-56}	2.0×10^{-56}	-0.0073607981128415757

φ_g	2.5×10^{-57}	3.6×10^{-56}	-0.0427973775860188253
$\pi/2$	1.0×10^{-56}	7.6×10^{-57}	-0.0476710589083126482
$5/2(\alpha=2/5)$	1.6×10^{-55}	1.5×10^{-52}	0.02905767071065043833
$29/10(\alpha=10/29)$	5.1×10^{-56}	5.6×10^{-53}	0.01105442640041502692
$101/100$	8.0×10^{-59}	3.2×10^{-58}	-0.0048550798217175446

The 14 v9 values of Table 7 agree with the closed form to 4.4×10^{-27}. The v9 sentence "we claim no elementary closed form" is therefore superseded for the CR mode and qualified for the F mode.

26 v10: the ghost normalizations from tree data

Theorem 26.1 (Derivation of λ_{CK} and λ_{CB})**.** *Use the dictionaries of Sections 15 and 20, with signs $s_i=1$, $s_c=-1$, $s_{\bar c}=1$; the K-dictionary $K^{00}=-\mathrm{LL}_{\tau\tau}$, $K^{03}=-2\mathrm{i}\,\mathrm{LL}_{\tau z}$, $K^{33}=\mathrm{LL}_{zz}$; and $q_0=\mathrm{i}q_\tau$, $q_3=q_z$. Then the following three tree-level operator identities hold on all $m=0$ atoms and all nine test momenta $(q_\tau,q_z)\in\{1,2,3\}^2$:*

$$\begin{aligned}
&\textit{engine } K_2c=\rho\,\delta_c h, && \delta_c h_{mn}=p_m c_n+p_n c_m, && \rho=-\mathrm{i},\\
&\textit{engine } E(-q)h=\epsilon\,(2-\delta_{mn})E^{mn}_{\mathrm{cat}}(h), && && \epsilon=1,\\
&\textit{engine } C_2^{\mathsf T}(-q)b=\tau\,Y_b, && Y_b=-2\eta F_L^{\mathsf T}b, && \tau=-\mathrm{i}/2.
\end{aligned}$$

With the eval normalizations of the engine ($\Gamma_{\mathrm{sym}}=4\Gamma$ on HH and 2Γ on CB and CK), the hc row becomes $8\rho B+2\lambda_{\mathrm{CK}}\epsilon G=0$. Theorem 10.3 gives $B+\lambda_{\mathrm{pred}}G=0$, with $\lambda_{\mathrm{pred}}=N/(\mathrm{i}A_{\mathrm{GH}})=-2\mathrm{i}$ (Input 10.2). Hence

$$\lambda_{\mathrm{CK}}=\frac{4\rho\lambda_{\mathrm{pred}}}{\epsilon}=-8.$$

The bc row becomes $2\lambda_{\mathrm{CB}}\Gamma_{\mathrm{D06}}+2\lambda_{\mathrm{CK}}\tau\Gamma_{\mathrm{D06}}=0$ by the ghost identity $\hat B_{\mathrm{D06}}=-2\eta F_L\hat\Gamma_{\mathrm{D13}}$ (Theorem 10.1). Hence

$$\lambda_{\mathrm{CB}}=-\tau\lambda_{\mathrm{CK}}=-4\mathrm{i}.$$

Proof. `code/v10/lambda_derive.py` evaluates the engine operators of `code/v8/apex_exact.py` (K_2 at q; E and $C_2^{\mathsf T}$ at $-q$, as in the row construction of `apex_certificate.py`). Each ratio is single-valued on every nonzero component, with no mismatch (`logs/v10/lambda_derive.json`). The two row reductions are linear algebra on these ratios. □

Remark 26.2. The derivation uses no loop value. It does use the dictionary signs, which v9 selected by exact solvability on loop data. The values -8 and $-4\mathrm{i}$ are therefore no longer free fitted parameters, but they are not independent of the v9 sign choice. The relation $\lambda_{\mathrm{CK}}=\lambda_{\mathrm{pred}}\lambda_{\mathrm{CB}}$ noted in v9 is explained by the numerical coincidence $4\rho/\epsilon=-4\mathrm{i}=\lambda_{\mathrm{CB}}$, i.e. $-\tau\lambda_{\mathrm{CK}}=4\rho/\epsilon$ with $\lambda_{\mathrm{CK}}=(4\rho/\epsilon)\lambda_{\mathrm{pred}}$.

27 v10: chamber ranks without exceptional points

`code/v10/minor_poly.py` picks, at one angle per chamber, three nonsingular maximal minors with α-free row tags. The atom key is (C,n,j,i) with $j=(m-\varepsilon|a|)/2$. The program then re-evaluates the minors at 11–15 angles per chamber (`logs/v10/minor/<tag>/`).

Theorem 27.1 (Rank inside chambers)**.** *Assume the a priori bound (D): every ST row entry is a polynomial of degree ≤ 2 in $1/\alpha$ on each chamber. This is argued from the construction: E carries two derivative factors, K_2 one, and each is linear in $1/\alpha$. It is not separately machine-checked. Then:*

(a) On CR $(\frac{1}{4},\frac{1}{3}),(\frac{1}{3},\frac{1}{2}),(\frac{1}{2},\frac{2}{3}),(\frac{2}{3},1)$ *and on F* $(\frac{1}{4},\frac{1}{3}),(\frac{1}{3},\frac{1}{2})$ *($N=34$), all three minors are the same constants at every sampled angle,* $12451552249753947340 8\,\mathrm{i}$, $-73477495818488684950134624092160000$ *and* $-40591041450572623824965628592128 00$*, and their entries are α-independent. The rank is therefore 26 at* every α *in these chambers.*

(b) On F $(\frac{1}{2},\frac{2}{3})$ *and* $(\frac{2}{3},1)$ *($N=38$), minor 2 equals* $-400569487999071945641108176896000000000\,\mathrm{i}$ *at all 28 sampled angles. Its only α-dependent entries lie in one column, so it is affine in that column and constant by (D). Hence the rank is ≥ 29 everywhere. Minors 0 and 1 vary, with 13 and 15 distinct values, and are never zero.*

(c) On F above $\frac{1}{2}$, $\dim Z(\alpha)=9$ *at 16 angles (`code/v10/cocycle9.py`). Every entry fits an affine function of $1/\alpha$ on 8 points per chamber. Write the 9 Cramer kernel vectors of minor 2 at $t=1/\alpha$. The residual $W(t)$ of the remaining rows has degree ≤ 6 by (D) and vanishes at 8 points, so it vanishes identically. Hence the generic rank is 29, and the rank equals 29 at every α: at most 29 by semicontinuity, at least 29 by (b).*

(d) The intersection $\bigcap_\alpha Z(\alpha)$ over the 8 angles of each chamber is 8-dimensional. Eight cocycles are constant vectors. The ninth exists at every angle, but its direction depends on α and is rational in α by Cramer's rule.

This replaces the v9 range "between 29 and 30" by exactly 29, conditional on (D). It also answers the v9 open item "exceptional set inside chambers": the set is empty, conditional on (D).

28 v10: the transverse-jet extension of the slice catalogue

Construction. `code/v10/jet_replay.py` executes the numerator section of the frozen v7/v8 replay with one change. The external transverse momenta p_1, p_2 are kept as symbols instead of being set to zero. The bubble denominator gains $-p_\perp^2$, and the loop momentum stays on the longitudinal slice ($l_\perp = 0$). This is a declared coverage limitation, and the status remains ALGEBRAIC_SLICE_ONLY. The UV pole is then an exact polynomial in $q_0, q_3, p_\pm$. It is stored per key and channel pair, by transverse degree and helicity weight (`logs/v10/jet/jet_<G>.json`). No frozen file is edited.

Checks. (i) At order $p_\perp^0$, $\sum_s \mathrm{pole}_{ss}^{(0)} z_s$ reproduces the v8 catalogue on every computed key: 684 key–branch checks with 0 mismatches. (ii) In D04 the UNIT sum is $-\kappa^2(q_0^2 - q_3^2 - p_\perp^2)/4\pi$, which is flat covariance restored by the jet terms.

Census. Summed over the 9 graphs, the jet entries are:

- same-channel $(2,0)$: 128;
- mixed $(2,0)$: 84;
- same $(2,\pm 2)$: 152 each;
- mixed $(2,\pm 2)$: 70 each;
- same $(1,\pm 1)$: 252 each;
- mixed $(1,\pm 1)$: 206 each.

D12 has none.

Definition 28.1 (Declared jet dressing and schemes)**.** A same-channel, weight-0 coefficient of p_+p_- on channel s is dressed by the r^2 moment d_s. In scheme A (the Abel subtraction of Theorem 5.2), $d_0 = \mathcal{D}_0$, $d_1 = \mathcal{D}_\alpha$, $d_2^{\mathrm{F}} = \mathcal{D}_{2\alpha-1}$ and $d_2^{\mathrm{CR}} = \mathcal{D}_{2\alpha}$. In scheme Z, each mode's $\mathrm{Tr}[r^{2w}e^{-sA_\mu}]$ is continued from $\Re w < -\frac{1}{2}$ to $w=1$, and the modes are summed by Hurwitz

zeta. This mirrors the role of $\mathcal{C}_b$ at constant profile. It is a reduction contract in the sense of Assumption 7.1, not a theorem.

Proposition 28.2 (Scheme Z; executed)**.** *Per mode,* $\mathrm{Tr}[r^2 e^{-sA_\mu}]^Z = s\,\mu(\mu^2-1)/3$*, from the Mellin closed form* $(2s)^w\Gamma(w+1+\mu)\Gamma(-\frac{1}{2}-w)/(2^{w+2}\sqrt{\pi}\,\Gamma(\mu-w))$*. This form matches quadrature for* $w<-\frac{1}{2}$ *to* 10^{-17} *or better. Its only poles are at half-integer* w*, so* $w=1$ *is regular. The mode sum gives*

$$\mathcal{D}_b^Z = \mathcal{D}_b + \frac{11\alpha}{180} \quad \textit{(exactly, all } b\textit{)},$$

so the scheme dependence is a bulk term linear in α*. Flipping one mode* $\nu\mapsto-\nu$ *reproduces* $\Delta\mathcal{D}=2\nu(1-\nu^2)/3$ *of Proposition 5.3.*

Proposition 28.3 (Dressed same-channel jets; executed)**.** *In scheme A, all dressed same-channel values vanish at* $\alpha=1$ *(F), and CR=F at* $\alpha=\frac{1}{2}$*, on every key (`logs/v10/jet/jet_analysis.json`). Examples: D05a* $(0,0)$ *gives* $\kappa^2(\alpha-1)(2\alpha-1)/(6\pi\alpha)$ *(CR) and* $\kappa^2(\alpha-1)(2\alpha-1)(\alpha^2-3\alpha+1)/(6\pi\alpha^3)$ *(F), each times* p_+p_-*. D11* $(1,1)$ *gives* $\kappa^2(\alpha^2-1)(11\alpha^2+1)/(2880\pi\alpha^3)$ *in scheme A and* $-\kappa^2(10\alpha^2+1)/(2880\pi\alpha^3)$ *in scheme Z. Of the 74 keys with weight-0* r^2 *content (CR), 38 are complete. The other 36 (all of D01, D06, and 9 of D03) also need mixed pairs and are* PARTIAL*.*

The D05a value is a slice-jet coefficient under the declared dressing. It is not compared with the earlier retained statement "D05a part = 0". That statement came from a different reduced model (a large-radius count without the slice), and no supersession is claimed.

The mixed r^2 trace. Term-by-term integration gives

$$T_1(x;\mu,\nu) = \tfrac{1}{4}(4x(1-x))^2\tfrac{\Gamma(h+2)\Gamma(h+1)}{\Gamma(\mu+1)\Gamma(\nu+1)}x^\mu(1-x)^\nu F_4(h+2,h+1;\mu+1,\nu+1;x^2,(1-x)^2)$$

with $h=(\mu+\nu)/2$. The double series has power divergences $N^{1/2-j}$ and no logarithms. Write $F_4(h+2,h+1)=(h+1+\theta)(h+\theta)F_4(h,h+1)/(h(h+1))$, with θ the Euler operator, and apply Bailey's reduction [2] off the boundary, $\lambda=1-t^2$. The finite part is then $h(h+1)\phi_0-2(h+1)\phi_2+2\phi_4$ in the Taylor coefficients of $\Phi(t)$ (`code/v10/t1_fp.py`). It agrees with an independent shell-by-shell zeta continuation to 10^{-19}–10^{-24} (`logs/v10/jet/t1_fp_check.json`). It reproduces $\mu(\mu^2-1)/3$ exactly at $\mu=\nu$ for every x, and its endpoint values are $T_1(0)=\mu(\mu^2-1)/3$ and $T_1(1)=\nu(\nu^2-1)/3$. After x-integration, the shell continuation is *not* reliable, because of non-uniform endpoint asymptotics (`logs/v10/jet/mixed_r2.json`). It is used nowhere.

29 v10: consequences of the jets

D04. The constant density for the jet term is $\rho_{(1,0)}=\kappa^2/4\pi$ (`code/v10/jet_density_D04.py`). Under the declared rule, the D04 part of the second $\phi\phi$ value is $\frac{\kappa^2}{4\pi}M_{10}$, where $M_{10}=\sum_{k\ge0}\varphi(k/\alpha)+\sum_{j\ge1}\varphi(j/\alpha-1)$ is k-ζ-regulated and $\varphi(n)=\int_0^1 T_1(x;n+1,n)\,\mathrm{d}x$. It is branch independent, since only channels 0 and 1 enter. We have φ at 27 points (for example $\varphi(0)=-0.057829713287632955855 5$ and $\varphi(3/2)=2.327852247334682150 21$; `logs/v10/jet/phi/`). No closed form was found in the basis $\{n^j, n^j\psi_1(n), n^j\psi(n)\}$ (`code/v10/phi_fit.py`). The large-n runs needed for the mode sum were stopped for time. M_{10} is therefore *not evaluated*, and the D04 part is PARTIAL. A further caveat: the bare r^2 insertion between channels 1 and 0 omits the frame intertwining H_I, which the spin-weight-one channel may require. The candidate is therefore a declared-rule quantity, not a derived value.

The 20 jet-level Ward combinations, the 34- and 18-dimensional projections, and e_5, e_6. These need catalogue values on the engine's $m = 2$ and weight ± 1 atoms. The jet catalogue now supplies the weight-0 same-channel part (Proposition 28.3). It does not supply the following:

(a) the mixed weight-0 pairs (84 entries), which need M_{st} mode sums as for D04;

(b) weight ± 1 (412 mixed, 504 same entries). These need $k \to k \pm 1$ traces with an $r\, e^{\pm i\theta}$ insertion, at $w = \frac{1}{2}$, where scheme Z has a pole: a genuine logarithm;

(c) weight ± 2, which needs $k \to k \pm 2$ traces;

(d) a dictionary from jet keys to engine $m = 2$ atoms. Such a dictionary was not constructed.

No residual was evaluated, and none is reported as zero. Status: OPEN, with the obstructions (a)–(d).

30 v10: vector-key finite parts in a declared scheme

Proposition 30.1 (Executed)**.** *Per mode, take scheme Z with* $\mathrm{Tr}_{\mathrm{mode}}\, e^{-A_\mu} = -\mu/2$*; this is consistent with* $\int g = 1/(2(\mu+\nu))$*. Then for a pair density* ρ *with* $R = \int_0^x \rho$*,* $I_\rho = -R(1)\nu/2 - (\mu^2 - \nu^2)\int_0^1 Rg$*. Pair the modes at the same k (channel 0:* $|k|/\alpha$*; channel 1:* $|k+\alpha|/\alpha$*), and regulate the sum by* $|k|^{-\varepsilon}$ *(`code/v10/vector_fp.py`, `vector_combine.py`). Then, for D01+D03 in the v7 normalization:*

(a) the $1/\varepsilon$ *residues equal the v9 logarithms* $(\alpha/2)L_{\mathrm{key}}$ *of Proposition 22.2 to* $\le 1.1 \times 10^{-20}$ *at all four angles. For keys* $(1,1)$ *and* $(2,2)$ *the pole is* $-\alpha\kappa^2 P_L^2/(60\pi\varepsilon)$*;*

(b) the off-diagonal keys $(1,6), (2,8), (6,1), (8,2)$ *have finite part 0, because D01 and D03 cancel digit for digit;*

(c) the diagonal keys have finite part $+c(\alpha)\kappa^2 P_L^2$ *on* $(1,1), (2,2)$ *and* $-c(\alpha)\kappa^2 P_L^2$ *on* $(6,6), (8,8)$*, with c given in Table 11. They are branch independent.*

Table 11: Vector-key finite parts (declared k-ζ scheme).

α	$c(\alpha)$ (k-ζ finite part)
3/10	0.0206627201453
41/100	0.0204800603712
3/5	0.0203890448352
7071/10000	0.020476362628

The finite part depends on the scheme. Rescaling the regulator $|k| \to \lambda|k|$ shifts c by $(\alpha/60\pi)\log\lambda$, and a sharp cutoff differs from it by the $\zeta(-1), \zeta(0)$ terms. It is a declared-scheme value and not an apex coefficient.

31 v10: changes to v9 statements

- Table 7 discussion: "PSLQ was not attempted; no elementary closed form" is superseded by Corollary 25.3 (CR derived, F identified).
- Theorem 20.2: "two fitted normalizations" is now "two normalizations derived from tree operators" (Theorem 26.1).
- Theorem 21.2: "F($> 1/2$) between 29 and 30" is now "exactly 29 at every angle", conditional on (D) (Theorem 27.1).

- Section 22.1: obstruction (a), "the slice does not export jet densities", is removed by the jet replay. Obstruction (b) is made precise: the D04 jet term is purely mixed, and scheme Z regularizes the per-mode power divergence. The mode sum is still missing.
- Proposition 22.2: the finite parts are now computed in a declared scheme.

No v9 number was found to be in error. The frozen-tree digests before and after v10 are in Appendix D.

32 What v11 adds, and what it does not

v11 keeps every v10 statement [5], except those amended in Section 38. The new items and their status are:

Table 12: v11 verdicts (all executed; hashes in Appendix E).

Item	Status
Mixed r^2 trace evaluator	*Corrected.* The v10 evaluator used a Cauchy radius that is too large near the endpoints and for large h. Every v10 φ value is withdrawn (Section 33).
F mode I_{F}	*Derived* for $0 < \mu < 2$ (Theorem 34.1). The v10 PSLQ form is now a theorem there. As a by-product, $\Delta J = -(\mu-1)B/W_1$ is a single incomplete-Beta integral.
Degree bound (D)	*Machine-checked* by symbolic degree tracking on all 8 chambers and cross-checked against the exact engine (Proposition 35.1). Theorem 27.1 is now unconditional in its scope.
M_{10} (D04 mode sum)	*Executed* in two declared pairings (P0 same-k; P1, the unit angular-momentum shift used as the frame intertwiner H_I) and two regulators, at four angles (Table 13). The precision is set by the data, about 4–8 digits, not 30. The pairing and the regulator are *declarations*, so D04 stays PARTIAL.
Weight $\pm1, \pm2$ and mixed jet traces	OPEN (Section 37).
Jet-key $\to m = 2$ atom dictionary; 20 Ward combinations; 34-/18-dim maps; e_5, e_6	OPEN. Not constructed and not evaluated; no residual is reported.
Transverse loop momentum	OPEN; obstruction stated in Section 37.
Registry	Unchanged (258 ids); no new sheet directory written.

33 v11: correction of the v10 mixed-trace evaluator

The v10 evaluator `code/v10/t1_fp.py` extracts the Taylor coefficients ϕ_0, ϕ_2, ϕ_4 of $\Phi(t)$ by a Cauchy integral on $|t| = 0.04$. Here $\lambda = 1 - t^2$, and $T_1 = \frac{1}{4}(4x(1-x))^2 \frac{\Gamma(h+2)\Gamma(h+1)}{\Gamma(\mu+1)\Gamma(\nu+1)} x^\mu (1-x)^\nu \frac{h(h+1)\phi_0 - 2(h+1)\phi_2 + 2\phi_4}{h(h+1)}$. v11 finds two defects.

(i) The discriminant of the Bailey change of variables has a second root at $|t_2| = 2\sqrt{x(1-x)}/|1-2x|$. For $x \to 0, 1$, $|t_2|$ falls below 0.04, and the extracted coefficients are then wrong.

(ii) Even inside the disc, the integrand grows roughly like $e^{ch|t|}$. The extraction loses about 1.3 digits per unit of h: for $\mu = \nu = 7$, the exact diagonal value $\mu(\mu^2-1)/3$ is missed by 2.6×10^{-10}.

(iii) Near the endpoints the natural variable is $t/|t_2|$, and the same growth sets in at the scale $|t_2|/h$. With $r = |t_2|/4$ alone, T_1 is aliased for $h \gtrsim 60$, and the error does not depend on the working precision. For example, an intermediate v11 run gave $F(61, 60)$ below both endpoint values.

The v11 evaluator `code/v11/t1v11.py` uses $r = \min(0.25, |t_2|/4, 1/(1+h), |t_2|/(1+h))$ with 56 nodes, and `code/v11/m10_eval.py` adds two measures. The half $x > \frac{1}{2}$ is evaluated through the symmetry $T_1(x;\mu,\nu) = T_1(1-x;\nu,\mu)$ (checked to 25 digits), and the working precision is raised by 1.5 digits per decade near the endpoint. Tanh–sinh is capped at degree 4, since degrees 3, 4 and 5 agree to 3×10^{-23} relative. On the diagonal the evaluator reproduces $\mu(\mu^2-1)/3$ to $\le 10^{-24}$ at $\mu = 20, 60$ and $x = 0.05, 0.3$. Near the endpoints, halving the radius and raising the node count to 80 change T_1 by less than 10^{-14} relative at $h = 60.5$ and $h \approx 131$. Three intermediate v11 runs with incomplete fixes are superseded and feed no printed number (`logs/v11/M10/cache_superseded*/`). As a final sanity check, every cached $F(\mu,\nu)$ lies between its two endpoint values: true for 403 of 404 entries; the exception $F(1,0)$ (`logs/v11/M10/m10_sanity.json`) has both endpoint values zero, so its negative value is no contradiction.

Consequences: the corrected value is $\varphi(3/2) = 2.32745055013616982735$, while v10 quoted 2.32785224733468215021, an error of 4.0×10^{-4}. The corrected value is $\varphi(0) = F(1,0) = -0.057829713287632955855$; these agree to 1.2×10^{-17}, since the defect is mild at small h. Because the evaluator was defective, all 27 v10 φ values are withdrawn as a set, together with the v10 negative result "no closed form in $\{n^j, n^j\psi_1(n), n^j\psi(n)\}$" are withdrawn. The v10 mid-range checks ($x \in [0.3, 0.8]$, `logs/v10/jet/t1_fp_check.json`) and the diagonal identity are unaffected, and so are the same-channel jets, scheme Z and the vector finite parts, none of which use T_1 off the diagonal.

34 v11: derivation of the F mode

Theorem 34.1 (F mode; `code/v11/fmode_derive.py`). *Let $0 < \mu < 2$, $\nu = 2-\mu$, $w(x) = x^{\mu-1}(1-x)^{1-\mu}$, $W(x) = \int_0^x w$ and $W_1 = W(1) = \Gamma(\mu)\Gamma(2-\mu)$. Then:*

(a) $g(x;\mu,2-\mu) = \dfrac{W(x)\,(W_1 - W(x))}{4W_1\,x(1-x)\,w(x)}$. *Both ${}_2F_1$ factors of Proposition 19.1 are incomplete Beta functions [6], by ${}_2F_1(a, a-1; a+1; x) = a\,x^{-a}B_x(a, 2-a)$.*

(b) $I_{\mathrm{F}} = I_{\mathrm{CR}} + (\mu-1)B/W_1$ *and* $\Delta J = -(\mu-1)B/W_1$, *where* $B = \int_0^1 (3-2x)W^2/w\,\mathrm{d}x$. *The linear term* $\int(3-2x)W/w = \mu^{-1}\sum_n (n+1)!K_n/(\mu+1)_n$ *carries the same K_n as the CR mode, and so equals* $-I_{\mathrm{CR}}/(\mu-1)$.

(c) $(x(1-x)/w)' = (2-\mu-2x)/w$ *gives* $B = -2W_1E + (1+\mu)C$, *with* $E = \frac{1}{6} - \frac{\mu(\mu+1)}{12} + \frac{\mu(\mu+1)(\mu+2)}{72}$ *and* $C = \int_0^1 W^2/w$.

(d) The series for W/w, together with $\int_0^1 W x^m = W_1(1 - (\mu)_{m+1}/(m+2)!)/(m+1)$, reduces C/W_1 to the v10 Gamma sums G_2, G_3 plus two rational sums. The result is

$$\frac{C}{W_1} = \frac{(\mu-2)(5\mu-3)}{3} + (\mu-1)^2\Big[\psi_1(\mu-1) - \frac{\pi^2}{6}\Big].$$

(e) Hence $I_{\mathrm{F}}(\mu) = \frac{(\mu-1)(59\mu^3-57\mu^2-62\mu+48)}{36} - (\mu-1)^3(\mu+1)\frac{\pi^2}{6}$, *and the difference from the v10 PSLQ form simplifies to 0 in SymPy.*

Checks: C/W_1 agrees with direct quadrature to $\le 1.2 \times 10^{-30}$ at $\mu = 7/4, 4/3, 10/7$ (`logs/v11/fmode_derive.json`). The reduction (a)–(b) agrees with the v10 closed form to $\le 10^{-31}$ at $\mu = 7/4, 5/4, 3/2$ (`logs/v11/fmode_reduce.json`). The Euler constant and $\psi(\mu-1)$ cancel identically. Scope: $0 < \mu < 2$, that is $\alpha > \frac{1}{2}$, where $B_x(\mu, 2-\mu)$ converges. This is the range in which the F mode enters ΔJ.

35 v11: machine check of the degree bound (D)

Proposition 35.1 (`code/v11/degree_check.py`, `code/v11/degree_crosscheck.py`)**.** *In the frozen v8 engine, α enters the ST operators only through the derivative coefficients $(m \pm a)/2$, with $a = nt + s$, $t = 1/\alpha$ and $m = \varepsilon|a| + 2j$. The row basis is a pivot* selection *of candidates, with no α-dependent combination, and the candidate evaluation is multilinear in the operator outputs and polynomial in q. Replace t by a SymPy symbol and apply K_2, $E(-q)$ and C_2T to every unit atom up to* DM $= 6$ *on each of the 8 chambers (CR and F, c_1–c_4). The maximal degrees in t are then 1, 2 and 1, so (D) holds. The atom counts are c: 22–52, h: 54–130, b: 16–36 per chamber (`logs/v11/degree_check.json`). At $t = 1/\alpha_0$ the symbolic outputs equal the exact numeric engine on 3744 coefficients (CR c_4, F c_3, CR c_1), with 0 mismatches (`logs/v11/degree_crosscheck.json`).*

The hypothesis of Theorem 27.1 is therefore discharged. The chamber ranks (26 on the six $N = 34$ chambers, 29 on F $(\frac{1}{2}, \frac{2}{3})$ and $(\frac{2}{3}, 1)$, at every α in the chamber) now hold without condition, in the scope of the declared ST rows of the frozen engine.

Certificate status. (i) The v8 four-point exact rank certificate is unaffected by v11 and remains CLOSED_PENDING_INDEPENDENT_REVIEW in its v8 scope. (ii) The chamber-rank certificate is CLOSED_PENDING_INDEPENDENT_REVIEW in the following precise scope: the rank of the declared ST row matrix of the frozen v8 engine (DM $= 6$, 6×6 q-grid) is 26 at every α in the six $N = 34$ chambers, and 29 at every α in the two F chambers above $\frac{1}{2}$. Every ingredient is executed: exact minors at the sampled angles, the constancy and affinity arguments of Theorem 27.1, and the machine-checked (D). The certificate does not cover the identification with the interacting basis or the walls between chambers.

36 v11: the D04 mode sum M_{10}

Definitions. Take $F(\mu, \nu) = \int_0^1 T_1(x; \mu, \nu)\, \mathrm{d}x$ with the corrected evaluator (tanh–sinh, 30 digits; endpoint limits for $x < 10^{-22}$). Two pairings of channel 1 with channel 0 are declared:

- P0 (the v10 rule, same k): $(\mu, \nu) = (\frac{k+\alpha}{\alpha}, \frac{k}{\alpha})$ for $k \geq 0$, and $(\frac{j-\alpha}{\alpha}, \frac{j}{\alpha})$ for $j \geq 1$.
- P1 (the unit angular-momentum shift; v11's implementation of the frame intertwiner H_I, as a *declared* pairing): $(\frac{k+\alpha}{\alpha}, \frac{k+1}{\alpha})$ for $k \geq 0$, and $(\frac{|\alpha-k|}{\alpha}, \frac{|1-k|}{\alpha})$ for $k \geq 1$.

Both involve only channels 0 and 1, so both are branch independent; at $\alpha = \frac{1}{2}$, CR=F holds trivially. Two regulators are used: R_k, weight $|k|^{-\epsilon}$ (the v10 declaration), and R_h, weight $h^{-\epsilon}$ with $h = (\mu + \nu)/2$. The terms $k < 20$ are summed exactly. For $k \geq 20$ a QR least-squares fit $F \sim \sum_{i<12} c_i u^{3-i}$ on $k = 20, 22, \ldots, 56$ is summed through Hurwitz $\zeta(i - 3, u_{20})$. The u^{-1} term gives a pole c_4/ϵ (reported) and the finite part $-c_4\psi(u_{20})$, minus $c_4 \log(1/\alpha)$ for R_h. The error estimate is the spread over fit orders 10–13 and over fits on half the points. A power expansion without logarithms is an *ansatz* for the large-k asymptotics. It is supported by the fit residuals, not derived (`code/v11/m10_sum.py`, `logs/v11/M10/m10.json`).

Check at $\alpha = 1$. P1 pairs identical modes, and $F(\mu, \mu) = \mu(\mu^2 - 1)/3$ exactly. The machinery returns $-19/180$ under R_k (to 2.8×10^{-26}) and $11/180$ under R_h (to 2.2×10^{-26}). The latter is the scheme-Z flat value $\mathcal{D}_0^Z(1) = 11/180$ of Proposition 28.2. The difference $1/6$ is a pure regulator effect: the shift between the index k and the mode label changes the zeta values. It shows that M_{10} is *not* regulator independent, and that only R_h is consistent with scheme Z on the diagonal.

Table 13: M_{10} finite parts ($\kappa^2/4\pi$ times these is the D04 term under the stated declaration). Error = spread estimate; pole = coefficient of $1/\epsilon$.

α	pairing, reg.	M_{10} (FP)	error	pole
3/10	P0, R_h	0.1157438826	4.4×10^{-9}	-0.0075
3/10	P0, R_k	0.0973390865	9.0×10^{-10}	-0.0075
3/10	P1, R_h	0.0169054255	1.2×10^{-8}	0.0196605
3/10	P1, R_k	1.9509542118	4.4×10^{-7}	0.0196605
1/2	P0, R_h	-0.023918875	5.0×10^{-8}	-0.0125
1/2	P0, R_k	-0.048208215	1.2×10^{-8}	-0.0125
1/2	P1, R_h	-0.023918875	5.0×10^{-8}	-0.0125
1/2	P1, R_k	0.2017917849	2.2×10^{-7}	-0.0125
7071/10000	P0, R_h	-0.043222064	2.6×10^{-7}	-0.0176775
7071/10000	P0, R_k	-0.071445663	6.5×10^{-8}	-0.0176775
7071/10000	P1, R_h	0.0208317775	5.7×10^{-8}	-0.00387076
7071/10000	P1, R_k	-0.044268878	1.7×10^{-8}	-0.00387076
1	P0, R_h	-0.047837750	1.3×10^{-6}	-0.025
1	P0, R_k	-0.079087750	3.5×10^{-7}	-0.025
1	P1, R_h	0.0611111111	2.2×10^{-26}	$5.36872e-27$
1	P1, R_k	-0.105555555	2.8×10^{-26}	$2.19251e-27$

Reading of Table 13. The leading fitted coefficient agrees with the prediction $c_0 = \alpha^{-3}/3$ (for $u = k$ and for $u = \alpha h$) from $F \sim h(h^2-1)/3$ to relative 1.7×10^{-15} in every family (`logs/v11/M10/m10_leading.json`). This is the only analytic input to the asymptotics; the subleading coefficients are fitted, not derived. The intertwiner choice is not a small effect: $M_{10}^{\mathrm{P1}} - M_{10}^{\mathrm{P0}}$ under R_h is 3/10: -0.098838457; 7071/10000: 0.064053842. At $\alpha = \frac{1}{2}$ the P1 pairs $(2k+1, 2k+2)$ are, after reindexing, the P0 pairs of the negative family and vice versa, so the two pairings must agree under the label-based regulator R_h (not under R_k). They agree to 5.1×10^{-12}, a non-trivial check of the head/tail machinery on non-exact data. The P0 pole coefficient equals $-\alpha/40$ at all four angles and for both regulators, to 9.7×10^{-12} (identified numerically, not derived). The pole coefficients are nonzero in general; a declared minimal subtraction of the $1/\epsilon$ term is part of the finite-part definition.

Status of D04. Under a declared pairing and regulator, the D04 part of the second $\phi\phi$ value is $\frac{\kappa^2}{4\pi} M_{10}$, with M_{10} as in Table 13. Neither the pairing nor the regulator is derived. The difference between P0 and P1, and between R_k and R_h, measures the size of the undeclared choice. Status: PARTIAL (executed, scheme-dependent, precision about 4–8 digits). PSLQ in the basis $\{1, \alpha^{\pm 1}, \alpha^2, \alpha^{-2}, \alpha^{-3}, \zeta(3)/\alpha^2, \pi^2, \log\alpha\}$ with coefficients up to 1000: relations among the basis elements alone (coefficient of M_{10} zero; e.g. $1/\alpha^2 = 2/\alpha$ at $\alpha = \frac{1}{2}$) for 7071/10000 P0 Rk, 7071/10000 P0 Rh, 1/2 P0 Rk, 1/2 P0 Rh, 1/2 P1 Rk, 1/2 P1 Rh; these are basis degeneracies; relations involving M_{10} for 3/10 P0 Rk, 3/10 P0 Rh, 3/10 P1 Rk, 3/10 P1 Rh, 7071/10000 P1 Rk, 7071/10000 P1 Rh. With ten basis elements and a tolerance of about 10^{-8}, small-coefficient relations exist generically, so these are *not* credible and are not claimed. No closed form is claimed. (`logs/v11/M10/m10_pslq.json`).

37 v11: items left open, with precise obstructions

(a) *Weight* ± 1 *traces.* The $k \to k \pm 1$ traces with an $r\, e^{\pm \mathrm{i}\theta}$ insertion were not built in v11. The task is to split the scheme-Z pole at $w = \frac{1}{2}$ into its residue (the genuine, scheme-independent log coefficient) and the finite part (scheme dependent). Doing so needs the two-channel

finite-part evaluator at weight one: the analogue of T_1 with ${}_2F_1$ parameters shifted by $\frac{1}{2}$, and its own mode sum. That evaluator was not written. Heuristically, in the same-channel analogue the log arises from the z^{-1} term of $e^{-z}I_\mu(z) \sim (2\pi z)^{-1/2}[1-(4\mu^2-1)/(8z)+\dots]$, but this is not a computation of the mixed coefficient, and no number is claimed. Status: OPEN.

(b) *Weight* ± 2 *and the 84 mixed weight-0 entries.* The corrected evaluator applies to the weight-0 mixed entries. Each needs a mode sum like M_{10}, with a declared pairing, and none was run. Status: OPEN.

(c) *Dictionary, Ward combinations, projections,* e_5, e_6. No map from jet keys $(s,t;w)$ to the engine's $m=2$ atoms (C,n,j,i) was constructed. Without it, the 20 jet-level Ward combinations cannot be formed, and neither can the remaining 12 coordinates of the 18-dim map, the 34-dim projection, or e_5, e_6 separately. No residual was evaluated, and none is reported as zero. Status: OPEN.

(d) *Transverse loop momentum.* The jet catalogue evaluates the loop at transverse momentum on the slice. Going off the slice needs the $p_\perp$ dependence of every vertex in the catalogue, plus a $p_\perp$ integral in a declared scheme. The v8 atoms are polynomial in the slice momenta q only, and the catalogue exports no $p_\perp$ jets. That export is the missing piece. Status: OPEN.

38 Changes to v10 statements

(C1) Section 29, D04: the quoted values $\varphi(0) = -0.0578297132876329558555$ and $\varphi(3/2) = 2.32785224733468215021$, the 27 cached values in `logs/v10/jet/phi/`, and the negative closed-form search are *withdrawn*, because the evaluator was defective (Section 33). Corrected values are in Section 33. "M_{10} not evaluated" is superseded by Section 36.

(C2) Section 28: the claim that the closed finite-part evaluator for the mixed trace is valid for all $x \in (0,1)$ is restricted. With the v10 radius it is valid only where $0.04 < |t_2|$ and h is small. The v11 radius removes the restriction.

(C3) Proposition 25.2: "identified by PSLQ, not derived" becomes *derived* for $0 < \mu < 2$ (Theorem 34.1).

(C4) Theorem 27.1: "Assume (D) ... not separately machine-checked" is replaced by Proposition 35.1, so the theorem holds without condition. The v10 release items (xvii) and (xiv) lose "conditional on (D)".

(C5) The v10 artifact manifest hashed `CHECKPOINT_v10.md` one append early; the corrected hash is in Appendix E.

(C6) Open table: the rows "ΔJ", "Exact Span certificate" and "D04" are updated (Table 17).

No other v10 statement is changed.

39 What v12 adds, and what it does not

v12 keeps every v11 statement [4] except those amended in Section 45. The new items and their status are:

Table 14: v12 verdicts (all executed; hashes in Appendix F).

Item	Status

Closed form of the mixed r^2 mode functional $F(\mu,\nu)$	*Identified* and verified against all 414 quadrature values to 4.8×10^{-28} relative (Proposition 40.1). Not derived.
D04 pairing and frame intertwiner	*Derived* from the channel labelling and the constant density: $e^{\pm i\theta}$ shifts k by $\pm\alpha$, which is the channel offset itself. The pairing is the same-k rule P0, and P1 is excluded (Proposition 41.1).
Large-k asymptotics and the P0 pole	*Exact* given the closed form: only odd powers of $1/h$ appear, with $a_4(d)=d^2(d^2-4)/240$, so the pole is $-\alpha/40$ (Corollary 40.2).
Regulator	R_k, R_h, R_μ, R_ν differ by an explicit local counterterm (Proposition 42.1, verified to 10^{-39}). The XVI-P prescription (DRED, $\overline{\text{MS}}$, normalization N4) induces no cone mode regulator, so the choice is a normalization. D04 is unique once it is declared; v12 declares R_h.
M_{10}	40 digits at four angles (Table 15). Closed forms at $\alpha=1,\frac12$ found by PSLQ; none at $\frac{3}{10}$ or $\frac{7071}{10000}$.
Chamber walls $\alpha=\frac13,\frac12,\frac23$	Exact $\mathbb{Q}(i)$ certificates at the walls (Table 16).
Weight $\pm1,\pm2$ traces; 84 mixed weight-0 sums; dictionary; 20 Ward combinations; 34-/18-dim maps; e_5,e_6	OPEN (Section 44). No residual is reported.
Transverse loop momentum	OPEN; feasibility statement in Section 44.
Registry	Unchanged (258 ids, re-fetched 04:40 UTC); no new sheet directory.

40 v12: closed form of the mixed r^2 mode functional

Proposition 40.1 (Identified closed form; `code/v12/F_closed.py`)**.** *Let $h=(\mu+\nu)/2$ and $d=\mu-\nu$. Then, to the precision of every evaluation made,*

$$F(\mu,\nu)=\int_0^1 T_1(x;\mu,\nu)\,\mathrm{d}x=\frac{h^3}{3}+\Big(\frac{d^2}{6}-\frac13\Big)h+d^2(d^2-4)\,\Psi(h),$$

$$\Psi(h)=\frac{h^3}{48}+\frac{h^2}{96}-\frac{5h}{288}-\frac{1}{96}-\frac{h^2(h^2-1)}{48}\,\psi_1(h).$$

How it was found.

(a) Fits to the v11 cache at fixed d give $F\sim\sum_i a_{2i}(d)h^{3-2i}$: the odd coefficients are $\le10^{-17}$ (`code/v12/phi_asym_fit.py`).

(b) At $h=\frac32,\frac72,5$ and $d=\frac12,1,3$, the ratio $\big(F-\frac{h^3}{3}-(\frac{d^2}{6}-\frac13)h\big)/(d^2(d^2-4))$ is independent of d to 20 digits, and at $d=2$ the remainder vanishes to 10^{-29} (`code/v12/structure_test.py`).

(c) PSLQ puts Ψ in $\mathbb{Q}+\mathbb{Q}\pi^2$ at those h. For example, $\Psi(5)=(987-100\pi^2)/48$ and $\Psi(\frac32)=(448-45\pi^2)/1536$.

(d) Matching the asymptotic coefficients $1/240$ and $-1/840$ fixes the coefficient of ψ_1 as $-h^2(h^2-1)/48$. This predicts the next two coefficients, $1/840$ and $-1/440$; the fits confirm both.

Verification. The formula agrees with all 414 cached tanh–sinh values (v11 evaluator; h up to 190, d in $\{\frac12,0.414,1,2,\frac73,3,\dots\}$). The maximum differences are 4.5×10^{-22} absolute and 4.8×10^{-28} relative (`logs/v12/F_closed.json`). *Status.* The formula is identified and verified, not derived. A derivation would follow from a closed evaluation of the Appell F_4 finite part; v12 did not attempt one.

Corollary 40.2 (Asymptotics and the P0 pole)**.** *Given Proposition 40.1, the expansion $\psi_1(h)\sim h^{-1}+\frac12h^{-2}+\sum_{n\ge1}B_{2n}h^{-2n-1}$ gives $F\sim\frac{h^3}{3}+(\frac{d^2}{6}-\frac13)h+d^2(d^2-4)(\frac{1}{240h}-\frac{1}{840h^3}+\frac{1}{840h^5}-\frac{1}{440h^7}+\dots)$.*

The expansion has no logarithms and no even powers. In a family $h_k = k/\alpha + \sigma$, the coefficient of k^{-1} is $\alpha a_4(d)$: the binomial re-expansion of h^{3-j}, $j \le 3$, produces no k^{-1} term. Summed over the two families of a pairing, the pole is $2\alpha a_4(d) = \alpha d^2(d^2-4)/120$. For P0 ($d = 1$) this is $-\alpha/40$, the value seen numerically in v11. For P1 ($d = 1/\alpha - 1$) it reproduces the v11 numbers, for example $637/32400$ at $\alpha = \frac{3}{10}$.

The proof of the pole value is complete given Proposition 40.1, and only to that extent.

41 v12: the D04 pairing and frame intertwiner

Proposition 41.1 (Pairing)**.** *Assume (i) the v8 channel labelling: the spin-weight-s component with frame label k has Bessel order $|k + s\alpha|/\alpha$; and (ii) the v10 D04 density $\rho_{(1,0)} = \kappa^2/4\pi$, which is independent of θ in the helicity frame. Then:*

(a) On C_α, angular modes are $e^{\mathrm{i}k\theta/\alpha}$, $\theta \in [0, 2\pi\alpha)$. The frame rotation $e^{\pm\mathrm{i}\theta}$ multiplies a mode by $e^{\pm\mathrm{i}\alpha\theta/\alpha}$, so it shifts k/α by ± 1, that is, k by $\pm\alpha$. This is the channel offset in (i). The rotation is therefore already absorbed in the channel orders, and the intertwiner H_I acts as the identity on k.

(b) The helicity frame is the only frame defined globally on C_α, because of the holonomy $e^{2\pi\mathrm{i}s\alpha}$. A θ-independent insertion in it commutes with $-\mathrm{i}\partial_\theta$, so the $(1,0)$ trace is diagonal in k. This is pairing P0: $(|k+\alpha|/\alpha, |k|/\alpha)$ for all $k \in \mathbb{Z}$.

(c) The v11 pairing P1 ($k \to k+1$) would need the insertion $e^{\pm\mathrm{i}\theta/\alpha}$, a scalar angular harmonic that does not occur in $\rho_{(1,0)}$. P1 is excluded.

Scope: this derivation uses (i) and (ii) as stated. It does not re-derive the frozen vertex tensors.

42 v12: regulators, counterterm, and M_{10} to 40 digits

Proposition 42.1 (Local counterterm; `code/v12/counterterm.py`)**.** *Let $f(k) \sim \sum_i c_i k^{3-i}$ be a power expansion with no logarithms (by Corollary 40.2), and let $w_k = s(k+t)$. Then*

$$\mathrm{FP}\sum_k f(k)k^{-\epsilon} - \mathrm{FP}\sum_k f(k)w_k^{-\epsilon} = c_4 \log s + \sum_{m=1}^{4} \frac{(-1)^{m+1}t^m}{m}\, c_{4-m}.$$

Proof. $k^{-\epsilon} - w_k^{-\epsilon} = \epsilon k^{-\epsilon}[\log s + \log(1 + t/k)] + O(\epsilon^2)$. Both sums have only simple poles at $\epsilon = 0$, so the difference of finite parts is the residue of $\sum_k f(k)k^{-\epsilon}[\log s + \log(1 + t/k)]$, which is the coefficient of k^{-1}. □

For each pair of regulators, the formula matches the 40-digit differences to $\le 1.3 \times 10^{-39}$. In particular, $M_{10}^{R_h} - M_{10}^{R_\mu} = M_{10}^{R_h} - M_{10}^{R_\nu} = \alpha/32$ for P0. *Which regulator?* XVI-P's declared prescription is DRED with $\overline{\mathrm{MS}}$ and the normalization condition N4, under which the local ambiguity is fixed by normalization (`XVI-P/code/p5_entry/prescription_d11.py`, read only). It defines no mode regulator on the cone. The regulator choice is therefore exactly the choice of the local term in Proposition 42.1. v12 *declares* R_h: it is symmetric in $\mu \leftrightarrow \nu$, and it reproduces scheme Z on the diagonal (v11: 11/180 at $\alpha = 1$). Every other value follows from the counterterm.

M_{10}. The terms $k < K$ are evaluated exactly from Proposition 40.1. The tail $k \ge K$ uses the exact asymptotic series in the regulator variable, summed through Hurwitz ζ. Runs with $(K, J) = (300, 36)$ and $(600, 48)$ agree to $\le 2 \times 10^{-34}$ (`code/v12/m10_exact.py`, `logs/v12/m10_exact_dps40.json`). All values lie inside the v11 error bars.

Table 15: M_{10} (finite part, 32 digits shown) and the $1/\epsilon$ pole coefficient. P0 is the derived pairing (Proposition 41.1); P1 is shown only for comparison with v11.

α	pairing	reg.	M_{10}	pole
3/10	P0	R_h	0.11574388259581850963848639109243	-0.0075
3/10	P0	R_k	0.09733908656337398969381579445921	-0.0075
3/10	P0	R_μ	0.10636888259581850963848639109243	-0.0075
3/10	P0	R_ν	0.10636888259581850963848639109243	-0.0075
3/10	P1	R_h	0.01690542553874612785881404187544	0.019660493827…
3/10	P1	R_k	1.95095421184601838796068716967526	0.019660493827…
1/2	P0	R_h	$-0.02391887532526779349301220416248$	-0.0125
1/2	P0	R_k	$-0.04820821508226710986072760568071$	-0.0125
1/2	P0	R_μ	$-0.03954387532526779349301220416248$	-0.0125
1/2	P0	R_ν	$-0.03954387532526779349301220416248$	-0.0125
1/2	P1	R_h	$-0.02391887532526779349301220416248$	-0.0125
1/2	P1	R_k	0.20179178491773289013927239431928	-0.0125
7071/10000	P0	R_h	$-0.04322206410166229544037429665526$	-0.0176775
7071/10000	P0	R_k	$-0.07144566327268729892016394092357$	-0.0176775
7071/10000	P0	R_μ	$-0.06531893910166229544037429665526$	-0.0176775
7071/10000	P0	R_ν	$-0.06531893910166229544037429665526$	-0.0176775
7071/10000	P1	R_h	0.02083177759556464749795711927522	$-0.00387075577\ldots$
7071/10000	P1	R_k	$-0.04426887899089171952349912499067$	$-0.00387075577\ldots$
1	P0	R_h	$-0.04783775065053558698602440832496$	-0.025
1	P0	R_k	$-0.07908775065053558698602440832496$	-0.025
1	P0	R_μ	$-0.07908775065053558698602440832496$	-0.025
1	P0	R_ν	$-0.07908775065053558698602440832496$	-0.025
1	P1	R_h	0.06111111111111111111111111111111	0.0
1	P1	R_k	$-0.10555555555555555555555555555555$	0.0

D04. With the derived pairing P0 and the declared regulator R_h, the D04 part of the second $\phi\phi$ value is $\frac{\kappa^2}{4\pi}M_{10}^{\mathrm{P0,R_h}}$, with a pole $-\frac{\kappa^2}{4\pi}\frac{\alpha}{40\epsilon}$. At the certified angles, $M_{10}^{\mathrm{P0,R_h}}(\frac{3}{10}) = 0.11574388259581850963848639109243$ and $M_{10}^{\mathrm{P0,R_h}}(\frac{7071}{10000}) = -0.04322206410166229544037429665527$. Status: PARTIAL, for two reasons: the closed form is identified rather than derived, and the regulator is a declared normalization. Beyond that, the value is unique.

Closed forms (PSLQ, 40 digits, coefficients ≤ 1600; `code/v12/m10_pslq.py`).

$$M_{10}^{R_h}(1) = \frac{1-40\log 2-20\gamma}{800}, \qquad M_{10}^{R_k}(1) = -\frac{6+10\log 2+5\gamma}{200},$$
$$M_{10}^{R_h}(\tfrac{1}{2}) = \frac{1-40\log 2-20\gamma}{1600}, \qquad M_{10}^{R_k}(\tfrac{1}{2}) = -\frac{6+15\log 2+5\gamma}{400}.$$

$R_\mu = R_\nu = R_k$ at $\alpha = 1$, as Proposition 42.1 requires. At $\alpha = \frac{3}{10}$ and $\frac{7071}{10000}$, no relation was found in the bases $\{1, \zeta(3), \zeta(3)/\pi^2, \pi^2, \log 2\}$, $\{1, \zeta(3)/\pi^2, \log 2, \zeta'(-1), \zeta'(-3), \gamma\}$ or $\{1, \zeta(3)/\pi^2, \pi^2, \log 2, \log \alpha\}$. The large-coefficient outputs are spurious. For rational $\alpha = p/q$, the sum $\sum_k h_k^n \psi_1(h_k)$ is a Barnes-type double zeta value that reduces to Hurwitz data at arguments $j/(2p)$. That reduction was not carried out.

43 v12: exact certificates at the chamber walls

`code/v12/wall_certificate.py` is the v8 certificate program with only the output path changed. It was run at $\alpha = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}$ on CR and F, in exact $\mathbb{Q}(\mathrm{i})$ arithmetic, with DM = 6 and the 6×6 q-grid (`logs/v12/walls/`).

Table 16: Wall certificates: N = constrained basis size, rank = Ward-constrained count $N - \dim Z$, image-column rank, $\dim H^0$, seconds.

α	branch	N	rank	$\dim Z$	img. rank	$\dim H^0$	s
1/2	CR	42	33	9	40	2	687
1/2	F	42	33	9	40	2	678
1/3	CR	35	27	8	33	2	305
1/3	F	35	27	8	33	2	305
2/3	CR	35	27	8	33	2	304
2/3	F	39	30	9	37	3	545

Reading. At each wall the constrained basis is larger than in the adjacent chambers, because extra jets appear when Bessel orders coincide. The rank rises with it: 27 at $\frac{1}{3}$ (CR, F) and at $\frac{2}{3}$ (CR), against 26 in the chambers; 30 at $\frac{2}{3}$ (F), against 29; and 33 at $\frac{1}{2}$ on both branches. CR and F coincide exactly at $\alpha = \frac{1}{2}$, as they must. The cocycle dimension $\dim Z = N - \text{rank}$ is 8 at $\frac{1}{3}$ and at $\frac{2}{3}$ (CR), the $N = 34$ chamber value. It is 9 at $\frac{2}{3}$ (F), the value for F above $\frac{1}{2}$, and 9 at $\frac{1}{2}$ on both branches. At every wall, $B \subset Z$ and B lies in the candidate span. The exceptional set in $(\frac{1}{4}, 1)$ is therefore exactly the three walls, and the ranks there are certified.

Certificate status. The wall ranks are CLOSED_PENDING_INDEPENDENT_REVIEW in this precise scope: the exact rank of the declared ST row matrix of the frozen v8 engine (DM = 6, 6×6 q-grid, exact $\mathbb{Q}(\mathrm{i})$) at $\alpha = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}$ on CR and F, as tabulated. The method is the v8 complete-evaluation argument, unchanged.

44 v12: open items and precise obstructions

(a) *Weight $\pm 1, \pm 2$ traces and the 84 mixed weight-0 sums.* v12 wrote no weight-one evaluator. Proposition 40.1 supplies, for every channel pair (s, t) under the derived same-k pairing, the *constant-density* part of a mixed weight-0 entry in closed form. A low-cost path for the 84 entries therefore exists for those with constant densities. Entries whose densities depend on x need the moments $\int x^n T_1 \, \mathrm{d}x$, which are not covered by the closed form. Neither the per-entry densities nor the CR flip of channel 2 were tabulated, so no entry is evaluated. The split of the weight-± 1 traces into a scheme-independent log coefficient and a finite part is not computed. Status: OPEN.

(b) *Dictionary, 20 Ward combinations, 34-/18-dim maps, e_5, e_6.* These depend on (a) and on the jet-key to (C, n, j, i) dictionary, which was not constructed. No residual was evaluated. Status: OPEN.

(c) *Transverse loop momentum (feasibility).* A minimal construction needs three pieces. First, each catalogue vertex must be exported as a polynomial in $p_\perp$ up to the order needed: degree 4 for the hh sector, by the N4 power counting. Second, the $p_\perp$ integrals in DRED ($d - 2$ transverse dimensions) reduce these polynomials to Γ-function multiples of $s^{-(d-2)/2}$ times powers of the heat time. Third, the extra powers of s shift the Mellin variable of scheme Z, so the per-mode functionals change from $F(\mu, \nu)$ to its Mellin-shifted analogues. Nothing in this chain is beyond the present tools, but the first step, a $p_\perp$ export from the v8 slice engine, does not exist. Status: OPEN (feasible, not started).

45 Changes to v11 statements

(C1) Section 36: "a power expansion without logarithms is an ansatz" is now a consequence of Proposition 40.1, which is identified, not derived. The fitted tail is replaced by the exact tail, and the quoted precision of "4–8 digits" by 40 digits (Table 15). The v11 values lie inside their error bars.

(C2) Section 36: P1 was "v11's implementation of the frame intertwiner H_I". This is withdrawn: H_I acts as the identity on k, and P1 is excluded (Proposition 41.1).

(C3) Section 36: "the P0 pole coefficient equals $-\alpha/40$ (identified numerically, not derived)" becomes Corollary 40.2, which is exact given the closed form.

(C4) Section 36: "M_{10} is not regulator independent" is sharpened. The regulators differ by the explicit local counterterm of Proposition 42.1.

(C5) Section 36: "PSLQ: no closed form" is superseded at $\alpha = 1, \frac{1}{2}$ (Section 42).

(C6) The certificate scope of Section 35, which excluded the walls, is supplemented by Section 43.

No other v11 statement is changed.

46 Open obligations and the exact release

Table 17: Open obligations after v12. Each status is stated exactly; a missing value is never zero.

Obligation	v12 status (evidence)	What is still missing
v6 hh branch difference	Restored (v9, Proposition 19.2).	Still conditional on Assumption 7.1.
ΔJ (TT branch difference)	**Closed form, both modes derived** (CR v10; F v11 for $0 < \mu < 2$, Theorem 34.1).	A declared scheme for absolute TT finite parts.
Ghost normalizations	**Derived (v10)** from tree operators (Theorem 26.1).	Independence from the v9 dictionary sign choice.
26 Ward combinations	PARTIAL. The 6 constant-profile combinations hold (v9), now without fitted parameters.	The 20 jet-level combinations: obstructions (a)–(c) of Section 37. No residual is evaluated.
34-dimensional projection; 18-dim engine map	PARTIAL (12 of 34; 6 of 18). Same-channel weight-0 jet values are now available (Proposition 28.3).	Mixed and weight-$\pm1, \pm2$ jet values; a jet-key to $m = 2$ atom dictionary.
e_5, e_6 separately	OPEN (excluded at constant profile, v9).	Jet data along the kernel direction.
D04 part of second $\phi\phi$ value	PARTIAL (v12). Pairing P0 derived (Proposition 41.1); regulators differ by an explicit counterterm (Proposition 42.1); M_{10} to 40 digits under the declared R_h (Table 15).	A derivation of the closed form of F (Proposition 40.1); a derived (not declared) regulator normalization.

Obligation	v12 status	Still missing
Transverse-jet catalogue	PARTIAL (v10, Section 28); loop momentum on the slice.	Transverse loop momentum (no $p_\perp$ jets exported); the 84 mixed weight-0 mode sums; weight-±1 residue/finite-part split ($w = \frac{1}{2}$); weight ±2. Mixed evaluator corrected in v11 (Section 33); constant-density mixed weight-0 sums now have a closed-form path (Proposition 40.1); transverse feasibility in Section 44.
Exact Span certificate	CLOSED_PENDING_ INDEPENDENT_REVIEW on the four points (v8). Chamber ranks 26 (six $N = 34$ chambers) and 29 (F above $\frac{1}{2}$) at every α: CLOSED_PENDING_ INDEPENDENT_REVIEW in the scope of Section 35; (D) machine-checked (Proposition 35.1). Walls $\frac{1}{3}, \frac{1}{2}, \frac{2}{3}$: exact ranks, CLOSED_PENDING_ INDEPENDENT_REVIEW (Section 43).	Identification with the interacting basis; independent review.
Evanescent coefficients	CONDITIONAL on slice-DRED (v8); excluded from the frozen $\tilde{g}$ rule (v9).	Transverse $\tilde{G}$ components.
Vector-type keys	Finite parts computed in a declared k-ζ scheme (Proposition 30.1).	A physical scheme; profile Ward partners.
Branch input	CR and F remain inputs.	A derived graviton operator.

In addition, graph identification still needs (R2), (R4), and (R5), and the undisposed parts of (R3), for every relevant graph. The D01 Mellin prescription obligation is explicit in Corollary 6.4. The general joint-apex and (H_{reg}) statements remain open.

Theorem 46.1 (Scoped v12 release)**.** *Under the displayed definitions and source qualifications:*

(i) *The normalized heat coefficients $\mathcal{C}_b^{\mathrm{F}}$ and $\mathcal{D}_b$, their specified singular-channel shifts, and the orbifold identities are established.*

(ii) *A complete graph satisfying Assumption 7.1 has the apex residue* (7.1)*.*

(iii) *The D01 composite has the local obstruction* (6.1)*; its specified Mellin continuation is* (6.2) *and is exactly zero at $z = 0$. The Mellin value does not select the graph prescription.*

(iv) *The declared coefficient catalogue satisfies the ghost identity and the constant-profile hh/Kc row. With the stated tree input, $\lambda = -2\mathrm{i}$ is fixed before loop fitting. The complete lattice branch difference is* (9.2)*.*

(v) *These rows do not determine the full interacting basis, a physical tensor-core limit, or a complete Slavnov–Taylor theorem on the cone.*

(vi) *The continuum/preparation/instrument objects in* (12.1) *remain distinct. The record-only recovery bound applies to a separately declared finite interface, not to a continuum residue. The spectral heat coefficients are not thermodynamic heat.*

(vii) *Image and echo-window functionals are not identified with the apex residue; Example 3.3 is an explicit negative control.*

(viii) *None of the obligations in Table 17 is set to zero or discharged by the companion comparisons.*

(ix) (v8, executed.) *Theorem 14.1 holds on its stated finite scope; on that scope the status is* CLOSED_PENDING_INDEPENDENT_REVIEW.

(x) (v8, executed, conditional on Assumption 7.1.) *The hh catalogue sum satisfies the two hh-supported Ward combinations exactly (Proposition 15.1); four of the six exact values are fixed in the stated basis; six of the eighteen hh engine coordinates are mapped.*

(xi) (v8, amended in v9.) *The hidden D01 mixed-channel term of Proposition 16.2 is pure TT and vanishes on the lattice (Proposition 19.2); the v6 hh branch difference is restored, subject only to Assumption 7.1. The evanescent coefficients of Section 17 are conditional on the declared prescription and are excluded from the frozen $\tilde{g}$ rule (Proposition 22.1).*

(xii) (v9, executed.) *The mixed trace has the closed form of Proposition 19.1; the TT branch difference ΔJ is finite and computed (Table 7); absolute TT values carry the IR logarithm of Proposition 19.3.*

(xiii) (v9, executed, exact on four points.) *The CB/CK dictionaries and all six constant-profile Ward functionals hold with two fitted α-independent normalizations; e_5, e_6 are not separately determined at constant profile (Theorem 20.2).*

(xiv) (v9.) *The generic ST rank over $\mathbb{Q}(\alpha)$ is as in Theorem 21.2 on the chambers of Proposition 21.1; the exceptional set inside chambers is open.* (Amended in v10: empty, conditional on (D); item (xviii).)

(xv) (v10, executed.) *ΔJ has the closed form of Corollary 25.3: the CR mode is derived, and the F mode is identified by PSLQ and verified, not derived.*

(xvi) (v10, executed.) *$\lambda_{\mathrm{CK}} = -8$ and $\lambda_{\mathrm{CB}} = -4\mathrm{i}$ follow from tree operator ratios and the v9 dictionaries (Theorem 26.1).*

(xvii) (v10; unconditional since v11.) *The rank is constant inside chambers: 26 on $N = 34$ chambers and 29 on F above $\frac{1}{2}$ (Theorem 27.1; (D) is Proposition 35.1).*

(xviii) (v10, partial.) *The slice jet catalogue, the scheme-Z r^2 moments, and the vector-key finite parts in a declared scheme are as in Sections 28 and 30. The D04 part, the jet Ward combinations and e_5, e_6 remain open.*

(xix) (v11, executed.) *I_{F} is derived for $0 < \mu < 2$ (Theorem 34.1), and $\Delta J = -(\mu - 1)B/W_1$.*

(xx) (v11, executed.) *The degree bound (D) is machine-checked (Proposition 35.1). The chamber-rank certificate is* CLOSED_PENDING_INDEPENDENT_REVIEW *in the scope stated in Section 35, and the v8 four-point certificate remains* CLOSED_PENDING_INDEPENDENT_REVIEW *in its v8 scope.*

(xxi) (v11, partial.) *The mixed-trace evaluator is corrected (Section 33). M_{10} is evaluated in declared pairings and regulators (Table 13), so D04 is scheme dependent. The weight $\pm 1, \pm 2$ traces, the dictionary, the 20 jet Ward combinations, the projections, e_5, e_6 and the transverse loop momentum remain open (Section 37).*

(xxii) (v12, identified and verified.) *The closed form of $F(\mu, \nu)$ (Proposition 40.1) and its consequences, Corollary 40.2 and the 40-digit M_{10} (Table 15).*

(xxiii) (v12, derived under stated assumptions.) *The D04 pairing is P0 (Proposition 41.1). The regulators differ by the local counterterm of Proposition 42.1. D04 is* PARTIAL *only because the closed form is identified and the regulator R_h is a declared normalization.*

(xxiv) (v12, executed.) *The wall ranks of Table 16 are* CLOSED_PENDING_INDEPENDENT_REVIEW *in the scope of Section 43.*

The release status is

O.1 on the cone = `NOT_CLOSED`,	`bulk_symmetry_restored` = `false`.

Proof. Items (i)–(iii) are Theorems 3.2, 4.1, 5.2, 5.4, 7.2, 6.1, and 6.2, with Proposition 5.3 and Corollary 6.4. Item (iv) is Theorems 10.1 and 10.3, Input 10.2, and Proposition 9.3.

Proposition 11.1, the identity classification in Section 10, and the explicit missing data prove the limitation in (v). Proposition 12.1, Theorem 12.2, and the separate hypotheses of (12.3) establish (vi). Example 3.3 and the distinct definitions establish (vii). Table 17 records the remaining obligations.

Items (xix)–(xxi) are Theorem 34.1, Proposition 35.1 and Section 36. Items (xv)–(xviii) are Corollary 25.3, Theorems 26.1, 27.1 and Propositions 28.2, 28.3, 30.1. Items (xii)–(xiv) are Propositions 19.1, 19.3, 21.1 and Theorems 20.2, 21.2. Items (ix)–(xi) are Theorem 14.1, Propositions 15.1 and 16.2, and the executed records of Sections 15 and 17.

The complete O.1 requirements are not consequences of these restricted statements. The status is non-closure, not a universal no-go theorem for a future completion. □

Submission boundary

v8 adds executed evidence, and every executed statement is listed with its hash in Appendix B. The listed runs are the replay, the exact certificates, the maps, the slice and evanescent runs, and the Duhamel corroboration. v8 was compiled with pdfLaTeX. It does not claim a proof-assistant check, an independent referee review, physical acquisition, a new graph evaluation beyond the declared catalogue, or a general-α theorem from finitely many exact points. Nothing was pushed or posted.

The exact rank statement is CLOSED_PENDING_INDEPENDENT_REVIEW only on the scope of Theorem 14.1. Every other new item is PARTIAL, CONDITIONAL, or OPEN, as listed in Table 17. The present conditional statements must not be relabelled as unconditional graph theorems.

v9 adds the items of Section 18, each hashed in Appendix C. It claims no closed form for ΔJ or for the D04 part, no derivation of the two ghost normalizations, no proof that $\pi_K B$ is independent of α beyond the four points, and no identification of the exceptional set inside chambers. No independent review was started; nothing was pushed, posted or written to the site.

v10 adds the items of Section 24, each hashed in Appendix D. It does not claim a derivation of the F mode, a machine check of the degree bound (D), any jet-level Ward residual, the D04 value, or a physical scheme for the vector finite parts. No scope is labelled `CLOSED_PENDING_INDEPENDENT_REVIEW`. No independent review was started; nothing was pushed, posted or written to the site.

v11 adds the items of Section 32, each hashed in Appendix E, and withdraws the v10 φ values (Section 38). Two scopes carry `CLOSED_PENDING_INDEPENDENT_REVIEW`: the v8 four-point rank certificate and the chamber ranks of Section 35. Nothing else does. v11 claims no derived pairing or regulator for D04, no weight-±1 log coefficient, no jet Ward residual, and no value of e_5, e_6. No independent review was started; nothing was pushed, posted or written to the site.

v12 adds the items of Section 39, each hashed in Appendix F. Three scopes carry `CLOSED_PENDING_INDEPENDENT_REVIEW`: the v8 four-point rank certificate, the chamber ranks of Section 35, and the wall ranks of Section 43. Nothing else does. v12 claims no derivation of the closed form of F, no derived regulator, no weight-±1 log coefficient, no jet Ward residual, and no value of e_5, e_6. No independent review was started; nothing was pushed, posted or written to the site.

A Embedded verification and replay program

A.1 Use and acceptance conditions

Save the listing below as `cone_v7.py`. It uses Python and SymPy; numerical quadrature is not required for its exact checks. The listing is ASCII Python. It is included directly with `listings`, so typesetting needs neither shell escape nor an external code file.

Example commands:

```
python3 cone_v7.py selftest

# New logs with explicit coverage:
python3 cone_v7.py logs /path/to/logs

# Historical nonzero-only logs: comparison, not coverage proof:
python3 cone_v7.py logs /path/to/logs --legacy

# Optional replay; frozen engine is an external required input:
python3 cone_v7.py replay D01 CR \
  --engine /path/to/XVI-P/code/p5_entry --out /path/to/newlogs

python3 cone_v7.py tree \
  --engine /path/to/XVI-P/code/p5_entry

# Redirect outside the hashed tree:
python3 cone_v7.py manifest /path/to/archive > manifest.json
```

What `selftest` means. It checks the displayed finite algebraic identities. It does not independently derive the catalogue from vertices.

What `logs` means. Without `--legacy`, every compared coordinate must have an explicit evaluated-key status. With `--legacy`, missing zero entries are reported as coverage-uncertified. A successful legacy comparison is not an execution or graph certificate.

What `replay` means. It rebuilds the historical zero-transverse-momentum algebraic reduction from a supplied frozen engine. It verifies the coarse projector assignment, records all mixed pairs, and excludes an entire affected physical-branch key. For UNIT it retains all channel pairs. The output is explicitly marked `ALGEBRAIC_SLICE_ONLY`; it does not establish (R2)–(R5). The graph prefactors are declared source conventions, not a new automatic authentication of Q02.

What is not embedded. The frozen `amplitude_rules.py`, `feynman.py`, `dred.py`, vertex arrays, Q02 record, and historical JSON logs are not reproduced here. Missing dependencies fail rather than being replaced by zero.

Listing 1: Consolidated exact checker and coverage-aware optional replay.

```
#!/usr/bin/env python3
"""Scoped cone v7 checker.

Status:
  Source supplied for review; no execution is asserted by the paper.

selftest:
  Exact identities of the printed formulas, not a vertex rebuild.

logs:
  Compare explicit retained coordinates. New logs require coverage.
```

```
  --legacy permits polynomial comparison with uncertified zero coverage.

replay:
  Frozen-engine zero-transverse-momentum algebra only.
  Entire physical-branch keys with mixed poles are quarantined.
  UNIT retains all pairs. This is NOT an R2--R5 certificate.

tree:
  Check the declared tree normalization using a supplied engine.

manifest:
  Byte hashes with explicit path serialization. This is not the
  historical find/sort/xargs tree-hash algorithm.
"""
import argparse
import ast
import hashlib
import importlib
import json
import sys
import time
from pathlib import Path

import sympy as sp

I = sp.I
Q = sp.Rational
alpha = sp.Symbol("alpha", positive=True)
kappa, q0, q3 = sp.symbols("kappa q0 q3")
c0, c1, c2 = sp.symbols("c0 c1 c2")
P2 = q0**2 - q3**2
ETA = (1, -1, -1, -1)
HC = ((0, 0), (0, 1), (0, 2), (0, 3), (1, 1),
      (1, 2), (1, 3), (2, 2), (2, 3), (3, 3))
LAT = (0, 3, 9, 4, 7)
DIAGNOSTIC = (0, 3, 9, 4, 7, 5)
GRAPHS = ("D01", "D02", "D03", "D04", "D05a",
          "D06", "D11", "D12", "D13")


def zero(expr, label):
    v = sp.factor(sp.cancel(sp.expand(expr)))
    if v != 0:
        raise AssertionError(label + ": " + str(v))


def no_duplicate_pairs(pairs):
    out = {}
    for k, v in pairs:
        if k in out:
            raise ValueError("duplicate JSON key: " + str(k))
        out[k] = v
    return out


def read_json(path):
    with Path(path).open(encoding="utf-8") as f:
        return json.load(f, object_pairs_hook=no_duplicate_pairs)


def write_json(path, data):
    path = Path(path)
    path.parent.mkdir(parents=True, exist_ok=True)
    with path.open("w", encoding="utf-8") as f:
        json.dump(data, f, indent=2, sort_keys=True)
        f.write("\n")


def key_parse(text):
    obj = ast.literal_eval(text)
    if not (isinstance(obj, tuple) and len(obj) == 2
            and all(type(x) is int and 0 <= x < 10 for x in obj)):
        raise ValueError("invalid component key: " + text)
    return obj
```

```
# Restricted expression parser: no eval, attributes, function calls,
# comprehensions, or implicit imports. Serialized logs are arithmetic.
NAMES = {"alpha": alpha, "kappa": kappa, "q0": q0, "q3": q3,
         "I": I, "pi": sp.pi}


def expression(text):
    if len(text) > 100000:
        raise ValueError("expression exceeds declared input bound")

    def walk(n):
        if isinstance(n, ast.Expression):
            return walk(n.body)
        if isinstance(n, ast.Constant) and type(n.value) is int:
            return sp.Integer(n.value)
        if isinstance(n, ast.Name) and n.id in NAMES:
            return NAMES[n.id]
        if isinstance(n, ast.UnaryOp):
            if isinstance(n.op, ast.UAdd):
                return walk(n.operand)
            if isinstance(n.op, ast.USub):
                return -walk(n.operand)
        if isinstance(n, ast.BinOp):
            a, b = walk(n.left), walk(n.right)
            if isinstance(n.op, ast.Add):
                return a + b
            if isinstance(n.op, ast.Sub):
                return a - b
            if isinstance(n.op, ast.Mult):
                return a * b
            if isinstance(n.op, ast.Div):
                return a / b
            if isinstance(n.op, ast.Pow):
                if not (b.is_Integer and abs(int(b)) <= 100):
                    raise ValueError("undeclared power")
                return a**b
        raise ValueError("unsupported symbolic syntax")

    return sp.expand(walk(ast.parse(text, mode="eval")))


def C(a, b):
    return (1/a - a)/12 - b*(1-b)/(2*a)


def D(a, b):
    B2 = b*b - b + Q(1, 6)
    B4 = b**4 - 2*b**3 + b*b - Q(1, 30)
    return -11*a/180 + B2/(3*a) - B4/(6*a**3)


def physical_cs(branch):
    a = alpha
    z0 = (1-a*a)/(12*a)
    z1 = (5*a*a-6*a+1)/(12*a)
    if branch == "CR":
        z2 = (23*a*a-12*a+1)/(12*a)
    elif branch == "F":
        z2 = (23*a*a-36*a+13)/(12*a)
    elif branch == "UNIT":
        return (sp.Integer(1),)*3
    else:
        raise ValueError("unknown branch")
    return z0, z1, z2


def put(out, key, val):
    val = sp.expand(val)
    if key in out:
        zero(out[key]-val, "inconsistent duplicate " + str(key))
    out[key] = val
```

```
def symmetric(d):
    out = {}
    for (a, b), v in d.items():
        put(out, (a, b), v)
        put(out, (b, a), v)
    return out


def tensor_images(d):
    # Exchange transverse coordinates 1 and 2, then transpose slots.
    exch = {1: 2, 2: 1, 4: 7, 7: 4, 6: 8, 8: 6}
    out = dict(d)
    for (a, b), v in d.items():
        put(out, (exch.get(a, a), exch.get(b, b)), v)
    return symmetric(out)


def catalogue(cs):
    z0, z1, z2 = cs
    A, B = z0+2*z1+4*z2, -z0+2*z1
    R, T, V = 3*z0+4*z1, 4*z0+3*z1, z1-z0
    W = (z0+2*z1+z2)/2
    k16 = kappa**2/(16*sp.pi)
    out = {}
    out["D04"] = {(0, 0): -kappa**2*z0*P2/(4*sp.pi)}
    out["D05a"] = {(0, 0): -kappa**2*(z2-z0)*P2/(4*sp.pi)}
    out["D11"] = symmetric({
        (4, 4): z0*P2*k16/2,
        (4, 7): z0*P2*k16/2,
        (7, 7): z0*P2*k16/2})
    d03 = {
        (0, 0): -z0*P2, (0, 9): -z0*P2, (9, 9): -z0*P2,
        (3, 3): 4*z0*P2,
        (0, 4): z0*(q0**2-3*q3**2),
        (3, 4): 4*z0*q0*q3,
        (4, 9): -z0*(3*q0**2-q3**2),
        (4, 4): -(z0+4*z1)*P2,
        (4, 7): -z0*P2, (5, 5): -8*z1*P2}
    d01 = {
        (0, 0): z0*P2, (0, 9): z0*P2, (9, 9): z0*P2,
        (3, 3): -4*z0*P2,
        (4, 4): (z0-4*z1+z2)*P2,
        (4, 7): (z0+z2)*P2,
        (3, 4): -4*(z0+2*z2)*q0*q3,
        (4, 9): A*q0**2-B*q3**2,
        (0, 4): -B*q0**2+A*q3**2,
        (5, 5): -8*z1*P2}
    out["D03"] = {k: v*k16 for k, v in tensor_images(d03).items()}
    out["D01"] = {k: v*k16 for k, v in tensor_images(d01).items()}
    d02 = {
        (1, 1): q3**2, (2, 2): q3**2, (5, 5): P2,
        (6, 6): q0**2, (8, 8): q0**2,
        (1, 6): -q0*q3, (2, 8): -q0*q3,
        (0, 4): -q3**2/2, (0, 7): -q3**2/2,
        (3, 4): q0*q3, (3, 7): q0*q3,
        (4, 7): -P2/2,
        (4, 9): -q0**2/2, (7, 9): -q0**2/2}
    out["D02"] = {k: kappa**2*W*v/sp.pi
                  for k, v in symmetric(d02).items()}
    out["D12"] = {}
    out["D06"] = {
        (0, 0): I*kappa**2*R*P2/(8*sp.pi),
        (3, 3): I*kappa**2*R*P2/(8*sp.pi)}
    out["D13"] = {
        (0, 0): -2*T*q0, (0, 3): 2*V*q3,
        (3, 0): -R*q3, (3, 3): R*q0,
        (9, 0): -2*V*q0, (9, 3): 2*T*q3}
    out["D13"] = {k: I*kappa**2*v/(16*sp.pi)
                  for k, v in out["D13"].items()}
    return out


def invariant_data():
```

```
    x = sp.symbols("x00 x03 x33 xf")
    y = sp.symbols("y00 y03 y33 yf")

    def tr(v):
        return v[0]-v[2]

    def php(v):
        return q0**2*v[0]-2*q0*q3*v[1]+q3**2*v[2]

    dot = x[0]*y[0]-2*x[1]*y[1]+x[2]*y[2]
    S = (P2*tr(x)*tr(y), P2*dot,
         php(x)*tr(y)+tr(x)*php(y),
         P2*(x[3]*tr(y)+tr(x)*y[3]),
         php(x)*y[3]+x[3]*php(y), P2*x[3]*y[3])
    idx = {0: 0, 3: 1, 9: 2, 4: 3, 7: 3}
    return x, y, S, idx


def structure_table(cs):
    a, b, c = cs
    W = (a+2*b+c)/2
    return {
        "D01": (-a/16, a/8, 0, -a/8-b/4-c/2,
                a/4+c/2, a/4-b/2+c/4),
        "D02": (0, 0, 0, W, -W, -W),
        "D03": (a/16, -a/8, 0, 3*a/8, -a/4, -a/4-b/2),
        "D11": (0, 0, 0, 0, 0, a/8),
        "D12": (0, 0, 0, 0, 0, 0)}


def bilinear(d, x, y, idx):
    return sp.expand(sum(v*x[idx[a]]*y[idx[b]]
                         for (a, b), v in d.items()
                         if a in idx and b in idx))


def einstein_coordinates(hv):
    p = (q0, 0, 0, q3)
    pu = tuple(ETA[i]*p[i] for i in range(4))
    h = sp.zeros(4)
    for a, (m, n) in enumerate(HC):
        h[m, n] = h[n, m] = hv.get(a, 0)
    tr = sum(ETA[i]*h[i, i] for i in range(4))
    ph = [sum(pu[r]*h[r, n] for r in range(4)) for n in range(4)]
    pph = sum(pu[n]*ph[n] for n in range(4))
    E = sp.Matrix(4, 4, lambda m, n:
        P2*h[m, n]-p[m]*ph[n]-p[n]*ph[m]+p[m]*p[n]*tr
        +(ETA[m] if m == n else 0)*(pph-P2*tr))
    for n in range(4):
        zero(sum(pu[m]*E[m, n] for m in range(4)), "E transverse")
    return {a: (2-int(m == n))*ETA[m]*ETA[n]*E[m, n]
            for a, (m, n) in enumerate(HC)}


def check_catalogue_algebra():
    cat = catalogue((c0, c1, c2))
    x, y, S, idx = invariant_data()
    table = structure_table((c0, c1, c2))
    for g, coeff in table.items():
        lhs = bilinear(cat[g], x, y, idx)
        rhs = kappa**2/sp.pi*sum(a*b for a, b in zip(coeff, S))
        zero(lhs-rhs, "structure " + g)

    variables = x+y+(q0, q3)
    polys = [sp.Poly(sp.expand(s), *variables) for s in S]
    mons = sorted(set(m for p in polys for m in p.monoms()))
    matrix = sp.Matrix([[p.coeff_monomial(m) for p in polys]
                        for m in mons])
    if matrix.rank() != 6:
        raise AssertionError("six-structure independence failed")

    h00, h03, h33, f = sp.symbols("h00 h03 h33 f")
    g0, g3 = sp.symbols("ghost0 ghost3")
    hv = {0: h00, 3: h03, 9: h33, 4: f, 7: f}
```

```
    dh = {0: 2*q0*g0, 3: -q0*g3+q3*g0,
          9: -2*q3*g3, 4: 0, 7: 0}
    E = einstein_coordinates(hv)
    bterm = 0
    for g in ("D01", "D02", "D03", "D11", "D12"):
        bterm += sum(v*dh.get(a, 0)*hv.get(b, 0)
                     for (a, b), v in cat[g].items())
    gterm = sum(v*E[a]*({0: g0, 3: g3}[n])
                for (a, n), v in cat["D13"].items())
    core = kappa**2*(c1-c0)*P2*f*(q0*g0+q3*g3)/sp.pi
    zero(bterm+core/2, "hh gauge variation")
    zero(gterm-I*core/4, "antifield contraction")
    lam = 1/(I*Q(1, 2))  # Declared tree input N=1, A_GH=1/2.
    zero(lam+2*I, "a-priori lambda arithmetic")
    zero(bterm+lam*gterm, "ST row")

    # Ghost equation, separately for each upper ghost column.
    for n in (0, 3):
        Y00 = cat["D13"].get((0, n), 0)
        Y03 = cat["D13"].get((3, n), 0)
        Y33 = cat["D13"].get((9, n), 0)
        F0 = q0*(Y00+Y33)/2-q3*Y03
        F3 = q0*Y03-q3*(Y00+Y33)/2
        zero(cat["D06"].get((0, n), 0)+2*F0, "ghost row 0")
        zero(cat["D06"].get((3, n), 0)-2*F3, "ghost row 3")

    # Endpoint and branch checks are polynomial checks only.
    for branch in ("CR", "F"):
        vals = catalogue(physical_cs(branch))
        if branch == "F":
            for d in vals.values():
                for v in d.values():
                    zero(v.subs(alpha, 1), "F flat catalogue")
    cr = catalogue(physical_cs("CR"))
    fr = catalogue(physical_cs("F"))
    for g in GRAPHS:
        for key in set(cr[g]) | set(fr[g]):
            zero((cr[g].get(key, 0)-fr[g].get(key, 0))
                 .subs(alpha, Q(1, 2)), "CR=F at half")
    bhcr = sum((bilinear(cr[g], x, y, idx)
                for g in table), sp.Integer(0))
    bhf = sum((bilinear(fr[g], x, y, idx)
               for g in table), sp.Integer(0))
    nu = 2-1/alpha
    zero(bhcr-bhf+nu*kappa**2*P2*x[3]*y[3]/(4*sp.pi),
         "complete branch difference")

    if len(cat["D02"]) != 23:
        raise AssertionError("D02 entry count")
    zero(sum(table[g][0] for g in table), "longitudinal S1")
    zero(sum(table[g][1] for g in table), "longitudinal S2")
    return {"catalogue_algebra": "PASS",
            "tree_normalization": "DECLARED_INPUT_NOT_REBUILT"}


def check_heat_and_mellin():
    u = sp.Symbol("u", positive=True)
    b = sp.Symbol("b")
    T = sp.cosh((Q(1, 2)-b)*u/alpha) / (
        2*sp.sinh(u)*sp.sinh(u/(2*alpha)))
    bulk = alpha/(2*(sp.cosh(u)-1))
    cser = sp.series(T-bulk, u, 0, 2).removeO()
    zero(sp.expand(cser).coeff(u, 0)-C(alpha, b), "C coefficient")

    weighted = -2*sp.diff(T, u)/sp.sinh(u)
    wbulk = alpha/(sp.cosh(u)-1)**2
    wser = sp.expand(sp.series(weighted-wbulk, u, 0, 2).removeO())
    for k in range(1, 5):
        zero(wser.coeff(u, -k), "weighted pole cancellation")
    zero(wser.coeff(u, 0)-D(alpha, b), "D coefficient")
    nu = 2-1/alpha
    zero(D(alpha, 2*alpha)-D(alpha, 2*alpha-1)
         -2*nu*(1-nu**2)/3, "weighted flip")
    zero(D(alpha, 2*alpha).subs(alpha, 1), "weighted CR flat")
```

```
382
383     # Exact finite checks in Q[zeta_N], not numerical nsimplify.
384     x = sp.Symbol("x")
385     for N in range(2, 9):
386         cyclo = sp.cyclotomic_poly(N, x)
387         for j in range(N):
388             acc = 0
389             for k in range(1, N):
390                 inv = sp.invert((x**k-1)**4, cyclo, x)
391                 phase = x**(((j+2)*k) % N) + x**(((2-j)*k) % N)
392                 acc = sp.rem(acc+Q(2, N)*phase*inv, cyclo, x)
393             target = D(Q(1, N), Q(j, N))
394             zero(sp.rem(acc-target, cyclo, x), "cyclotomic moment")
395
396     # Finite rational algebra underlying the closed Mellin proof.
397     t, n = sp.symbols("t nu")
398     ratio = (t*t-n*n)/(2*t+1)
399     bracket = (t+1)*ratio+n*(n+t)
400     zero(bracket-t*(t+n)*(t+n+1)/(2*t+1),
401          "Mellin recurrence reduction")
402     # Gamma reflection gives M'_z(0)=-(nu+1)/(2s).
403     return {"heat_moments": "PASS",
404             "finite_cyclotomic_checks": "PASS",
405             "Mellin_rational_reduction": "PASS",
406             "Mellin_integral_and_continuation":
407                 "ANALYTICAL_PROOF_IN_PAPER"}
408
409
410 def retained_keys(g):
411     if g in ("D04", "D05a"):
412         return [(0, 0)]
413     if g == "D06":
414         return [(a, b) for a in (0, 3) for b in (0, 3)]
415     if g == "D13":
416         return [(a, b) for a in LAT for b in (0, 3)]
417     if g in ("D01", "D03"):
418         return [(a, b) for a in DIAGNOSTIC for b in DIAGNOSTIC]
419     return [(a, b) for a in range(10) for b in range(10)]
420
421
422 def compare_logs(root, legacy):
423     root = Path(root)
424     report = {}
425     for branch in ("CR", "F"):
426         expected = catalogue(physical_cs(branch))
427         for g in GRAPHS:
428             path = root/("spec_" + g + "_" + branch + ".json")
429             data = read_json(path)  # Missing files are errors.
430             if data.get("graph") != g or data.get("branch") != branch:
431                 raise ValueError("graph/branch metadata mismatch: " + str(path))
432             got = {key_parse(k): expression(v)
433                    for k, v in data[
434                        "Gamma_pole_coeff_of_1/(4-d)"].items()}
435             status = data.get("key_status", {})
436             unverified = []
437             for key in retained_keys(g):
438                 skey = str(key)
439                 if not legacy:
440                     st = status.get(skey)
441                     if st not in ("COMPUTED_ZERO_SLICE",
442                                   "COMPUTED_NONZERO_SLICE"):
443                         raise ValueError("missing/excluded coverage "
444                                          + str(path) + " " + skey)
445                     if st == "COMPUTED_NONZERO_SLICE" and key not in got:
446                         raise ValueError("nonzero status without value")
447                     if st == "COMPUTED_ZERO_SLICE" and key in got:
448                         zero(got[key], "zero status with nonzero value")
449                 elif key not in got:
450                     unverified.append(skey)
451                 zero(got.get(key, 0)-expected[g].get(key, 0),
452                      "catalogue mismatch " + g + " " + skey)
453             report[g+"_"+branch] = {
454                 "polynomial_comparison": "PASS",
455                 "coverage": ("LEGACY_UNCERTIFIED" if legacy
```

```
                            else "EXPLICIT_SLICE_COVERAGE"),
                "absent_legacy_keys": unverified,
                "graph_identification": "R2_R5_NOT_CERTIFIED"}
    return report


def load_engine(folder):
    root = Path(folder).resolve(strict=True)
    for name in ("amplitude_rules.py", "feynman.py", "dred.py"):
        if not (root/name).is_file():
            raise FileNotFoundError(root/name)
    sys.path.insert(0, str(root))
    AR = importlib.import_module("amplitude_rules")
    if Path(AR.__file__).resolve().parent != root:
        raise RuntimeError("import resolved outside declared engine")
    return AR


def projector_check(AR):
    Pi = AR.pinv_sparse()
    M = sp.Matrix(10, 10, lambda a, b: sp.sympify(Pi.get((a, b), 0)))
    P0 = sp.zeros(10)
    for a in (0, 3, 9):
        P0[a, a] = 1
    for a in (4, 7):
        for b in (4, 7):
            P0[a, b] = Q(1, 2)
    P1 = sp.zeros(10)
    for a in (1, 2, 6, 8):
        P1[a, a] = 1
    P2m = sp.eye(10)-P0-P1
    projectors = (P0, P1, P2m)

    def ctype(a):
        m, n = HC[a]
        count = int(m in (0, 3))+int(n in (0, 3))
        return {2: 0, 1: 1, 0: 2}[count]

    def channel(a, b):
        ca, cb = ctype(a), ctype(b)
        if ca == cb == 1:
            return 1
        if ca == cb == 2:
            return 2
        return 0

    for s, P in enumerate(projectors):
        zero_matrix = P*P-P
        if any(sp.simplify(v) != 0 for v in zero_matrix):
            raise AssertionError("projector is not idempotent")
        PM = P*M
        for a in range(10):
            for b in range(10):
                want = M[a, b] if channel(a, b) == s else 0
                zero(PM[a, b]-want, "coarse helicity marker")
    return Pi, channel


def laurent_coefficient(expr, x, power):
    expr = sp.cancel(expr)
    num, den = sp.fraction(expr)
    dp = sp.Poly(sp.expand(den), x)
    if len(dp.terms()) != 1:
        raise ValueError("not a Laurent polynomial")
    (deg,), cf = dp.terms()[0]
    np = sp.Poly(sp.expand(num), x)
    return sp.expand(np.coeff_monomial(x**(deg+power))/cf) \
        if deg+power >= 0 else sp.Integer(0)


def replay(AR, graph, branch, outdir):
    """Historical flat momentum-slice algebra, with honest coverage."""
    start = time.time()
    F = AR.F
```

```
    if tuple(F.HC) != HC:
        raise ValueError("independent-coordinate order mismatch")
    Pi, channel = projector_check(AR)
    Gh = AR.ghost_num()
    V = F.vertices()
    kap = F.KAPPA
    LS, PS = sp.symbols("l0:4"), sp.symbols("p0:4")
    names = {**{r: LS[r] for r in range(4)},
             **{4+r: PS[r] for r in range(4)}}
    u, v = sp.symbols("u0:3"), sp.symbols("v0:3")
    slice_sub = {LS[1]: 0, LS[2]: 0, PS[1]: 0, PS[2]: 0,
                 PS[0]: q0, PS[3]: q3}

    def route(vt, routing):
        out = {}
        for key, poly in F.vt_route(vt, routing).items():
            e = poly.to_sympy(names) if hasattr(poly, "to_sympy") else poly
            # Polynomial substitution commutes with the finite contractions.
            out[key] = sp.expand(sp.sympify(e).subs(slice_sub))
        return out

    def pw(a, b, marker):
        return sp.sympify(Pi[(a, b)])*marker[channel(a, b)]

    def gw(a, b, marker):
        return sp.sympify(Gh[(a, b)])*marker[0 if a in (0, 3) else 1]

    N = {}
    def add(key, value):
        N[key] = N.get(key, 0)+value

    dens = "bubble"
    if graph == "D01":
        vt = V["V3_grav"]
        va = route(vt, [(0, 1), (-1, 0), (1, -1)])
        vb = route(vt, [(0, -1), (1, 0), (-1, 1)])
        for (a, c, e), pa in va.items():
            for (b, d, f), pb in vb.items():
                if (c, d) in Pi and (e, f) in Pi:
                    add((a, b), pa*pb*pw(c, d, u)*pw(e, f, v))
        pref = sp.sympify(AR.base_rules()["graphs"]["D01"]["S_f"]) \
            *(I*kap)**2*I**(2*vt.phase)*I**2

    elif graph == "D03":
        vt = V["V_cb_c_h"]
        va = route(vt, [(-1, 0), (1, -1), (0, 1)])
        vb = route(vt, [(-1, 1), (1, 0), (0, -1)])
        for (rho, mu, a), pa in va.items():
            for (sg, nu, b), pb in vb.items():
                if (mu, sg) in Gh and (nu, rho) in Gh:
                    add((a, b), pa*pb*gw(mu, sg, u)*gw(nu, rho, v))
        pref = -(I*kap)**2*I**(2*vt.phase)

    elif graph == "D11":
        vt = V["V_h_phi_phi"]
        va = route(vt, [(0, 1), (-1, 0), (1, -1)])
        vb = route(vt, [(0, -1), (1, 0), (-1, 1)])
        for (a, _, _), pa in va.items():
            for (b, _, _), pb in vb.items():
                add((a, b), pa*pb*u[0]*v[0])
        pref = Q(1, 2)*(I*kap)**2*I**(2*vt.phase)*I**2

    elif graph == "D04":
        vt = V["V_h_phi_phi"]
        va = route(vt, [(-1, 0), (0, 1), (1, -1)])
        vb = route(vt, [(1, 0), (0, -1), (-1, 1)])
        for (c, _, _), pa in va.items():
            for (d, _, _), pb in vb.items():
                if (c, d) in Pi:
                    add((0, 0), pa*pb*pw(c, d, u)*v[0])
        pref = (I*kap)**2*I**(2*vt.phase)*I**2

    elif graph == "D06":
        vt = V["V_cb_c_h"]
```

```
        va = route(vt, [(0, -1), (1, 0), (-1, 1)])
        vb = route(vt, [(-1, 0), (0, 1), (1, -1)])
        for (rho, mu, a), pa in va.items():
            for (sg, nu, b), pb in vb.items():
                if (mu, sg) in Gh and (a, b) in Pi:
                    add((rho, nu), pa*pb*gw(mu, sg, u)*pw(a, b, v))
        pref = (I*kap)**2*I**(2*vt.phase)*I

    elif graph == "D13":
        rules = AR.base_rules()
        vk, vg = AR.K_vertex(rules), V["V_cb_c_h"]
        va = route(vk, [(0, -1), (1, 0), (-1, 1)])
        vb = route(vg, [(-1, 0), (0, 1), (1, -1)])
        for (ap, mu, c), pa in va.items():
            for (rho, nu, d), pb in vb.items():
                if (mu, rho) in Gh and (c, d) in Pi:
                    add((ap, nu), pa*pb*gw(mu, rho, u)*pw(c, d, v))
        r = rules["graphs"]["D13"]
        pref = sp.sympify(r["ext_sign"]*r["S_f"]*r["loop_sign"]) \
            *(I*kap)**2*I**(vk.phase+vg.phase)*I

    elif graph == "D02":
        dens = "tadpole"
        vt = V["V4_grav"]
        va = route(vt, [(0, 1), (0, -1), (1, 0), (-1, 0)])
        for (a, b, c, d), pa in va.items():
            if (c, d) in Pi:
                add((a, b), pa*pw(c, d, u)*v[0])
        pref = Q(1, 2)*(I*kap**2)*I**vt.phase*I

    elif graph == "D12":
        dens = "tadpole"
        vt = V["V_hh_phi_phi"]
        va = route(vt, [(0, 1), (0, -1), (1, 0), (-1, 0)])
        for (a, b, _, _), pa in va.items():
            add((a, b), pa*u[0]*v[0])
        pref = Q(1, 2)*(I*kap**2)*I**vt.phase*I

    elif graph == "D05a":
        dens = "tadpole"
        vt = V["V_hh_phi_phi"]
        va = route(vt, [(1, 0), (-1, 0), (0, 1), (0, -1)])
        for (c, d, _, _), pa in va.items():
            if (c, d) in Pi:
                add((0, 0), pa*pw(c, d, u)*v[0])
        pref = Q(1, 2)*(I*kap**2)*I**vt.phase*I
    else:
        raise ValueError("unknown graph")

    # Exact circle average via a Laurent variable z=e^(i theta).
    z, w = sp.symbols("z w", nonzero=True)
    cosine = (z+1/z)/2
    sine = (z-1/z)/(2*I)
    lm0, lm3 = I*cosine/w, sine/w
    l2 = -1/w**2
    den = l2 if dens == "tadpole" else \
        l2*((lm0-q0)**2-(lm3-q3)**2)

    def pole(expr):
        e = sp.expand(expr.subs({LS[0]: lm0, LS[3]: lm3}))
        if e == 0:
            return sp.Integer(0)
        ser = sp.series(e/den, w, 0, 3).removeO()
        g2 = sp.expand(ser).coeff(w, 2)
        average = laurent_coefficient(g2, z, 0)
        # i/(4 pi^2) times 2 pi times the circle average.
        return sp.cancel(I*average/(2*sp.pi))

    if graph in ("D04", "D05a"):
        domain = [(0, 0)]
    elif graph == "D06":
        domain = [(a, b) for a in range(4) for b in range(4)]
    elif graph == "D13":
        domain = [(a, b) for a in range(10) for b in range(4)]
```

```
    else:
        domain = [(a, b) for a in range(10) for b in range(10)]

    cs = physical_cs(branch)
    values, states, mixed = {}, {}, {}
    for key in domain:
        e = sp.expand(N.get(key, 0))
        pair_poles = {}
        if e != 0:
            for mon, cf in sp.Poly(e, *u, *v).terms():
                iu = [i for i in range(3) if mon[i]]
                iv = [i for i in range(3) if mon[3+i]]
                if len(iu) != 1 or len(iv) != 1 \
                        or sum(mon[:3]) != 1 or sum(mon[3:]) != 1:
                    raise ValueError("unexpected channel-marker degree")
                s, t = iu[0], iv[0]
                if dens == "tadpole":
                    t = s
                pv = pole(cf)
                pair_poles[(s, t)] = pair_poles.get((s, t), 0)+pv
        pair_poles = {st: sp.factor(vv)
                      for st, vv in pair_poles.items()
                      if sp.simplify(vv) != 0}
        bad = {str(st): str(vv) for st, vv in pair_poles.items()
               if st[0] != st[1]}
        skey = str(key)
        if bad:
            mixed[skey] = bad
        if bad and branch != "UNIT":
            states[skey] = "EXCLUDED_MIXED_CHANNEL"
            continue
        total = 0
        for (s, t), pv in pair_poles.items():
            total += pv if branch == "UNIT" else pv*cs[s]
        value = sp.factor(sp.cancel(-I*pref*total))
        if value == 0:
            states[skey] = "COMPUTED_ZERO_SLICE"
        else:
            states[skey] = "COMPUTED_NONZERO_SLICE"
            values[skey] = str(value)

    result = {
        "graph": graph, "branch": branch,
        "status": "ALGEBRAIC_SLICE_ONLY",
        "prescription_status": "R2_R3_R4_R5_NOT_CERTIFIED",
        "prefactor_status": "DECLARED_FROZEN_SOURCE_CONVENTIONS",
        "pref": str(pref),
        "key_status": states,
        "mixed_channel_poles_before_prefactor": mixed,
        "Gamma_pole_coeff_of_1/(4-d)": values,
        "seconds": time.time()-start}
    path = Path(outdir)/("spec_"+graph+"_"+branch+".json")
    write_json(path, result)
    return {"written": str(path), "status": result["status"],
            "excluded_keys": sum(s == "EXCLUDED_MIXED_CHANNEL"
                                 for s in states.values())}


def tree_normalization(AR):
    F = AR.F
    if tuple(F.HC) != HC:
        raise ValueError("coordinate order mismatch")
    vt = F.grav_vertex(2)
    ks = [F.ksym(j) for j in range(2)]
    names = {4*j+r: ks[j][r] for j in range(2) for r in range(4)}
    p = (q0, 0, 0, q3)
    h00, h03, h33, f = sp.symbols("h00 h03 h33 f")
    # De Donder equations on h11=h22=f.
    F0 = q0*(h00+h33+2*f)/2-q3*h03
    F3 = q0*h03-q3*(h00+h33-2*f)/2
    solutions = sp.solve((F0, F3), (h00, h03), dict=True)
    if len(solutions) != 1:
        raise AssertionError("generic de Donder solve failed")
    sol = solutions[0]
```

```
    hv = {0: sol[h00], 3: sol[h03], 9: h33, 4: f, 7: f}
    Eu = einstein_coordinates(hv)
    ratios = []
    for a in range(10):
        val = 0
        for b in range(10):
            poly = vt.T.get((a, b))
            if poly is None:
                continue
            e = I**vt.phase*poly.to_sympy(names)
            sub = {ks[0][r]: p[r] for r in range(4)}
            sub.update({ks[1][r]: -p[r] for r in range(4)})
            val += e.subs(sub)*hv.get(b, 0)
        val = sp.cancel(val)
        den = sp.cancel(Eu[a])
        if den == 0:
            zero(val, "tree zero-coordinate comparison")
        else:
            ratios.append(sp.cancel(val/den))
    if not ratios:
        raise AssertionError("tree check had no nonzero probe")
    N = ratios[0]
    for r in ratios[1:]:
        zero(r-N, "tree normalization not constant")
    zero(N-1, "declared N")
    Agh = sp.sympify(F.A_GH)
    zero(Agh-Q(1, 2), "declared A_GH")
    zero(N/(I*Agh)+2*I, "tree-predicted lambda")
    return {"N": str(N), "A_GH": str(Agh),
            "lambda_pred": str(N/(I*Agh)),
            "scope": "TREE_NORMALIZATION_ON_DECLARED_DD_CLASS"}


def manifest(root):
    root = Path(root).resolve(strict=True)
    if not root.is_dir():
        raise ValueError("manifest root is not a directory")
    files = []
    for p in sorted(root.rglob("*"),
                    key=lambda p: p.relative_to(root).as_posix()):
        if p.is_symlink():
            raise ValueError("symlink requires a separate policy: " + str(p))
        rel = p.relative_to(root)
        if ".pytest_cache" in rel.parts or p.suffix == ".pyc":
            continue
        if p.is_file():
            digest = hashlib.sha256(p.read_bytes()).hexdigest()
            files.append((rel.as_posix(), digest))
    tree = hashlib.sha256()
    for rel, digest in files:
        name = rel.encode("utf-8")
        tree.update(len(name).to_bytes(8, "big"))
        tree.update(name)
        tree.update(bytes.fromhex(digest))
    return {"serialization":
                "sorted UTF8 relative names; 8-byte length; name; raw SHA256",
            "exclusions": [".pytest_cache path component", "*.pyc"],
            "symlinks": "REJECTED",
            "files": dict(files), "tree_sha256": tree.hexdigest()}


def main():
    p = argparse.ArgumentParser()
    sub = p.add_subparsers(dest="command", required=True)
    sub.add_parser("selftest")
    lp = sub.add_parser("logs")
    lp.add_argument("root")
    lp.add_argument("--legacy", action="store_true")
    rp = sub.add_parser("replay")
    rp.add_argument("graph", choices=GRAPHS)
    rp.add_argument("branch", choices=("CR", "F", "UNIT"))
    rp.add_argument("--engine", required=True)
    rp.add_argument("--out", required=True)
    tp = sub.add_parser("tree")
```

```
    tp.add_argument("--engine", required=True)
    mp = sub.add_parser("manifest")
    mp.add_argument("root")
    a = p.parse_args()
    if a.command == "selftest":
        result = {**check_heat_and_mellin(), **check_catalogue_algebra(),
                  "execution_claim":
                      "THIS_OUTPUT_EXISTS_ONLY_IF_THE_USER_RUNS_THE_PROGRAM",
                  "O1": "NOT_CLOSED",
                  "bulk_symmetry_restored": False}
    elif a.command == "logs":
        result = compare_logs(a.root, a.legacy)
    elif a.command == "replay":
        result = replay(load_engine(a.engine), a.graph, a.branch, a.out)
    elif a.command == "tree":
        result = tree_normalization(load_engine(a.engine))
    else:
        result = manifest(a.root)
    print(json.dumps(result, indent=2, sort_keys=True))


if __name__ == "__main__":
    main()
```

A.2 Reproducibility obligations not replaced by this listing

A full computational submission should additionally provide:

(1) immutable frozen input files and their actual byte manifests;

(2) the graph inventory and the complete coordinate/domain coverage ledger;

(3) exact contracted spectral expressions before evaluation at zero transverse spectral parameter;

(4) the regulated trace reduction, with boundary defects;

(5) the UV/IR separation and meromorphic interchange certificate;

(6) source normalization and the complete relevant ST functional;

(7) the full basis projection and exact rank/span certificates;

(8) software versions, commands, process return codes, raw outputs, failed runs, and replay-versus-rebuild declarations.

A successful run of the embedded program would not, by itself, provide items 3–5 or 7. v8 ran it (Section 13.2) and separately supplies item 7 on a finite scope (Section 14); items 3–5 are not supplied.

B v8 artifacts, hashes and rerun commands

Paths are relative to `/workspace/XVI-O1-CONE/` on the audit box. Long jobs were run detached with `setsid` and logged. The environment is Python 3.13.5, SymPy 1.14.0, with `LC_ALL=C.UTF-8`. The frozen trees were hashed before the first v8 run and after the last one; the before/after values are identical: XVI-P: yes, XVI-Q: yes, XVI-R: yes. A full manifest is `logs/v8/artifact_manifest_v8.sha256`; the TEX source and PDF are hashed in `SHA256SUMS_v8`, outside this document.

Rerun commands.

```
cd /workspace/XVI-O1-CONE
python3 code/v8/cone_v7.py selftest
python3 code/v8/cone_v7.py logs logs/v4 --legacy
```

```
bash code/v8/run_replay.sh   # 27 runs
python3 code/v8/cone_v7.py logs logs/v8/replay
python3 code/v8/ward_scan_radial.py 0.85
python3 code/v8/apex_exact_run.py 7071/10000 CR
python3 code/v8/apex_certificate.py 3/10 CR      # also 3/10 F, 7071/10000 CR|F
python3 code/v8/k_support.py 3_10_CR             # per tag
python3 code/v8/hh_map.py 3/10 CR                # per (alpha, branch)
python3 code/v8/exact_values.py 3_10_CR 3_10_F 7071_10000_CR 7071_10000_F
bash code/v8/run_slice.sh && bash code/v8/run_slice2.sh
python3 code/v8/evanescent_combine.py
python3 code/v8/duhamel_mode_check2.py
python3 code/artifacts_v8.py && python3 code/build_v8.py
pdflatex -halt-on-error XVI-01-CONE-v8.tex   # twice
```

The certificate runs take 5–10 minutes each and stay below 2 GB of memory. The box limit is about 7 GB, so they were run at most three at a time.

Table 18: SHA-256 of v8 artifacts.

File	SHA-256
Base source (read-only) (1 files)	
XVI-01-CONE-v7.tex	d0ba7b9cdbf535d1296a74146011c1cb1269ff53148d3a96add64574873d5132
v8 programs (23 files)	
code/v8/apex_certificate.py	14a49b42dcb7c1fb53a69825a75a9a6dcfa36ad4f88f7f8ca80e95f47818b1ca
code/v8/apex_certificate_la.py	803c363e5aa3fd32824286bf5f4da891067c2a6e91c80705acf14178787cfb21
code/v8/apex_exact.py	ba6b0946e84e9f2fb3a60a24d38a3e4343e3887ae6c79c385b88e5c0db4ec46b
code/v8/apex_exact_run.py	db8f950dd31e012044776bddc3bd594aa2ededead73794459b71301321e23d48
code/v8/cone_v7.py	55b197552eb4c400c526cd39e4964a4cc8ec1fb89cf45dd41b1336dd42ba9b07
code/v8/duhamel_mode_check.py	b7a1183a11c4f0219b700a7d6ca0056f63aa8fc9efd3012d89a156d8310794f5
code/v8/duhamel_mode_check2.py	b010c66de54a5b7fde6bfc8efd42a4d618533f65390337ee0cdb8d250b698aaa
code/v8/evanescent_combine.py	e967018847f31c26448a832381c558fb01dde75dfb36bf60402e8873c8e58a0c
code/v8/exact_values.py	4b4ad1682afb5629f24df91b97a9a905a980ba1fa034fa060c087dfe9f7bfae3
code/v8/gq.py	a48693860dde5d3d0b244ca0b13c00cea1988849a45a3b16fe39d1cc7e1eae3b
code/v8/hh_map.py	372b3f01f084e1bb8091c933d3d979a445b445febc6610694048a7253c850cee
code/v8/k_support.py	471f02fa05515d76d4fd94b57b38811bdcafac450d0cd900e9466dfa076aa94f
code/v8/make_apex_exact.py	2f4e74c6c3ce1c0d92470f7babbb27f0b9d53515cd390270d352303cdc6da6fa
code/v8/make_apex_exact_run.py	c8e9abbb5402d35f1c97dafce62215c94d0bfae44e3ba6b266bedc23902fcb18
code/v8/slice_engine.py	296bfb8541ded14d27f1050cf8602693fb1a6f4c7c99cdd378b57eaed40069ed
code/v8/slice_run.py	2b46ef8d8c6dc8c9085725eccdf0d89aad609098ea913e57ba55f9a92b2e0688
code/v8/test_slice.py	75acc5a9c6deea06be6623038e745ddbff481bb12ae7b2a1dc23d65f4279cf70
code/v8/ward_scan_radial.py	453a8f3dfd959465b5722a710485fea655b4265af5cca914e7d09499049fafca
code/v8/run_replay.sh	c459fd5759b7cdec921d12fdd27ad3241a118c5c12361541e23b3916e07ef968
code/v8/run_slice.sh	2ff93bab2df9006a0054b71f39b4f055f475eb2c230fc9d740bec708b64e8d7b
code/v8/run_slice2.sh	bd47fcb81bdfa021554c3d667088249b3973ac8dde448239d696955778e90251
code/build_v8.py	c83d29a050937ab65347255ece236c47fc3ef8f78063f50710391532f2bf1263
code/artifacts_v8.py	adb9abcb3d879cdb7eb32e1956c16ae21ada684b5bb3aabcb0b4c8ab0f7f7766
v8 source fragments (9 files)	
v8src/abstract.tex	f5d9e2b3fd61a71d616d41562c68ba2d1bec5537f5c069b29dcaa0abb5c7d130
v8src/boundary.tex	33dd134192a6b257f9065161aba3ebf2df4238cb834ebb9dd066465c57e40fa3
v8src/opentable.tex	03a4908ff49b8f49d0ad402839aafcd70ad891e57a8a20f01b54322758fbe4bb
v8src/response.tex	7dd77765cbdc314c760cebc8eaa66839f0a4c329cb706075098a0399802d9336
v8src/sec_evan.tex	47e4623fa8bc6200026cf430642b705674154a95d40e472bc37ad95a17673422
v8src/sec_map.tex	83e8eb0f2e6b6b24ee812bd5924a0da5344c98bd34a7d091fdafed4ef2d0d1f8
v8src/sec_mixed.tex	23807ea310beb231d885e08bed37aebbe69014c601b91becbc7d08b0f57fb1ed
v8src/whatv8.tex	7d654d0008959917d3dee81e41930b6e68ad7bc058f9897b3fa6c621becc021b
v8src/bibmarks.json	12e6130eabfd85e6fc1841c039149ad8688cd32a66f934b606825675f3cc4cd3

File	SHA-256
v7-program runs (4 files)	
logs/v8/v7_logs_legacy.json	0674a32ac36a26c0a485c5591fb006bc4739e4159c6019912fb2860c37b1ca4f
logs/v8/v7_logs_replay.json	856162e1d1ba8365213ee7af80e1a661a02535b0512d011d99aa5a353d8871de
logs/v8/v7_selftest.json	5d0668fc426d17a17cb0b058f78ce919532f7003c0d8d548f0440a4bb10277ef
logs/v8/v7_tree.json	acf9699d11beaca179da32ffab88ee57366f1cbe1da8ea48aa8e7876538d5391
Replay (27 files)	
logs/v8/replay/spec_D01_CR.json	acce791a88e9204de5691ef10b411cc9809625ea5fa1cfe2afb0eb21c54b6003
logs/v8/replay/spec_D01_F.json	79a84353204e8e9e50fc91d924b4c4da91bab831d5a10e98ffe74dc118f13f03
logs/v8/replay/spec_D01_UNIT.json	5649acd95b4f21ca4a564945e3eb35cc968a051c0b6e5f29934e5384b5cbea8c
logs/v8/replay/spec_D02_CR.json	7f4a1d36ff5f1723abe33324b654b07cb5e51bde054f82c7e9235c3abe5decd50
logs/v8/replay/spec_D02_F.json	99bb7ba60647bc744d2da6007717972772910937 9e857bbf3f9f8ff466035062
logs/v8/replay/spec_D02_UNIT.json	fd11e5c3f5272aa46177f9adcd2cf56fd30b31b1f28ff2f6118f6afdc5959dbb
logs/v8/replay/spec_D03_CR.json	2bec5983c7c649fe57e29452e4819847e581d9a4ff059c7df4f50127034f14c6
logs/v8/replay/spec_D03_F.json	d3f142193c21fb2bcb45bb44692cba7cd09a59131cfc2557ee084cbf7aa23abb
logs/v8/replay/spec_D03_UNIT.json	3e1e4792fb8ec91035b5ef3c9bb6343255c46d1f099c28750a56ef3044e6f24c
logs/v8/replay/spec_D04_CR.json	a2151dc64d8159bd2183dd4fb84b9fa69b2500ccb8e959282279fae9ba2d3d71
logs/v8/replay/spec_D04_F.json	542b00df420c2d5bb8cc884b8103cc9fa705d3e60a06458bcbd0ef8f5da2ae19
logs/v8/replay/spec_D04_UNIT.json	512cdf20a2a5df3a0ef7df19acd867b5c70e65a53ca3c57b34adf14b36cb7446
logs/v8/replay/spec_D05a_CR.json	179e86ad7825cfebf3a48ab4e670b639518fbab389513f9fd220bc458e734317
logs/v8/replay/spec_D05a_F.json	4f5cbe0ae2b9a9965780a290bc3652e960c356383716794b659f2549f94e12ea
logs/v8/replay/spec_D05a_UNIT.json	ca56b30462aa4fca5a73bac77870b9f55e377f43fe9cd7afe17d1ce0ecdd105f
logs/v8/replay/spec_D06_CR.json	62ef7ca5c49060ff207c284497ff1eaf60e7214227e0164e5e8fb13a6fb0d1f0
logs/v8/replay/spec_D06_F.json	43575824758f99a429e9980f88bb9441687a209a0e7f6f4b0172a039e554ed2b
logs/v8/replay/spec_D06_UNIT.json	e0967ee852a1f35946ea020ca4afc39b83f961bcae556b0ff6f890493dd97916
logs/v8/replay/spec_D11_CR.json	05f947251f8a3d1cee8c721e998a5e431c71fe87d75abb76881aac32d99698d5
logs/v8/replay/spec_D11_F.json	a350cacd428979ff494fdc2b7d0069a12a895520420344259fee3252bac0b0a5
logs/v8/replay/spec_D11_UNIT.json	b9c70d687aee15bd7c9f53697fb04264da32cba89e4824c0eb5d1fe1400fe30a
logs/v8/replay/spec_D12_CR.json	6675f26bf142c5e549d2d4e4ed78640f30de99bbc2c8c3d57c6b5de4a19059fa
logs/v8/replay/spec_D12_F.json	356db7ea109fe8dfce4e29c5c67bf7c3a6c525bcb0fb7cb4c3e5f66b12674ee6
logs/v8/replay/spec_D12_UNIT.json	8c3d2c8aa6018bac44281d5d21aeb77cd35c53cb3663fabe1ea920da5196f6c0
logs/v8/replay/spec_D13_CR.json	794463632e6f4f6303047f8c94b12172874b60f1dce1c5d35bdfec781c6eb249
logs/v8/replay/spec_D13_F.json	ff97041eff19ac02e6009fd36f30748715f8856eaaa5dc6fef165d82ba78342f
logs/v8/replay/spec_D13_UNIT.json	d4041679d0a9c8b27237d1746a9c72dc661bba20b8c76591f8ab7a37570b7d9b
Ward scan (2 files)	
logs/v8/ward/wardscan_radial_0.85_fried.json	c802124092dc962b8fa7ccac419650eeae2a906b78b5ab5efbdd32bfa6675258
logs/v8/ward/wardscan_radial_0.85_geom.json	007709fdd124593107898379c9c318e188c6a926d67905ff2c3908d496ef57a3
Exact certificates and maps (17 files)	
logs/v8/cert/apex_cert_3_10_CR.json	001aa482da32b879365eaad7cd97e584b587895074354840f14b0ad94df53a96
logs/v8/cert/apex_cert_3_10_CR_vectors.json	a719330531cb070bea918f2f8a0b755e48e0100974756bc2aa932e779cff9319
logs/v8/cert/apex_cert_3_10_F.json	fc76e1a9d2fb95eb2256c5ee4a24353d8efce1c9e38942a67b65ca64d979f93b
logs/v8/cert/apex_cert_3_10_F_vectors.json	a719330531cb070bea918f2f8a0b755e48e0100974756bc2aa932e779cff9319
logs/v8/cert/apex_cert_7071_10000_CR.json	204c46e1b8ac59609072a060018373f67ac19e93717fbdc56b0547b863330a0b
logs/v8/cert/apex_cert_7071_10000_CR_vectors.json	a719330531cb070bea918f2f8a0b755e48e0100974756bc2aa932e779cff9319
logs/v8/cert/apex_cert_7071_10000_F.json	f3fd2f7010379d22b7236f92247f19c8a82a1eaa9947b7c7be27e9cdcff02af0
logs/v8/cert/apex_cert_7071_10000_F_vectors.json	ef66990df199ceca552c9986688ebfe82f3c1e050bedfe76b16a415ed40dc728
logs/v8/cert/exact_values.json	21aefbf917ccda4a9a70bc0bc8a1405ea2e556713bd6bd4aa104aeed757926e7
logs/v8/cert/hh_map_3_10_CR.json	ee9d20a4b630496ba5340c74e05ae16d2613d21a1b6cb05433b21f5cc1939894
logs/v8/cert/hh_map_3_10_F.json	de28306ff18ec729efd27eca9269fa6d75a2fef8839950c4f11978ae41c1c40e
logs/v8/cert/hh_map_7071_10000_CR.json	d6c9abe28830ccd985c69466ee1eca53344b69b27216ea3e8a8ef68100aebad5
logs/v8/cert/hh_map_7071_10000_F.json	e6f492d4cf6f19b6c9b75f3be96b13ac97a0382c23f8902f92261beb9f30742e
logs/v8/cert/k_support_3_10_CR.json	0e442beaef6ed47ef2e53bfc7f7f9034a83536a4cacda009f88e0fb5fbe22a45
logs/v8/cert/k_support_3_10_F.json	0e442beaef6ed47ef2e53bfc7f7f9034a83536a4cacda009f88e0fb5fbe22a45
logs/v8/cert/k_support_7071_10000_CR.json	0e442beaef6ed47ef2e53bfc7f7f9034a83536a4cacda009f88e0fb5fbe22a45
logs/v8/cert/k_support_7071_10000_F.json	0e442beaef6ed47ef2e53bfc7f7f9034a83536a4cacda009f88e0fb5fbe22a45
Exact sampled port (1 files)	
logs/v8/apex_exact/apex_7071_10000_CR_seed0.json	9b14e0b3f879676cdcd97d162f5507ff9fb9ae64026473b76c00da0d2ad8f4d1
Slice / evanescent / hidden pairs (11 files)	

File	SHA-256
logs/v8/slice/evanescent_combined.json	de6d4bf42a5477bb9c82a331d0a5d77447dee43f8b950ee1140aaf5832ef1054
logs/v8/slice/hidden_mixed.json	4f4c0cb1d3e8a0b53adcd0c94c39515f8d7b06b64f00c043de47d8a2b8731077
logs/v8/slice/slice_D01.json	0a55a654154890d9c8c50e70638d47f59340c05b8afa87842c6e9be2b492f6fb
logs/v8/slice/slice_D02.json	06e19f1c9e8f2acd0f4546c03e3b3e8a4273456c87b5304eab5551f23ef92e74
logs/v8/slice/slice_D03.json	c7eb6a6efcfc389f9d2439f2472b62b47ef0ef6201deff6f4ef92d968acf418d
logs/v8/slice/slice_D04.json	cc23ad207c516682715544fbafe2434dce83e84785d19e8383adeab0c1a833b7
logs/v8/slice/slice_D05a.json	5a6b5c35ae58f9197a75128b7e002ebee0a8688682b0d4e8f8414687861fa897
logs/v8/slice/slice_D06.json	cf0bcdad0d20b76076fcf327fafbbe5c5f2a6a568537e98fd7d4383e64d09786
logs/v8/slice/slice_D11.json	f4afed2f0745d95d685294001dab6e0099720733a6e624df8a5971336fcd5656
logs/v8/slice/slice_D12.json	acfa367bac579ab8f18c2bd731b3b3550a83aa1862722a4da2250774cdef88be
logs/v8/slice/slice_D13.json	eeb384f033363738b9bcd5d0160ee3d69299c8426c2c2fb35ee9b62d628226c6
Duhamel corroboration (2 files)	
logs/v8/duhamel_mode_check.json	2931da69cbc9873704391c64ac13299b41666f1ad59eeaeb67ac4eab63eb86db
logs/v8/duhamel_mode_check2.json	74bc8b04bc98559da3c5e3521d6b96d46e0056d5d09bd411c025b02c56b5fae7
Manifests, environment (5 files)	
logs/v8/manifest_XVIP.json	8dcd372a5977065d7f30e9698917c9b371faf5caa111a84bd9d27a2e4c2714a6
logs/v8/manifest_XVIQ.json	c33ef1abe8d6f63a28885a7d41bc70d27455cf45eba06a94697270d5cd0ab337
logs/v8/manifest_XVIR.json	44f17586766b4c13c088080143a99864b397dbc1893a7b39422855e88876fa17
logs/v8/tree_hashes_v8.json	a6b3246b70cf27eed7eb7d2f7f475903fbb8d1cd7199283ed1b90b5f42bceb8f
logs/v8/env.json	4a40f99a724e190dad2fa6e65c49b318415204377fa0723aa71128884aed714e

C v9 artifacts, hashes and rerun commands

Paths are relative to the project directory `XVI-01-CONE/` under `/workspace`; `intake/` abbreviates the companion intake directory `XVI-R-companion-intake/`. Long jobs were run detached with `setsid`, logged, and made resumable with per-item JSON caches, because the box runs out of memory at about 7 GB and is restored from time to time. Environment: Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0, `LC_ALL=C.UTF-8`. Frozen trees before = after: XVI-P: yes, XVI-Q: yes, XVI-R: yes. The v8 source and PDF are unchanged: yes. The modular scan comprises 192 result files, which are hashed individually in `logs/v9/scan/SHA256SUMS`. The full manifest is `logs/v9/artifact_manifest_v9.sha256`, and the TEX source and PDF are hashed in `SHA256SUMS_v9`. `code/v9/mixed02.py` is kept only because it produced the asymptotic coefficients. Its ${}_3F_2$ shortcut is unreliable at large parameter and is not used for any printed value.

Rerun commands.

```
cd /workspace/XVI-01-CONE
python3 code/v9/g_formula_check.py
python3 code/v9/mixed02.py   # -> logs/v9/asym02.json (~250 s)
setsid nohup python3 code/v9/hidden02_run.py \
   > logs/v9/hidden/run.log 2>&1 &
python3 code/v9/cbck_map.py 3/10 CR   # and 3/10 F, 7071/10000 CR|F
python3 code/v9/ck_free.py 3/10 CR    # and the other three points
python3 code/v9/e_solve.py
python3 code/v9/chamber_scan.py       # resumable cache
setsid nohup bash code/v9/scan_driver.sh \
   > logs/v9/scan/driver.log 2>&1 &
setsid nohup bash code/v9/scan_driver_rev.sh \
   > logs/v9/scan/driver_rev.log 2>&1 &
python3 code/v9/evan_match.py
python3 code/v9/vector_logs.py
python3 code/artifacts_v9.py && python3 code/build_v9.py
```

```
pdflatex -halt-on-error XVI-O1-CONE-v9.tex   # twice
```

Table 19: SHA-256 of v9 artifacts.

File	SHA-256
Base source (read-only) (1 files)	
XVI-O1-CONE-v8.tex	153e4e9dbcf116a138d0f3919607bed1f3646b1a8437f8e5cdbcc5dd686e85f8
v9 programs (15 files)	
code/v9/apex_scan.py	9e03be0cce4c825d42b995384f40846fea9fbac73677acb5e2dfb11edef36529
code/v9/cbck_map.py	68539cf07ee63910f5ddab8e3a309ce9a305070b69d9448cb3bacbe43123600e
code/v9/chamber_scan.py	499d77465702a3d471325b0abf81324a4545dd57a0f76bd775b7504aa917fe24
code/v9/ck_free.py	8cffb01060ba5899e3f1a4055dd0ecf4c1d8eefee915033f656135fe63ea79bf
code/v9/e_solve.py	c29ad9480e827430f572ef433425ed1123cd86fd98ce93d654d68142287be119
code/v9/evan_match.py	260a30e2fa25ba065f9ec141bb104e1b6f3ef018a381f74ebc1db55cf0e28005
code/v9/g_formula_check.py	1661f1bc5ec9c3aa4e66e5415630761390e3790e2d92ccb36f4534a284c29def
code/v9/hidden02_run.py	9bcafe3ad4254c68bfdaa95d9d11f0e6db8fc235bdbda39829c5c1668f130487
code/v9/mixed02.py	808a312ab7bfacb834e58e3d1b8988178e220845df162fd2014c19bd483d2530
code/v9/modefun.py	451c91de56316ab65955d0d439cf8f0ed40d63c1431d7e5dcea12149a050689c
code/v9/vector_logs.py	cd8a433208136e21180d8f14ed7a7e67e3cf31c5be9718192fca17c0c0973064
code/v9/scan_driver.sh	fbc160089288c70e1a201a9c63481cd74bc13c35b20fbbf27876c785c526851f
code/v9/scan_driver_rev.sh	e062603c4cdcd258400c4f932ff4583ce2bc1024bd3a75f6c81b2fbe839280dd
code/build_v9.py	6449baa74d39ee947f6c537f7adb3aa0d14565b422ee81362f818b7ac430af71
code/artifacts_v9.py	c5e51e8aa1ab4acaa8918e940872c9440abf81e9134a778867820cf40fd480e3
v9 source fragments (4 files)	
v9src/abstract.tex	1367810f9f3f59426a4c29b553724f4955113b48b99c71b140aab7e7e5142ea5
v9src/bib_v9.tex	0fc845a1e879e4863832c91b1c0e41d42068ba1507cdaaa0d32fb73d3734733b
v9src/opentable.tex	5cf973cdb2281f71777c962af3b6056e948ee9dfdbea28b9f2bc7bf2a4c6e477
v9src/sec_v9.tex	362ce962df3d6d5f29c9a547ee3d5da7da4c8a3a22b7d37981750806b1396cdb
Mixed trace and hidden pairs (16 files)	
logs/v9/g_formula_check.json	00a65b245b5217c9c77c05598404dc37334b218e5613aa2088ea74ad44f280aa
logs/v9/asym02.json	a3cc63868cdb683638e305f62dfca309e33de9e1046e65f70b6e49613a909abe
logs/v9/hidden/dJ_11_20.json	6c3d9cea71400827e43587ddd37bec984243ddbb35648662b0f5c5f4e7d12fd8
logs/v9/hidden/dJ_17_20.json	512aa07e728aeec0f47b10be0f4860a00d25d9021e0200ce67fd846ef9b89bf3
logs/v9/hidden/dJ_19_20.json	e7abcfd32ff1e0e0894484b05e13277f8340c2d0c5b1ef88b4799e8e71c1dce6
logs/v9/hidden/dJ_1_2.json	6e2ba8202b076ce5ec92c54e284c3ce7cfdbf2210ceff6c4a772dc4aac78bc25
logs/v9/hidden/dJ_2_3.json	761940238265555748db3130ed6f04ae20c33000936c38ec69c84b1336ea31ed9
logs/v9/hidden/dJ_2_5.json	601caae331eaf8283387fed0336e7fcdc0308dbb8211ccf2fa6d723c437eb573
logs/v9/hidden/dJ_3_4.json	3a0eb389c6a9337fb0b455aec786aae697edb21e760d87378ddfd808228b67e8
logs/v9/hidden/dJ_3_5.json	ece5ba7a75d11370982406ce2f0c3385d754b21308857c409a0e85dec28117e1
logs/v9/hidden/dJ_4_5.json	b9693cb7179baba41b508894675c27736167ee0ba694b70b975fbc9d999fcad8
logs/v9/hidden/dJ_7071_10000.json	29acf5b74f782ea5708a880eb9081d12a5ad9831da748428bdc749e6e1fac245
logs/v9/hidden/dJ_7_20.json	5b99ee97bac9315bb96959dd3de3c3ce3b34c734fddf169bb30550fd96b01fc7
logs/v9/hidden/dJ_99_100.json	fe6dcf8f07d79fb333b66b44f9898fd2335fd2820b36405f8c4ddc7bae5fabd9
logs/v9/hidden/dJ_9_10.json	33f5d6897ea5a3dd20f58437155b0b7bb62734038648fcf95ef17b11dabaa621
logs/v9/hidden/dJ_9_20.json	7ae4b36c77cbdae96e3d5d22d89f48ca882552947b263fb515d7459a6c330668
Ghost dictionary and exact values (9 files)	
logs/v9/cbck_map_3_10_CR.json	265f5928ebc1c396f0c7a55e7d5b115631966c95b9f12fef38bf61f5e2cddff9
logs/v9/cbck_map_3_10_F.json	0f5e005e2284889ca69c165ca2e225656c4e7c0cf47b41f2449ad129dab30e4d
logs/v9/cbck_map_7071_10000_CR.json	8c8f3fb53f10d9160c091206de68e8a1a7dca7544ae87a4a433e6e6b805fdf5b
logs/v9/cbck_map_7071_10000_F.json	c81a43ba76ad1387775b36a1d317e23bb6c4b030ded84ced361f754ac3cc5831
logs/v9/ck_free_3_10_CR.json	5fceb30dbe2e706dff67e8b717d44a8367b5114da9ed7e6d17907de73c55002b
logs/v9/ck_free_3_10_F.json	689499331b250796e9f48a458a928f0f379ccf2b9840ebf114a1d395599d8e98
logs/v9/ck_free_7071_10000_CR.json	4a2dbdb45f12039736b80afd150f92cb952148c6bf3ae367756e89121a4091b3
logs/v9/ck_free_7071_10000_F.json	ba3b72e7cd1718401711bca6ce8893b3fe67710e0b2d86e187c2230eb152c404
logs/v9/e_solve.json	99c4b03a828c4ec0560ee2178b11500293e2c9a283ec516d59e50ca974b8d8ee
Chambers and modular scan (3 files)	
logs/v9/chamber_cache2.json	8058faead811c4c2fd3d5cba97de65070dcb27c08447d22a2f62f2963e24e393

File	SHA-256
logs/v9/scan/SHA256SUMS	2f17bf26b9a0f3f989267752da85a21ddc8d23ae406f81943417d5ad034da4e5
logs/v9/scan/alphas.txt	fbdd209b93ebf0cd45dc65800bba9b24f7be32bc38ce5b1cfa0c0190e12a3239
Obstruction checks (2 files)	
logs/v9/evan_match.json	1a273d335cbf50dc9a264ba4bed7dcc4edd6f6c279e799f810ae3a579a6dca81
logs/v9/vector_logs.json	7531cd19399da49e3952ffcaaa70e76a96b70eb4bfa775b0395f4603ec22a6da
Registry (3 files)	
logs/v9/sheets_extract.txt	de32592311345e01e242f29888b51b1f4e55e2f78c9b0714391cdfc3fbe6d1b6
logs/v9/sheet_ICEx2_extract.txt	b2598ce64a6f6847598fe4896004ca6ad15a9a4263b6c7cc0e390e068e91635d
intake/new_oct18g/SHA256SUMS	4911a83afa44259453723f4a6b81da4b20fad1462642a0bcf8b3ccfbf9249267
Integrity, environment (7 files)	
logs/frozen_v9_before.sha	cb9c2a11921e376aa078f2081cea58b7c0a8a0363a1fd6976bf7272df89e7b5b
logs/v9/tree_hashes_v9.json	a2358223bf1146b2f118651cb82aad0c6721553eb0a64f91a3b5a80110a02ea7
logs/v9/v8_hashes_before.sha	3dd87be0833964d8d5d9302a2cecca67f0df333d207b829eb57c154cbfe8e5ba
logs/v9/v8_hashes_after.sha	3dd87be0833964d8d5d9302a2cecca67f0df333d207b829eb57c154cbfe8e5ba
logs/v9/earlier_versions_sha.json	7ea1fb72a1f61af6ec47b14a3e0441d2d13c65eefde6c48494ce519115431399
logs/v9/env.json	5633c0d1b2c59fd04977cd8dbe7742a0417a910c10290567be167e4de639c425
CHECKPOINT_v9.md	cf3b80bfdbb3340176ac0d147d1da6eb43a95238f5dbb550eedd890003e40df9

D v10 artifacts, hashes and rerun commands

Paths are relative to `/workspace/XVI-01-CONE/`. Long jobs ran detached with `setsid`, were logged, and were resumable through per-item JSON caches. Environment: Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0. Frozen trees before = after: XVI-P: yes, XVI-Q: yes, XVI-R: yes. Digests: XVI-P `ec965c15ae25cd3b...`, XVI-Q `0df9a0d10409d31a...`, XVI-R `57187bc6825f5c6a...`. The v9 source and PDF are unchanged: yes. The v1–v8 sources are unchanged: yes. The cached φ values (20 files) and the minor evaluations (106 files) are hashed in `logs/v10/jet/phi/SHA256SUMS` and `logs/v10/minor/SHA256SUMS`. The full manifest is `logs/v10/artifact_manifest_v10.sha256`; the TEX source and PDF are hashed in `SHA256SUMS_v10`. The large-n φ runs were stopped before completion; they feed no printed value.

Rerun commands.

```
cd /workspace/XVI-01-CONE
python3 code/v10/cr_mode_closed.py
python3 code/v10/f_mode_pslq.py
python3 code/v10/dJ_closed_verify.py
python3 code/v10/lambda_derive.py
setsid nohup bash code/v10/minor_driver.sh \
   > logs/v10/minor/driver.log 2>&1 &
python3 code/v10/minor_entries.py
python3 code/v10/cocycle9.py F_c4 7071/10000 7/10 3/4 \
   4/5 5/6 7/8 9/10 19/20   # F_c3 likewise
setsid nohup bash code/v10/jet_driver.sh > /dev/null 2>&1 &
python3 code/v10/jet_analyse.py; python3 code/v10/r2_scheme.py
python3 code/v10/t1_fp.py; python3 code/v10/t1_fp_check.py
python3 code/v10/jet_density_D04.py
setsid nohup bash code/v10/phi_driver.sh > /dev/null 2>&1 &
setsid nohup bash code/v10/vector_driver.sh > /dev/null 2>&1 &
python3 code/v10/vector_combine.py
python3 code/artifacts_v10.py && python3 code/build_v10.py
pdflatex -halt-on-error XVI-01-CONE-v10.tex   # twice
```

Table 20: SHA-256 of v10 artifacts.

File	SHA-256
Base source (read-only) (1 files)	
`XVI-O1-CONE-v9.tex`	`3a8dd43489cf29b96d953a544e5a6758feac16c307449f738ea235ce41c6f1b0`
v10 programs (26 files)	
`code/v10/cocycle9.py`	`c6d73ccf092b90a9afda8ba3b65b26ca1a387d746ad57dafc44c8e650c7f7543`
`code/v10/cr_mode_closed.py`	`f9339e76d370390523fc7213f21430cb5b2d4a7ba10f96841069836650ca78df`
`code/v10/dJ_closed_verify.py`	`26d96b59e31da187cc61c29919aab675fc4885b0364954a9e7a2c5bea51a0690`
`code/v10/f_mode_pslq.py`	`b99327b9686e07428c15037508931f3776acb4b50853eb17f6c0d3a39a8bfe77`
`code/v10/jet_analyse.py`	`f7618580da4559d7fa92e4aeecb5489dd910347cbf937248f538aa5a3528bfe7`
`code/v10/jet_density_D04.py`	`d25ea7a43495f6dfad1ce3b03542ee671d720ada7617ba472ffdcfbd40462bfd`
`code/v10/jet_replay.py`	`9b93ab0600ed67a6296b8e3374cad4cebf7e9bcf9c328f9cb14664ed454fcfde`
`code/v10/lambda_derive.py`	`94af1831e985bfdfc288cbed8e27e0d11c3e16b8be7e26e5ffb5e5b485908d98`
`code/v10/minor_entries.py`	`85655f6a8fd58dd70aa5cfe3ee736e51a5d3932c25804be9612d1f6900ee6043`
`code/v10/minor_poly.py`	`06456aa92632b8d75ab673850d83baccb856b38c994b632ecf1d49ef2fcbf640`
`code/v10/mixed_r2.py`	`a2776419626cc47100505424cccabd3dfa966c75d4174ac52a461bc699fcb60b`
`code/v10/mixed_r2_check.py`	`67a2856a2c8e8f2830f558dc6325d99aba81e0c5f50c802fa39924d498fac97d`
`code/v10/phi_fit.py`	`24ce224a5e9f4079f022f518e31af7061c5657eab361660fef100a80b835094f`
`code/v10/phi_mixed.py`	`b51174ecd2ae1216caf550b0d9e02a31e1c1d7cd634bdeca62227f8ab81f88ef`
`code/v10/r2_scheme.py`	`cd5360f41f048b3ad5c964f15d2fda9aca84e37ff8c2f08df1783a51b257a213`
`code/v10/t1_fp.py`	`8db8da98a37588fc30854ccfe04a6ed3c59b663108e157b9432c4435f3648a06`
`code/v10/t1_fp_check.py`	`4fcd4587a4d8a2447cedc2de8a88743232e852f591efed7485d8fd10419fed0d`
`code/v10/vector_combine.py`	`7c52024edbc55c226720c62e362d4028065fae39035803aff8d9a6771168e280`
`code/v10/vector_fp.py`	`1748e39a63b7376d0277a19d58717b7886046e7bf75dfe597ae50201f790cd08`
`code/v10/jet_driver.sh`	`4711578a57f8540d893b633d86decc55f2106523d06fe466881149cba8cd6b88`
`code/v10/minor_driver.sh`	`5489cf621510e8d23c22e9ae3a31ecc1b35f509f3e6db3664322a8bfdd1621e3`
`code/v10/phi_driver.sh`	`685ef1dcd00c971fa85ccf04769995ed13d629c0ca5e941c1c68ad8efd0abfda`
`code/v10/phi_driver2.sh`	`d06e442cc222ef8c43cc8d750956c062e28d8d16140e805e3bcbda24e665b163`
`code/v10/vector_driver.sh`	`dee6090a4b399f9193563d35e942830728c95b992a42a22a8bfb2175c955ef86`
`code/build_v10.py`	`75cb95c20748413a366029c40b9f0280cc90a425981fc304b80a7b7847e6c7f4`
`code/artifacts_v10.py`	`8de0cad6e1d9ad59cee1ee0b640ffdca7dcc9385e0c0660f305a8dcf0429932b`
v10 source fragments (4 files)	
`v10src/abstract.tex`	`6d4a42db29b130adfc18702497cf6eef0e8d654cc821c08d1549ea5cf7be0db3`
`v10src/bib_v10.tex`	`c648846e7462ed90a7a0907bc31bc748f7e95dd0358373e8342bf906372ebb7a`
`v10src/opentable.tex`	`060a3ade76c775984acbf220242f76498caca62a66829780e87951d6b7c316a6`
`v10src/sec_v10.tex`	`770e85953af1f47e47a387ffcf22156c4e2c737075530616b68a76555c63af4a`
Delta J closed form (11 files)	
`logs/v10/cr_mode_closed.json`	`939528c5aaf52273d12f32a570dc6ceed917a0ceb21988848f59c3fc5e265ad8`
`logs/v10/dJ_closed_verify.json`	`5159c9163551c83a17161d1814e7dff061d9d3448cd65a0f88e8712d036a6a59`
`logs/v10/fmode/3_2.json`	`25daa838dce8ce663f92ab61cdc38b49b5f2b6f61355977416734d1647222181`
`logs/v10/fmode/4_3.json`	`aca72774baf4e99f3855d1e5c2ac85118b7c50c7274f5f9c16e3e13f7dad9017`
`logs/v10/fmode/5_3.json`	`9dbd7ad598b5444481965b76ea81cf638c726376a074bc7ab57ea3a9ae2679cf`
`logs/v10/fmode/5_4.json`	`ab984d9a4a13b211ece98e64fc6be0b03b89cdefac36cd365ee9e0eb8ae540f6`
`logs/v10/fmode/6_5.json`	`ce0cc18323063200588745de47548e0aaa952dbee89ad04b0f1eb808475190f2`
`logs/v10/fmode/7_4.json`	`37e213fb00f05380f2d11d1c28d9711f75249261cdf9b86c76cd2c7f6becd092`
`logs/v10/fmode/7_5.json`	`445b722bd9d3f3729ccee37269bbe561386cff57c4be70da19217c3dae2d4945`
`logs/v10/fmode/8_5.json`	`4b464ff79c117543f91134fab5b185d5700a1fed5f10e941fad5397689bd5360`
`logs/v10/fmode/9_5.json`	`cb11bbd8860cbf9a186a701d2d870900f7a706ed7eb0eb71e106d50788ba2aef`
Ghost normalizations (1 files)	
`logs/v10/lambda_derive.json`	`b15ccf4c87d1d0dc0a27fea25af189ee441684f0ad06030f46ce24a7832decaf`
Chamber minors (19 files)	
`logs/v10/minor/CR_c1/pick.json`	`ef8ed9fdf3a5b5bfb1d57321bf2e431b6d0a058c62046811fbc29c2f0e86dc13`
`logs/v10/minor/CR_c2/pick.json`	`d6b21d7a8209f9e3ce179b7dbe1fbc20a3d8c4f374c9ff951ae27ecad084596f`
`logs/v10/minor/CR_c3/pick.json`	`a9fbb9bb784a70b5818a6423122f18877cb6719c4c0f6427c6d18e6dd73c8448`
`logs/v10/minor/CR_c4/pick.json`	`ff43cbf02a89e2081ce558e7c2b2c9d3f403c6621383926827a4ca0ae51772c6`
`logs/v10/minor/F_c1/pick.json`	`bd008573b6d427d752ae6136c5ad44f721ccd35f6315c21e85b44c597c003f67`
`logs/v10/minor/F_c2/pick.json`	`ade2b067e5fe3da91fc1ececd3ede26d86c6857f47ca814eb41c74d8a894b162`

File	SHA-256
logs/v10/minor/F_c3/pick.json	cb90901af66560cbdd497ceb8b0077ca7af89e4abd4e9cff54476d39e4851a2a
logs/v10/minor/F_c4/pick.json	a4af66a3b268f73ac168bb4613f584ba818ccf94087f448a015bcb4740d22c4b
logs/v10/minor/CR_c1/entries_check.json	761c25903f00a6612b18e786ee39f7e674e68abfeea8ab9162fcef35211e7c8f
logs/v10/minor/CR_c2/entries_check.json	348bc38ecc754a39a22dab2434f9dfa8f9508a4d967659b6e5e101fe4ae61385
logs/v10/minor/CR_c3/entries_check.json	1c4cae65bd188f4015c057a43ee109481089a79173cc270743a9997275f1b516
logs/v10/minor/CR_c4/entries_check.json	df135b94bd6e67cff2391c7cc666d80e138925aee88341abd72edbe25f7e606b
logs/v10/minor/F_c1/entries_check.json	aa2e6d90dd5e394a2b71bdf02aa82b8aafb361c9de26dd9188f7d250b35c24eb
logs/v10/minor/F_c2/entries_check.json	807c5572509ec6966b278ba622e105b56cc0777b7cbd912cb03badaaf227e9db
logs/v10/minor/F_c3/entries_check.json	f8fb788a64060d68b0a3c0b5123f5b26ea03cab8781715ed6ff6fbd3ff2d9f73
logs/v10/minor/F_c4/entries_check.json	a310782e9ef118d07c13f07d7dcc2b36142ab5e33726f2fed036fd51464dfeeb
logs/v10/minor/F_c3/cocycle9.json	377c62498a429585ea9f1b6629697fb0d9c9114f3344c211758df947165644d9
logs/v10/minor/F_c4/cocycle9.json	fbf314449c9bc32240477f4427087165efbcc7df931c4f29f1da60f099750a8d
logs/v10/minor/SHA256SUMS	ed4653f9ecb3ac7c2443db90827bb1f73af34e6ed720b8e0caad22182311ef6d
Transverse jets (17 files)	
logs/v10/jet/jet_D01.json	ffed1138c1f2fb5b554c865593a5e48da10d19ee6900922f17d21538a21457df
logs/v10/jet/jet_D02.json	35643ee8d7aa9459d629a8d19d8ab38318d6ce68a39ddc643ed8f9fdd874e11f
logs/v10/jet/jet_D03.json	531ba4cfa9741a5d999c19da05c1b09c453f6c61c8d0f7cceb0c91ca88caff00
logs/v10/jet/jet_D04.json	4c3d5a5460806bb341648bf9b9f367fa2faeb70837294238a80d8f38145a8d89
logs/v10/jet/jet_D05a.json	c695b8be7cdf20cb2d4d71ccfb524bffc5bf5a531056de7e959041736ab738bc
logs/v10/jet/jet_D06.json	04cebbf0bdc3eb40a9e254a8ab430026ff3b3b88773bbec70efcd00224adda6a
logs/v10/jet/jet_D11.json	7b68328a09b92eb169da152c5e9557be7e09f923798c05f50efd8ef9524ff7b6
logs/v10/jet/jet_D12.json	934e721f7e9ff41029c74b299aeba4e83960d42e02097a29e1bfeb64881d44ef
logs/v10/jet/jet_D13.json	f856bcfff7707c4ca5eccbf26aac233c79e7597aead51b1acec8d7dc7249cc88
logs/v10/jet/jet_analysis.json	5daf13a48ff23cf576f8f4b8f1d02d7524ea52163b14dc3d08615103bf0fad08
logs/v10/jet/r2_scheme.json	7f6492a3ee94d407ddcaa10fffa39b6d52462aa699274decab531a24b4112131
logs/v10/jet/t1_fp_check.json	5beac73b57965a11d7ebe68397bb32d1cbbeb84a5866955f53c24fb764b4d168
logs/v10/jet/t1_fp_same.json	9b9385cdae53f49af5e845e24e1003cfa3d0e9e3e4c9f1655d6e0dcd4f6bed8b
logs/v10/jet/mixed_r2.json	b17bd5d3f8788e573c7c4e9cd5b8c283746a129fb18a4d200cbcc7844bd03675
logs/v10/jet/mixed_r2_check.json	3416ad37869d05fcbc93748e3cb1b8037bc6f72356900dcdbd66b99288e4997b
logs/v10/jet/density_D04.json	f8e9f35864582aec6338493bd81b1c6e66308cac09d0f76a77f03f9e8b14336f
logs/v10/jet/phi/SHA256SUMS	a65d81990623d6593aa653c3e4d7219596865e45245fed47ccc2df874a7b8c41
Vector finite parts (5 files)	
logs/v10/vector_fp/3_10.json	911f52f839f4be0cd2f80bb7b2c2df76b31810accf3727e3927f5087883ebcc4
logs/v10/vector_fp/3_5.json	c138cb911e3e051a2b1013eb4c5273f3af486875c28277f50538621fc6be60bb
logs/v10/vector_fp/41_100.json	43619f5aa07f5362d8d9531b2c6405d16b25c7fc7887259a07711e13ed21ce17
logs/v10/vector_fp/7071_10000.json	2fb16311db70b19ce6b1cba7646c9263ec444348340c256e4294e345de7c791f
logs/v10/vector_fp/combined.json	920a24127826fda08054402b1255615c7f7bc31c0a63a7fba7c7f020fafc24c1
Integrity, environment (6 files)	
logs/frozen_v10_before.sha	cb9c2a11921e376aa078f2081cea58b7c0a8a0363a1fd6976bf7272df89e7b5b
logs/v10/tree_hashes_v10.json	ddf510d823e0af9de28430bdf7be42309eaab04b1d6b9a6c87b39b8e9735626d
logs/v9_hashes_before_v10.sha	97840e3bd1107e61deb848ac53de97a8aa68275b16517f2b17579b465df2dc74
logs/v10/earlier_versions_sha.json	c9d46ce4bbae98a1c22a08a7f91ba24f3ac242f054ee673fbe6d0a7286bbc43c
logs/v10/env.json	f3c3b679dd87e7e87a4a5670b7e7dede9a62e737459a7a2a21d8fc562df914fa
CHECKPOINT_v10.md	4c01e1e654d8f96797988e7208517f82e7259ce2d5bf774b07b863924c3ee04c

E v11 artifacts, hashes and rerun commands

Paths are relative to `/workspace/XVI-01-CONE/`. Long jobs ran detached with `setsid`, were logged, and were resumable through per-item JSON caches. Environment: Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0. Frozen trees before = after: XVI-P: yes, XVI-Q: yes, XVI-R: yes. Digests: XVI-P `ec965c15ae25cd3b...`, XVI-Q `0df9a0d10409d31a...`, XVI-R `57187bc6825f5c6a...`. The v9 and v10 sources and PDFs are unchanged: yes. The v1–v8 sources are unchanged: yes. The cached $F(\mu, \nu)$ values (404 files) are hashed in `logs/v11/M10/cache_SHA256SUMS`; the superseded first run (Section 33) is kept in `logs/v11/M10/cache_superseded/` and feeds no printed value. The full manifest is `logs/v11/artifact_manifest_v11.sha256`; the

TEX source and PDF are hashed in `SHA256SUMS_v11`.

Manifest fix. The v10 manifest hashed `CHECKPOINT_v10.md` one append before its final state (`4c01e1e654d8f967...`); its current hash is `8bb24346c6f7d991...` (`logs/v11/checkpoint_v10_fix.json`). In v11 the last append to `CHECKPOINT_v11.md` precedes this hashing, and nothing is appended afterwards.

Rerun commands.

```
cd /workspace/XVI-01-CONE
python3 code/v11/degree_check.py; python3 code/v11/degree_crosscheck.py
python3 code/v11/fmode_reduce.py; python3 code/v11/fmode_derive.py
python3 code/v11/m10_jobs.py > logs/v11/M10/jobs.txt
setsid nohup bash code/v11/m10_driver.sh > /dev/null 2>&1 &
python3 code/v11/m10_sum.py
python3 code/artifacts_v11.py && python3 code/build_v11.py
pdflatex -halt-on-error XVI-01-CONE-v11.tex   # until stable
```

Table 21: SHA-256 of v11 artifacts.

File	SHA-256
Base source (read-only) (1 files)	
`XVI-01-CONE-v10.tex`	`d21c8d84b12b53e50a78227d86dbff204293748d7712ced02e11b08b983b8d25`
v11 programs (12 files)	
`code/v11/degree_check.py`	`c5c0c48ef4e5f2750235f8da59daa62e157d177250c556b275b6dfbfa93d7362`
`code/v11/degree_crosscheck.py`	`09139a504eaf3af98adeacfa3a8ae344455cbf6c133b58b4604d2e73e47d505e`
`code/v11/fmode_derive.py`	`d16e4f844f54237fde609d74360fb7d81a1169e98b71cfbe96cf378423866a5f`
`code/v11/fmode_reduce.py`	`1bcf052a6baf0894b8b14cdd765d88a4933314cd8f595f26c86212c0f45b839c`
`code/v11/m10_eval.py`	`d01c93f6689f8844efcf5acfca47f7ccaef33325b749b916e96ecb3f60ef904c`
`code/v11/m10_jobs.py`	`43f70dfbc390768faaf2573f5c8f4575b3373ea83ac2cca06e4431466e083b25`
`code/v11/m10_post.py`	`5b8a6aab7f5d46d4e913b70ae6d93702911326c99c6a31416c290b987189b7d9`
`code/v11/m10_sum.py`	`1595ff5aa5ed8e3fe300e71ffaf4514c680cdd3ea44e00ba0ece13c33d5abce3`
`code/v11/t1v11.py`	`4660ac0960ab9dfe2e92a01ce2c7900aabb9c295834e60306965f4abdb167dd1`
`code/v11/m10_driver.sh`	`627f6ff0aa57b4e6aed21f248af15e5e5dec784eea05f9105377da2558756596`
`code/build_v11.py`	`a71bde039d87789829333866ab7a3bf31db9d1fa966dfd1500936e629c7bd3fc`
`code/artifacts_v11.py`	`8645d9bbfd275047c2a149914f9efc5f94511284f1147916d79a91a3e15385ae`
v11 source fragments (4 files)	
`v11src/abstract.tex`	`82c1899a7e301fab8022a089f012d160b59576044a6294d86f0918b588453435`
`v11src/bib_v11.tex`	`60374aa56074952db5b9c95fb5965153933e35644ab481052613996402eaa4f9`
`v11src/opentable.tex`	`7451352004052c4ef79a1736ea0872691ea47162ac07312215aefec1e0655205`
`v11src/sec_v11.tex`	`7431c14dfd2eb064a6738b10fbe11f65cd7a9190a25bed923f0414663add9685`
F mode derivation (2 files)	
`logs/v11/fmode_reduce.json`	`6289543dec1700c1b7944344756742677dc6a7e21cd417ed9f03de1e76a46dbc`
`logs/v11/fmode_derive.json`	`9b897c1cdd37263e49544181583dc633f2c9c3e365df8da26b0065a3c26402f6`
Degree bound (D) (2 files)	
`logs/v11/degree_check.json`	`229f265c1dd566a0b02ccf5173434762b70ae03f162a162680bf45dbe299d35a`
`logs/v11/degree_crosscheck.json`	`2901c5d65f674a6a5124a8f600061bd63e0a1efb17ebfad5d943fdef9c985641`
Mode sum M10 (9 files)	
`logs/v11/M10/jobs.txt`	`5378b8e118373db999c3f5611d7054ef425cc2c5e6a875a7b1178b8034010829`
`logs/v11/M10/m10.json`	`f58b460f79bc0fbfbbf7196be0e923585586bdaa47ad67058684c0651071980b`
`logs/v11/M10/cache_SHA256SUMS`	`4e63c1d913104cd121a044955d1dd4670371ffb88c5ae4f64ee0ec10e1818c71`
`logs/v11/M10/phi_corrected.json`	`cd4cc6653e3ef9b4385e4f8d8323efa578ebfe71055c70e1984167424ef06558`
`logs/v11/M10/m10_text.tex`	`a1a666abe117a79edc09c2ed15c5625a8e3e356b52e89345dfafbc9e2f75c3ee`
`logs/v11/M10/jobs_dedup.txt`	`72b5560949a87642c8a44ebb56044881a7c31961cbfc6bdff99f0fc88560c561`
`logs/v11/M10/m10_leading.json`	`173abaf669a892c4d963bb16080299f95c9f7a55a46ce96c878b04ff85b9e7ad`
`logs/v11/M10/m10_pslq.json`	`2e35be6bc699295dc53dda73fb79bec1efcb4dabf51b2d8b49333b3f362627e4`

File	SHA-256
logs/v11/M10/m10_sanity.json	081018c7db3a61b81ed5ac4e47ece9f3e490a1cc49e926dd70872cc4550e7e49
Registry (1 files)	
logs/v11/registry_oct19.sha	ba94230ae35fbb1c4456568f0c3787d01310ef19195d5373d5821c5309d91538
Integrity, environment (7 files)	
logs/frozen_v11_before.sha	cb9c2a11921e376aa078f2081cea58b7c0a8a0363a1fd6976bf7272df89e7b5b
logs/v11/tree_hashes_v11.json	ddf510d823e0af9de28430bdf7be42309eaab04b1d6b9a6c87b39b8e9735626d
logs/v10_hashes_before_v11.sha	1ca9d9369867da38fee1e2458b4e2d94fe277505ab39c3f84392c7b7985ee63a
logs/v11/earlier_versions_sha.json	111da13ad1339049d8fcaeaecf12373f70553db4f0cbe749e707a77aef0e7523
logs/v11/checkpoint_v10_fix.json	cd7ab2d3abbebc904e1d32fbdfa78f0e4418ab42884ceba36fda28abb86c6860
logs/v11/env.json	f3c3b679dd87e7e87a4a5670b7e7dede9a62e737459a7a2a21d8fc562df914fa
CHECKPOINT_v11.md	0bdb0d051f8b5ad8acb8effd0f8a5c357d86b2357a6fd3738eb1044cb530447c

F v12 artifacts, hashes and rerun commands

Paths are relative to /workspace/XVI-O1-CONE/. Long jobs ran detached with setsid and were logged. Environment: Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0. Frozen trees before = after: XVI-P: yes, XVI-Q: yes, XVI-R: yes. Digests: XVI-P ec965c15ae25cd3b..., XVI-Q 0df9a0d10409d31a..., XVI-R 57187bc6825f5c6a.... The v9, v10 and v11 sources and PDFs are unchanged: yes. The v1–v8 sources are unchanged: yes. The full manifest is logs/v12/artifact_manifest_v12.sha256; the TEX source and PDF are hashed in SHA256SUMS_v12. The last append to CHECKPOINT_v12.md precedes this hashing.

Rerun commands.

```
cd /workspace/XVI-O1-CONE
python3 code/v12/phi_asym_fit.py; python3 code/v12/structure_test.py
python3 code/v12/F_closed.py
python3 code/v12/m10_exact.py 40; python3 code/v12/counterterm.py
python3 code/v12/m10_pslq.py
setsid nohup bash code/v12/walls_driver.sh > /dev/null 2>&1 &
python3 code/v12/make_tables.py
python3 code/artifacts_v12.py && python3 code/build_v12.py
pdflatex -halt-on-error XVI-O1-CONE-v12.tex   # until stable
```

Table 22: SHA-256 of v12 artifacts.

File	SHA-256
Base source (read-only) (1 files)	
XVI-O1-CONE-v11.tex	863bb08efc04416464d5628041d98ec7478015315b4c804cacc6b260181f203e
v12 programs (11 files)	
code/v12/F_closed.py	0456cb703049723257452addde7e3783799d113deca58085b661b7a0c419ce00
code/v12/counterterm.py	7342f223780372c078030725bc1d11960f119b4615c340fb72c2e24ab3ce22bc
code/v12/m10_exact.py	202e65ba9d96c8b93b91a305061e845cc0a03e006e1a9525bcb242076876d773
code/v12/m10_pslq.py	74df977c35cc874c14b09ac4413ea56a4c22f745daea393afc140a02797de7e4
code/v12/make_tables.py	c34407b1fdce5641ce96b36bb878e0534c0cc82ad4371e8cbdb79c4fdce16e53
code/v12/phi_asym_fit.py	7fc2ba6ecddf002e4c7b5e5536014a8b7aad9c082a61ebe53ec7e86ca4923037
code/v12/structure_test.py	93ed4137164b4514f64c991e3d254cb53c6c5757b689f1b5312bacfd7ab1aa8c
code/v12/wall_certificate.py	3f6e849b3abc9e9a838511693944ba7cb74914730d6c658189b10ae9bec5609d
code/v12/walls_driver.sh	29de260c64a06ea0d926f0f74904eb90f8ea2dd0f057ea2656ed5e09a7dd5c7b
code/build_v12.py	aaeddb313b8530955ccb27be7eb4d869a45969ff44cf19a5647bf2b42906fb4d
code/artifacts_v12.py	1a14fb8922da23c875b74b57db326beae92a6200d466284ee1dfe3309ea5c134
v12 source fragments (10 files)	

File	SHA-256
v12src/abstract.tex	35a9eaa1bee58b57325ddd2ced0a1766b71034a8fe7b65caad11b314961ccf3e
v12src/bib_v12.tex	b50296bc16b71c6f30c0e46371fd6e0b51342d3899c4e0447c18b92c3dcc1d01
v12src/m10_table.tex	b53cd6e0767926f7d17c0dcca0cfc4b675a78746c0c3af1467c57fd134f9940d
v12src/opentable.tex	d8c870ee5a528d43f1b9a8f38394b718246d8849396343bc70df294857eb164f
v12src/release_items.tex	02a0ceb5cb2863352d4851f12f522f8904f9a8fb14345c566f25ee264c6fb3ad
v12src/scope.tex	6cc18e30fd344499325571b3e2ef2cbbdc217c0d113747a5f7eff80c5dd16062
v12src/sec_v12.tex	b037e2e87c0f55ffb24335c0c7a4a8684e2ebfaa5c04f2699829b3b84cda8bf5
v12src/subtitle.tex	af9a7488e1b84118cab4b33c11275717ef6c147f75fa36e630cc95e24bfbb284
v12src/walls_table.tex	69a47164e45e335d8fd835271391c8a0ab58dcd12ee4070d7999b25a70abcd21
v12src/walls_text.tex	949dbad49645b19e8788aa09057247367a7915c3e402c371fb6c1aa5ae17c65b
Closed form and M10 (5 files)	
logs/v12/F_closed.json	52555dbfa415868a37e5d028282ddefd4bba54b887256c0f7a830dcf36ae31de
logs/v12/structure_test.json	1d547e416e764d165eb735eccb5108acd4ce1ef290a677b61a53b9682ae94d6d
logs/v12/m10_exact_dps40.json	e3289ebaebf2f4489950862d17a57820aea4bfae3adaa77b9fb5fac0d42f2e27
logs/v12/counterterm.json	edcb0e4f321a645b8d66ae7570305a3322e27bcedb13959f15537572841b4af0
logs/v12/m10_pslq.json	45975a515bc74bb28632ac648909dd8af68129880056a2c083790aa6e7f9f308
Wall certificates (19 files)	
logs/v12/walls/apex_cert_1_2_CR.json	1bd0bc1bb1b437d1f76b90328b672f79a3bcda95843cf4fc9cda565e058b4e57
logs/v12/walls/apex_cert_1_2_CR_vectors.json	801d520ac941843d8d728449035e904740c5b200dbd715898544a16af8a6ef14
logs/v12/walls/apex_cert_1_2_F.json	c64cf4905b4234b3d083c4b0b8a97509950d34ae49b87348fc2be9ff6dbb57a8
logs/v12/walls/apex_cert_1_2_F_vectors.json	801d520ac941843d8d728449035e904740c5b200dbd715898544a16af8a6ef14
logs/v12/walls/apex_cert_1_3_CR.json	fc1ac350a8b2b6cff9bf59f6bc549a8df90c6683373b0627c518b051b10c62e3
logs/v12/walls/apex_cert_1_3_CR_vectors.json	5a822527a80a1825993c7e1b506820040e94a73cdce924a3958281e35318f304
logs/v12/walls/apex_cert_1_3_F.json	a1936f1bfc569d6884c5d4924d7735c9f6e3eded4473f1f5cb602e8aa4a1412a
logs/v12/walls/apex_cert_1_3_F_vectors.json	5a822527a80a1825993c7e1b506820040e94a73cdce924a3958281e35318f304
logs/v12/walls/apex_cert_2_3_CR.json	a8ef79e8533f9e701dbdce05c852bb55febdeb5b1793e3ffe7c44c1c7b2a87d4
logs/v12/walls/apex_cert_2_3_CR_vectors.json	5a822527a80a1825993c7e1b506820040e94a73cdce924a3958281e35318f304
logs/v12/walls/apex_cert_2_3_F.json	9c91707d46277ebc95d7cdd15f80361929ce522cd2e9ff73d7c68f0126e6f983
logs/v12/walls/apex_cert_2_3_F_vectors.json	9a60468c4977ac366c0a0154237887c9ec10a5f03a5802dcbd1d1e6493cfb590
logs/v12/walls/driver.log	5c91bc5fb3c4c9ab3cc8ceee39df4d1ff7bb892e5eb34ed8f809873d8f1dd969
logs/v12/walls/run_1_2_CR.log	6743f05ef33f0e5533c6658d58c844bd0aa482e82945900266ebd70e9808ceb2
logs/v12/walls/run_1_2_F.log	e6ccb77c4ae29bb0bb3bafd306c0558b5945b8e76b45184cd7f71106f3b53cd4
logs/v12/walls/run_1_3_CR.log	eacceb0421c58b556a319afbb873531c2f58299023d02520cd98c5965c14ea0e
logs/v12/walls/run_1_3_F.log	1a262adc22c9a96f7030764301f02860a7e2cbf88f46d1ef49cce04436035cdc
logs/v12/walls/run_2_3_CR.log	62622c79c276713f0b69f8c498df5ef80592b7897678f2a311e1ec1a5e2ca1ea
logs/v12/walls/run_2_3_F.log	e0b50e18a7cde49e044e7e09403a2bf7e97e82aca92cd696ea1160befafee5cf
Registry (1 files)	
logs/v12/registry_oct19b.sha	828d9afa530fa1dd0628ee751730b88bc2645a01f41adee08f0561e8d5e777e3
Integrity, environment (6 files)	
logs/frozen_v12_before.sha	cb9c2a11921e376aa078f2081cea58b7c0a8a0363a1fd6976bf7272df89e7b5b
logs/v12/tree_hashes_v12.json	ddf510d823e0af9de28430bdf7be42309eaab04b1d6b9a6c87b39b8e9735626d
logs/v11_hashes_before_v12.sha	b8d08e5c00ece0b8988b6df73a0294246700b6ee14241b2614e364e0af92fa9c
logs/v12/earlier_versions_sha.json	93b6c79020f3ad890c9878cc20307aa56a913d0a456afd262b5b1beb67b22a0f
logs/v12/env.json	f3c3b679dd87e7e87a4a5670b7e7dede9a62e737459a7a2a21d8fc562df914fa
CHECKPOINT_v12.md	64621e13a2f2765c05724debe7f5d481c93d51499e77ea3f5eff361aaeaf375a

G Source, version, and compilation declarations

Supplied material. The v6 manuscript and the Python-listing manuscript were supplied in full as text. Statements about their historical executions are attributed to those sources. In v8 the frozen XVI-P/XVI-Q engine files were read and replayed from read-only copies; nothing in those trees was edited (Section 13.6). v7 was read in full.

Companion source scope. The I-B and I-B-G public records support the host and test-domain comparisons cited above. The other Cheung companion descriptions are cited at the version

and locator level supplied with the revision material. Their full manuscripts and proofs are not represented as independently audited here. In particular, the source-qualified tensor-core and XVI-O1-ST comparisons are not silently upgraded to local proofs.

Primary analytical sources. The Bessel representation, standard special-function identities, and self-adjoint functional calculus are identified inputs. The normalizations, subtractions, Mellin reduction, and finite catalogue algebra used in the conclusions are written out above.

Code status. v7: the embedded program is newly consolidated source. v8: it was extracted unchanged and executed, together with the v8 programs in `code/v8/`; return codes, logs and SHA-256 values are in Appendix B. Software: Python 3 with SymPy (versions in `logs/v8/env.json`). No proof-assistant check is asserted. v9: programs in `code/v9/`, logs in `logs/v9/`, hashes in Appendix C; mpmath is additionally used for the 30-digit mode functionals. v10: programs in `code/v10/`, logs in `logs/v10/`, hashes in Appendix D. v11: programs in `code/v11/`, logs in `logs/v11/`, hashes in Appendix E. v12: programs in `code/v12/`, logs in `logs/v12/`, hashes in Appendix F.

Typesetting. This is a single-file pdfLaTeX source with an internal bibliography and inline listings. It requires no external figure, bibliography database, code file, or shell escape for typesetting. v12 was compiled with

```
pdflatex -halt-on-error XVI-O1-CONE-v12.tex   # repeated until stable
```

and the final log was checked for unresolved references and citations.

Authorship and approval. The author line follows the supplied manuscript. This revision does not independently authenticate authorship, archive provenance, or author approval. No independent review of v8–v12 has taken place; nothing was pushed or posted. Final submission requires the author's approval of the claims, scope, versions, and computational artifacts.

Standing non-claims. The cone, deficit, holonomy, extension, and CR branch remain prescribed inputs. No Einstein evolution, physical tensor Hessian, spacetime emergence, gravitational evaporation channel, or inherited characteristic quantum coefficient is constructed. XVI-N is not discharged; XV-B is not opened; H2/I3/J2 are not completed by this paper.

References

Marks (v12): * read in full; ‡ confirmed online (registry fetches of 18 October 2026, re-checked twice on 19 October 2026, `https://enter.grok.me/browse`; for the v9 sheets: source fetched, abstract and every theorem statement read); † from memory or from a supplied description only.

[1] * N. Cheung, *Scoped O.1 on the Prescribed Cone, Version 9*, 18 October 2026. Source `XVI-O1-CONE-v9.tex` (SHA-256 `3a8dd434...`); the base of this revision.

[2] † W. N. Bailey, *Generalized Hypergeometric Series*, Cambridge Tracts 32 (1935), reduction formulae for Appell's F_4.

[3] † H. R. P. Ferguson, D. H. Bailey and S. Arno, Analysis of PSLQ, an integer relation finding algorithm, *Math. Comp.* **68** (1999) 351–369.

[4] * N. Cheung, *Scoped O.1 on the Prescribed Cone, Version 11*, 19 October 2026. Source `XVI-O1-CONE-v11.tex` (SHA-256 `863bb08e...`); the base of this revision.

[5] * N. Cheung, *Scoped O.1 on the Prescribed Cone, Version 10*, 18 October 2026. Source `XVI-O1-CONE-v10.tex` (SHA-256 `d21c8d84...`); the base of this revision.

[6] † NIST Digital Library of Mathematical Functions, §8.17 (incomplete Beta function) and §15.4 (special values of ${}_2F_1$).

[7] * N. Cheung, *Scoped O.1 on the Prescribed Cone, Version 8*, 18 October 2026. Source `XVI-O1-CONE-v8.tex` (SHA-256 `153e4e9d...86e85f8`); the base of this revision.

[8] ‡N. Cheung, *I-B-G-S-B: Positive tensor-core selection and validated conical-kernel error bounds*, Enter Calculus Companion; registry source `new_oct18f/all/I-B-G-S-B.tex` (SHA-256 `a3ef5d15...29864`). Locators: `thm:selection`, `prop:warddefect`, `prop:curvaturetable`. Cited input only.

[9] ‡N. Cheung, *I-B-ECHO-v2: Diffracted echoes and finite supplied-report instruments*, Enter Calculus Companion; registry source (SHA-256 `9c3bfb55...7a745`). Cited input only.

[10] ‡N. Cheung, *X-J-ST-2: Charged scalar reports and a Ward-compatible free photon reference*, Enter Calculus Companion; registry source (SHA-256 `3d0bae7f...08e81`). Cited input only.

[11] ‡N. Cheung, *Hawk05-ST: Finite Enter/Page Proxies on a Prescribed Conical Spacetime*, registry source (SHA-256 `2cbd6c9f...b7f8e`). Not used for O.1.

[12] ‡N. Cheung, *I-C-v2: The Shredder Interface*, registry source (SHA-256 `6684c952...5d187`). Not used for O.1.

[13] ‡N. Cheung, *I-C-Ex-v2: Sharp Access Bounds, Residual Capacity*, registry source (SHA-256 `8051b3e1...15a6f6`). Not used for O.1.

[14] ‡N. Cheung, *I-C-Ex2: Eight Further Extensions of the Shredder Cut*, registry identifier new on 18 October 2026, stored in `new_oct18g/` (SHA-256 `69833777...2647d`). Not used for O.1.

[15] * N. Cheung, *Scoped O.1 on the Prescribed Cone, Version 7*, 18 October 2026. Source `XVI-O1-CONE-v7.tex` (SHA-256 `d0ba7b9c...5d132`); the base of this revision.

[16] * N. Cheung, *Scoped O.1 on the Cone, v6: Executed Apex Records, a Closed Constant-Profile Slavnov–Taylor Row, Weighted Apex Coefficients, and Ward-Identity Boundaries*, 18 October 2026. Supplied complete source `XVI-O1-CONE-v6.tex`. Historical execution and manifest claims are attributed to this source, not newly authenticated here.

[17] * N. Cheung, *XVI-O1-CONE: Python scripts (v1–v6)*, 18 October 2026. Supplied complete listing source `XVI-O1-CONE-scripts.tex`. The listings state that frozen XVI-P/Q files and run logs are external dependencies.

[18] ‡N. Cheung, *XVI-P*, Enter Calculus Companion, unpublished manuscript and computational records. Frozen p22 contract as identified in the supplied cone audit; later p23 excerpt is a comparison, not a silent contract change. Named engine files: `code/p5_entry/amplitude_rules.py`, `feynman.py`, and `dred.py`. Relevant supplied locators include `rem:o1`, `rem:p5`, and `rem:qds`.

[19] ‡N. Cheung, *XVI-Q*, Enter Calculus Companion, unpublished records. Supplied Q02 graph-normalization description and separately scoped Q11 release. Named comparison artifacts include `Q02_bulk_graphs.json` and `P18_D11.json`. v8 inspected and replayed named engine files from a read-only copy.

[20] †N. Cheung, *XVI-R*, frozen comparison record identified in the supplied cone audit. No theorem from its unavailable contents is imported here.

[21] ‡N. Cheung, *I-B: A Finite Enter Spacetime: Calibrated Destructive Time Readout on a Prescribed Conical Product*, version 1, Enter Calculus Companion. Relevant locators: `thm:instrument`, `prop:not-source`, `prop:no-return`, and `prop:record-residual`. Public manuscript record: https://enter.grok.me/browse/I-B.

[22] ‡N. Cheung, *I-B-G: Intrinsic and Holonomy-Twisted Green Operators on a Prescribed Conical Spacetime*, version 2, Enter Calculus Companion. Relevant locators: `prop:nonGH`, `thm:green`, `thm:speed`, and `rem:compact-exact`. Public manuscript record: https://enter.grok.me/browse/I-B-G.

[23] ‡N. Cheung, *I-B-G-S: Intrinsic Spin-1 and Spin-2 Green Operators on the Prescribed Conical Spacetime*, version 2, Enter Calculus Companion. Supplied version-qualified excerpts, especially `thm:spin1`, `thm:spin2`, `def:geo`, and `thm:selection`. Free source-test compatibility is not an interacting loop theorem.

[24] ‡N. Cheung, *I-B-G-S-B*, Enter Calculus Companion. Supplied source descriptions of positive tensor-core selection and exceptional source-test defects. Relevant supplied locators: `cor:exclude`, `thm:contrast`, and `prop:warddefect`. A complete independent tensor-core proof is not asserted here.

[25] ‡N. Cheung, *I-B-VP: Vacuum Polarization of the Friedrichs Ground State on a Prescribed Conical Spacetime*, version 3 and supplied predecessor passages, Enter Calculus Companion. Relevant supplied locators: `lem:infrared` and `thm:friedrichs`. One-point/compact-test comparison only.

[26] ‡N. Cheung, *I-B-VP-1: Prescribed Apex Finite Parts on a Conical Enter Spacetime: Wick Square, Stress Tensor, and Charged Current*, version 3, October 2026. Supplied source-qualified descriptions of one-point extension freedom and Ward conditions.

[27] ‡N. Cheung, *I-B-VP-4*, Enter Calculus Companion. Supplied frozen-source Maxwell-response comparison. The current and its Maxwell response are distinct objects; no interacting cone Ward theorem is imported.

[28] ‡N. Cheung, *I-B-VP-5-Ex1-B*, version 1, 18 October 2026. Supplied excerpts on heat normalization, local stress running, continuity of complete finite parts, and non-universality. Relevant locators: `thm:running`, `thm:continuity`, `thm:equalradius`, and `thm:trace`.

[29] ‡N. Cheung, *I-B-VP-5-Ex2: Smooth Flux-Core Selection on a Prescribed Cone*, version 1, 18 October 2026. Supplied excerpts on twisted scalar Friedrichs selection and critical-window Dirac selection.

[30] ‡N. Cheung, *I-B-VP-5-Ex2-B*, version 1, 18 October 2026. Supplied excerpts on exhausted-annulus Mosco arguments, endpoint continuity, and finite-core qualifications.

[31] ‡N. Cheung, *I-B-ECHO-v2: Diffracted Echoes and Finite Supplied-Report Instruments*, version 2, October 2026. Supplied source-qualified comparisons: `thm:readout`, `thm:core`, and `cor:nonreconstruction`. No continuum acquisition or loop-residue identity is imported.

[32] ‡N. Cheung, *I-C-v2: The Shredder Interface: Destructive Readout, Black-Hole Analogues, and Quantum Computers*, version 2, October 2026. Supplied excerpts and source descriptions. Theorems A–F separate referenced recovery, classical histories, conditional thermal accounting, complete classical access, endpoint underdetermination, and persistent classical output.

[33] ‡N. Cheung, *En09-A: Realization of Relational Time by Destructive Enter on a Clock Register*, version 1, Enter Calculus Companion. Supplied excerpts, especially `thm:main`. Complete reference-preserving instrument and consumed-clock conditioning; no continuum acquisition is inferred.

[34] ‡N. Cheung, *En07-A*, *En07-B*, and *En07-C*, version 1 source records, Enter Calculus Companion. Supplied complete-complement and classical-output comparisons. Model-specific channels and physical endpoint assumptions are not imported.

[35] ‡N. Cheung, *X-J-ST-2*, Enter Calculus Companion. Identifier and `prop:noacquisition` cited from the supplied revision memorandum. No complete manuscript or additional theorem is asserted to have been inspected.

[36] ‡N. Cheung, *Oai290-A*, version 1, Enter Calculus Companion. Supplied excerpts distinguish whole-code recovery, named observable targets, and declared geometric models. The companion identifier does not make an external catalogue family a premise of this apex calculation.

[37] ‡N. Cheung, *XVI-O1-ST-v5*, Enter Calculus Companion. Source-qualified sign-integrity witness described in the supplied revision memorandum: D01 sixth coordinate $-4/3$ versus $+4/3$, with difference $(8/3)e_6$. Full underlying artifacts were not supplied for this revision.

[38] ‡*Hawk05-A*, *Hawk05-B*, and *Hawk05-C*, version 1 supplied manuscript records, attributed to N. Cheung in the supplied source list. Finite operational recovery, Page–Sen proxy, and conditional extrapolation comparisons only. Their authorship and full artifacts are not independently authenticated here; no gravitational Page theorem is imported.

[39] ‡N. Cheung, *XVI-N*, Enter Calculus Companion. Supplied version-qualified boundary, including the identifier `thm:massless-obstruction`. No absent theorem is reconstructed or discharged here.

[40] ‡N. Cheung, *XIV-H*, *XIV-I*, and *XIV-J*, version 1 source records, Enter Calculus Companion. Supplied historical H2 ending, narrower I2 construction, and unresolved full I3/J2 characteristic-map obligations.

[41] ‡N. Cheung, *XV-B*, version 2, Enter Calculus Companion. Supplied technical-interface descriptions. The technical manuscript is distinguished from the unopened downstream quantum-admission exit.

[42] †D. V. Fursaev, "The heat kernel expansion on a cone and quantum fields near cosmic strings," *Classical and Quantum Gravity* **11** (1994), 1431. DOI: `10.1088/0264-9381/11/6/008`. arXiv:`hep-th/9309050`. The full-opening-angle convention and equations (2.11)–(2.15) fix the comparison in Remark 5.5.

[43] †D. V. Fursaev and G. Miele, "Cones, spins and heat kernels," *Nuclear Physics B* **484** (1997), 697–723. DOI: `10.1016/S0550-3213(96)00631-1`. arXiv:`hep-th/9605153`.

[44] †D. V. Vassilevich, "Heat kernel expansion: user's manual," *Physics Reports* **388** (2003), 279–360. DOI: `10.1016/j.physrep.2003.09.002`. arXiv:`hep-th/0306138`.

[45] †F. W. J. Olver et al., editors, *NIST Digital Library of Mathematical Functions*, Chapter 10, especially Sections 10.27, 10.29, 10.32, and 10.43. Modified-Bessel relations, recurrences, and integral transforms. `https://dlmf.nist.gov/10`.

[46] †J. Dereziński and S. Richard, "On Schrödinger operators with inverse square potentials on the half-line," *Annales Henri Poincaré* **18** (2017), 869–928. DOI: `10.1007/s00023-016-0520-7`. arXiv:`1604.03340`.

[47] †B. S. Kay and U. M. Studer, "Boundary conditions for quantum mechanics on cones and fields around cosmic strings," *Communications in Mathematical Physics* **139** (1991), 103–139. DOI: `10.1007/BF02102731`.

[48] †D. Stöckinger, "Regularization by dimensional reduction: Consistency, quantum action principle, and supersymmetry," *Journal of High Energy Physics* **2005**(03), 076. DOI: 10.1088/1126-6708/2005/03/076. arXiv:hep-ph/0503129.

[49] †M. Horodecki, P. W. Shor, and M. B. Ruskai, "Entanglement breaking channels," *Reviews in Mathematical Physics* **15** (2003), 629–641. DOI: 10.1142/S0129055X03001709. Preprint title: "General Entanglement Breaking Channels," arXiv:quant-ph/0302031.

[50] †M. A. Nielsen, "A simple formula for the average gate fidelity of a quantum dynamical operation," *Physics Letters A* **303** (2002), 249–252. DOI: 10.1016/S0375-9601(02)01272-0. arXiv:quant-ph/0205035.

[51] †D. Reeb and M. M. Wolf, "An improved Landauer principle with finite-size corrections," *New Journal of Physics* **16** (2014), 103011. DOI: 10.1088/1367-2630/16/10/103011. arXiv:1306.4352.